

# Masked, Not Missing: Mid-Depth Retrieval Collapse in T5-Family Encoders

Anonymous ACL submission

## Abstract

Practitioners often read intermediate layers of off-the-shelf encoders, check them for anisotropy with mean pairwise cosine, and correct the embedding space post hoc. We test this practice on medieval Latin canon law, where a scholarly edition labels which manuscript copies (witnesses) share a source. In the raw T5 encoders LaTa, PhilTa and mT5-base, 26 of 36 layers rank same-source pairs near chance, although their first and last layers retrieve well. Mean pairwise cosine gives no reliable warning of this collapse, while the share of variance on the first principal component is high at every collapsed layer. The signal is masked, not missing: a projection fit on training embeddings alone brings all 100 layers of our six-encoder panel to AUROC 0.962–0.987. Four of the five raw T5-family encoders we test, these three and T5-v1.1-base, collapse, and all four share the T5 v1.1 layout; encoder-only siblings pre-trained on the same corpora and the original T5-base do not. Removing about three principal directions restores every collapsed layer, while zeroing the ten largest coordinates restores none. After the projection the six encoders route witnesses to their source or to “new” about equally well, but no frozen encoder at its train-selected layer beats a character n-gram reference.

## 1 Introduction

Off-the-shelf encoders are a common tool for finding reused text in digital-humanities collections (e.g., [Manjavacas et al., 2019](#); [Schelb et al., 2026](#)). Practitioners often read an intermediate layer, because intermediate layers can outperform the final one ([Skean et al., 2025](#)). They check the space for anisotropy, usually with the mean cosine between embeddings ([Ethayarajh, 2019](#)). They then correct it, for example by removing the top principal components ([Mu and Viswanath, 2018](#)) or by whitening ([Su et al., 2021](#)). This practice assumes that the

check warns when a layer retrieves poorly. Mean cosine is already known to be a poor isotropy measure ([Rudman et al., 2022](#)); we ask whether it, or any label-free statistic, warns at the layers where retrieval fails.

Medieval Latin suits this test. The Carolingian Canon Law project (CCL; [Firey, 2009](#)) keys each transcribed manuscript unit to the source it copies, with expert labels that no model shaped. Our testbed holds 1,705 *witnesses*, manuscript copies of 840 sources. We evaluate two tasks: *ranking* pairs of witnesses by cosine, scored by AUROC, and *routing* each witness to its source or to “new”. Because witnesses copy their source’s wording, a character n-gram reference ranks pairs at 0.987 AUROC, so a retrieval failure points to the representation. Latin also offers T5 and encoder-only models from the same authors and pretraining corpora (Section 3).

Three raw T5 encoders, trained with no embedding objective, fail at mid-depth (Figure 1). In LaTa, PhilTa and mT5-base ([Xue et al., 2021](#)), 26 of 36 layers rank same-source pairs at AUROC 0.50 to 0.66, near chance. Their first and last layers still retrieve: LaTa ranks at 0.934 at layer 1 and 0.938 at layer 12. The three embedding-trained encoders of our panel stay at or above 0.806 at every layer. The usual check, mean pairwise cosine, gives no reliable warning in either direction. The share of variance on the first principal component (top-PC share) is high at every collapsed layer: necessary for collapse in our models, but not sufficient.

The signal is masked, not missing. All-but-the-Top (ABTT; [Mu and Viswanath, 2018](#)) removes the mean and the top principal components from every embedding. We fit it on training embeddings only, with no parameter update. It brings all 100 layers of the six panel models to AUROC 0.962 to 0.987, the 26 collapsed layers included. The projection is linear and sees no test label, so the ranking it recovers is already present in each collapsed layer.

Controls show which models collapse. The encoder-only siblings LaBERTa and PhilBERTa never fall below 0.826 and 0.883. Within the T5 family, T5-v1.1-base collapses on the same Latin text (lowest AUROC 0.489), while the original T5-base (Raffel et al., 2020) never falls below 0.816. Four of the five raw T5-family encoders we test collapse, and all four share the T5 v1.1 layout: a gated-GELU feed-forward block (Shazeer, 2020) and untied input and output embeddings (Google Research, 2020). Across the ten models we test, the collapse tracks the T5 v1.1 layout; we do not isolate which change is responsible. T5-base also saw supervised tasks in pretraining, so this contrast does not separate layout from pretraining mix.

Inside the collapsed layers, about three directions mask the signal. Removing three principal components brings all 26 layers to AUROC 0.914 or above, and mT5-base keeps improving up to ten. Zeroing the ten largest coordinates restores none of them, although the first principal component (the dominant direction) sits mostly on such coordinates. In LaTa, pooling without the comma, the period and </s> restores every collapsed layer; in mT5-base, no set of up to 100 token types does.

This paper makes four contributions. (1) The first per-layer retrieval profile of T5 encoder states that we know of, each layer read on its own rather than averaged with another layer or concatenated with decoder states (Section 4). (2) A within-family contrast: T5-v1.1-base collapses on the same text where the original T5-base does not, and every collapsing model has the T5 v1.1 layout; encoder-only siblings on the same corpora do not collapse, and the high-share middle layers also appear on English text (Section 5). (3) A localization of the mask to about three principal directions, which zeroing the ten largest coordinates does not remove, with a token-level account in LaTa and none in mT5-base (Section 6). (4) A measure of what the projection gains for routing and where it stops, set against a character n-gram reference (Section 7). We will release the frozen testbed, split, evaluation code and character n-gram reference scores. We localize and repair the collapse; we do not explain why the T5 v1.1 layout produces it.

## 2 Related Work

**Anisotropy statistics.** Contextual representations occupy a narrow cone, and the usual check is the mean cosine of random pairs (Gao et al., 2019;

Ethayarajh, 2019). This statistic is not an appropriate measure of isotropy (Rudman et al., 2022). At the final layer, the variance share of the top principal direction predicts when rank-based similarity beats cosine (Parupudi, 2026), and effective rank moderately predicts retrieval quality (Rottach et al., 2025). Yet geometric statistics do not reliably predict downstream performance (Nastase and Merlo, 2025; Kulkarni et al., 2026). We relate both statistics to retrieval at every layer (Section 4).

**Outlier dimensions and the tokens that carry them.** One to three rogue dimensions dominate cosine similarity in pretrained transformers (Timkey and van Schijndel, 2021). In BERT they are tied to a few LayerNorm weights and to token frequency (Kovaleva et al., 2021; Puccetti et al., 2022). Clipping them improves mean-pooled sentence embeddings (Luo et al., 2021), though multilingual BERT is anisotropic without them (Rajae and Pilehvar, 2022). Decoder language models carry massive activations on starting and delimiter tokens (Sun et al., 2024). The original T5’s encoder carries outliers that grow with depth (Zhao et al., 2025), and outliers persist through fine-tuning (Rudman et al., 2023). In BERT most outliers sit on [SEP], the period and the comma (Bondarenko et al., 2023), and removing punctuation and special tokens before pooling improves sentence similarity (Fuster Baggetto and Fresno, 2022). We test these accounts on pooled T5 encoder states, comparing coordinate zeroing, direction removal and token dropping (Section 6).

**Layerwise analysis and post-hoc correction.** Prior T5 embedding results use the final layer (Ni et al., 2022), the first and last layers averaged (Huang et al., 2021), or encoder and decoder states concatenated (Godey et al., 2024). Post-hoc corrections remove the mean and top principal components (Mu and Viswanath, 2018), whiten (Su et al., 2021), or down-weight frequent tokens (Arora et al., 2017). Timkey and van Schijndel (2021) apply them at every layer of four models and find that rogue dimensions obscure representational quality; our reading of the collapse as masked follows theirs. Whitening beats zeroing outlier dimensions in a multilingual encoder (Hämmerl et al., 2023), centering alone barely changes embedding models (Ren et al., 2026), and after fine-tuning isotropy corrections become ineffective (Rajae and Pilehvar, 2021). For routing, the nearest precedent is out-of-distribution detection in supervised classi-

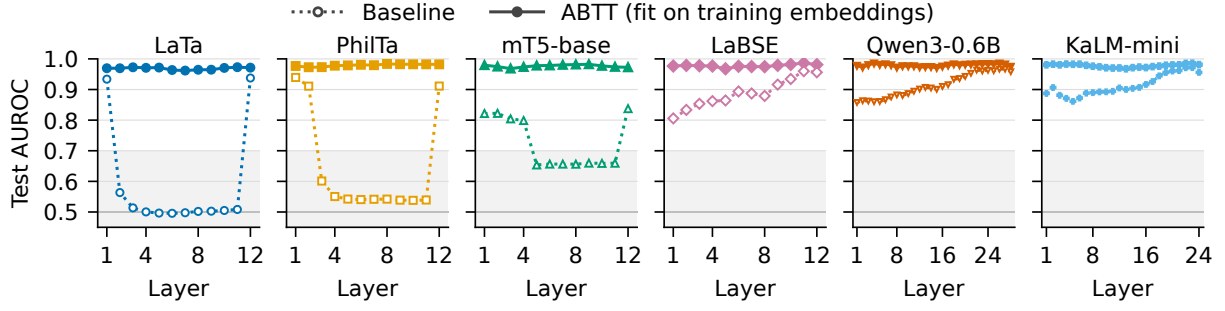


Figure 1: Test ranking AUROC at every layer of the six panel models. Dotted lines, open markers: baseline (mean-pooled hidden states). Solid lines: All-but-the-Top (ABTT: remove the mean and the top principal components), fit on training embeddings only. Gray band: AUROC below 0.70, the collapsed range; horizontal line: chance. After ABTT every layer scores 0.962 to 0.987.

fiers, where low-variance components carry much of the signal (Kamoi and Kobayashi, 2020; Podolskiy et al., 2021), and nearest-neighbor scores depend on feature geometry (Sun et al., 2022). Lexical methods match or beat dense retrievers on Latin intertextuality (Schelb et al., 2026), while dense encoders win on paraphrastic reuse (de la Selle, 2026). We use an existing projection, ABTT, as a probe.

### 3 Testbed and Protocol

**Corpus and labels.** The testbed comes from CCL (Firey, 2009), an open-access edition of early medieval, mostly ninth-century, canon-law manuscripts. A *witness* is one manuscript’s copy of a source text, kept as one file. Witnesses that copy the same source share a *directory*; the testbed holds 1,705 witnesses in 840 directories. The labels are expert-assigned by the project and checked by collation in its parallel display, so no model shaped them (Appendix A).

**Split and train-only rule.** We split the witnesses into 847 training and 858 test witnesses, and pairs form only within a split (Appendix B). This gives 565 positive training pairs and 596 positive test pairs. Of the test witnesses, 535 have a same-source partner in test (*existing*) and 323 do not (*new*). The layer, the routing threshold and the post-hoc correction are all chosen or fit on the training split. Each headline number (Table 3) reads a model at its layer of highest training AUROC for ranking, or of highest training routing accuracy (DirAcc@1, defined below) for routing (Table 20). The test split selects nothing.

**Tasks and metrics.** *Ranking* scores every pair of test witnesses by the cosine of their mean-pooled embeddings. We report AUROC with same-source pairs as positives. *Routing* is open-set: each test

witness goes to an existing directory or to “new”. Each test directory scores a witness by the highest cosine between the witness and any member. A pseudo-directory NEW scores a fixed threshold  $\tau$ , chosen to maximize training F1 at separating same-source from different-source pairs. The witness goes to the highest-scoring directory. DirAcc@1 counts a decision as correct if it names the labeled directory, or NEW for a new witness. Assignment accuracy scores only the existing-versus-new decision; always answering existing scores 62.4%.

A *collapsed layer* has baseline test AUROC below 0.70. The cutoff only describes layers; it selects no reported number. Two label-free statistics, computed on the training embeddings, describe each layer. The *mean pairwise cosine* is the usual anisotropy check (Ethayarajh, 2019). The *top-PC share* is the share of variance on the first principal component of the centered embeddings.

**Models.** Of the ten models in Table 1, six form the *panel*, which carries the full analysis over 100 layers. Three are raw T5 encoders with no embedding objective: the Latin LaTa (Riemenschneider and Frank, 2023d), its Ancient Greek, Latin and English sibling PhilTa (Riemenschneider and Frank, 2023a), and mT5-base (Xue et al., 2021). Three are embedding-trained: LaBSE (Feng et al., 2022), Qwen3-Embedding-0.6B (Qwen3-0.6B; Zhang et al., 2025) and KaLM-mini (Hu et al., 2025). Four raw controls complete the main text. Two are the encoder-only siblings LaBERTa and PhilBERTa. LaTa and LaBERTa were released by the same authors and were both trained on the Corpus Corporum (Riemenschneider and Frank, 2023d,c). PhilTa and PhilBERTa share authors and pretraining data (Riemenschneider and Frank, 2023a). The other two are English T5 encoders that share a tokenizer: T5-v1.1-base (Google Re-

| Model                                  | Type | L  | AUROC <sub>min</sub> ( $\ell$ ) | PC1 <sub>max</sub> | Coll. |
|----------------------------------------|------|----|---------------------------------|--------------------|-------|
| <i>Raw T5 encoders, T5 v1.1 layout</i> |      |    |                                 |                    |       |
| LaTa                                   | T5   | 12 | 0.496 (6)                       | 0.952              | 10    |
| PhilTa                                 | T5   | 12 | 0.538 (10)                      | 0.858              | 9     |
| mT5-base                               | T5   | 12 | 0.654 (5)                       | 1.000              | 7     |
| T5-v1.1-base                           | T5   | 12 | 0.489 (2)                       | 0.975              | 10    |
| <i>Raw T5 encoder, original layout</i> |      |    |                                 |                    |       |
| T5-base                                | T5   | 12 | 0.816 (11)                      | 0.545              | 0     |
| <i>Raw encoder-only siblings</i>       |      |    |                                 |                    |       |
| LaBERTa                                | Enc. | 12 | 0.826 (1)                       | 0.254              | 0     |
| PhilBERTa                              | Enc. | 12 | 0.883 (6)                       | 0.162              | 0     |
| <i>Embedding-trained</i>               |      |    |                                 |                    |       |
| LaBSE                                  | Enc. | 12 | 0.806 (1)                       | 0.524              | 0     |
| Qwen3-0.6B                             | Dec. | 28 | 0.858 (1)                       | 0.183              | 0     |
| KaLM-mini                              | Dec. | 24 | 0.861 (5)                       | 0.419              | 0     |

Table 1: The ten main-text models, read as mean-pooled hidden states with no correction. Type: T5 encoder (T5), encoder-only (Enc.) or decoder (Dec.; KaLM-mini runs with bidirectional attention). L: blocks read. AUROC<sub>min</sub>: lowest test ranking AUROC over layers, at layer  $\ell$ . PC1<sub>max</sub>: peak top-PC share of the training embeddings. Coll.: collapsed layers (AUROC below 0.70). Only the four raw T5 encoders with the T5 v1.1 layout (gated-GELU feed-forward, untied embeddings) collapse.

search, 2020) and the original T5-base (Raffel et al., 2020). We say a model has the *T5 v1.1 layout* when it has a gated-GELU feed-forward (Shazeer, 2020) and untied input and output embeddings (Google Research, 2020). LaTa, PhilTa, mT5-base and T5-v1.1-base have it; T5-base keeps a ReLU feed-forward and tied embeddings. We mean-pool the hidden states of every transformer block, dropping only padding and tokens that decode to no text, such as SentencePiece’s bare word-boundary piece; special tokens such as `</s>` and punctuation are kept. Mean pooling routes better than last-token pooling in all five models we checked (Appendix J). Sentence-T5 (Ni et al., 2022) and SPhilBERTa (Riemenschneider and Frank, 2023b) appear only in Table 7.

**Post-processing and lexical reference.** ABTT (Mu and Viswanath, 2018) subtracts the training mean and removes the top  $D$  principal components of the centered training embeddings from every vector. We choose  $D \in \{1, 2, 3, 5, 7, 10\}$  per layer by training DirAcc@1. ABTT is fit on training embeddings only and updates no model parameter; we also call it *the projection*. SIF frequency weighting (Arora et al., 2017) and reduced-rank whitening (Su et al., 2021) are baselines in Appendix K. The task is lexically solvable by design, because the witnesses of one source copy its wording. Our

*character n-gram reference*, TF-IDF over character 3–5-grams fit on training witnesses, reaches 0.987 AUROC and 89.9 DirAcc@1 (Table 19), the level the projection approaches.

## 4 Collapse by Depth

This section asks whether the middle layers of our encoders retrieve, and whether the usual anisotropy check warns when they do not. In three of the six panel models most middle layers rank near chance, and a projection fit on training embeddings restores them all: the signal is masked, not missing. Mean pairwise cosine does not flag these layers, while top-PC share is high at every one of them.

**LaTa, PhilTa and mT5-base fall to near chance at mid-depth.** Figure 1 gives baseline ranking AUROC at every layer of the six panel models. In LaTa, PhilTa and mT5-base, 26 of the 36 layers collapse, ranking at AUROC 0.50 to 0.66. They are layers 2–11 of LaTa, 3–11 of PhilTa and 5–11 of mT5-base. The next-lowest layer in the panel ranks at 0.799 (mT5-base, layer 4), so in the panel any cutoff from 0.66 to 0.79 gives the same split. A cutoff on training AUROC picks the same 26 layers, so no test score defines the collapse. The first and last layers still retrieve: LaTa ranks at 0.934 at layer 1 and 0.938 at layer 12. The embedding-trained models stay at or above 0.806 at every layer. Section 5 shows that not every T5 encoder collapses.

**A train-only projection restores ranking at every layer of the six panel models.** ABTT (Section 3) lifts all 100 layers of the panel to AUROC 0.962 to 0.987. The 26 collapsed layers are included (Figure 1, solid lines). Reduced-rank whitening, fit on the same training embeddings, ranks within 0.021 of ABTT at every layer (Table 29, Appendix K). The projection is linear and never sees a test label. The ranking it recovers must therefore already be present in each collapsed layer. We use ABTT to test the reading of Timkey and van Schijndel (2021), that post-processing reveals quality which a few dominant dimensions hide, at layers that rank at chance. The restoration holds for ranking only: routing after ABTT still varies by layer, from 75.6 to 90.2 DirAcc@1 (Appendix I).

**Mean pairwise cosine gives no reliable warning.** Figure 2a plots mean pairwise cosine, computed on training embeddings, against baseline test AUROC. It fails in both directions. The collapsed layers of LaTa, PhilTa and mT5-base have a lower mean

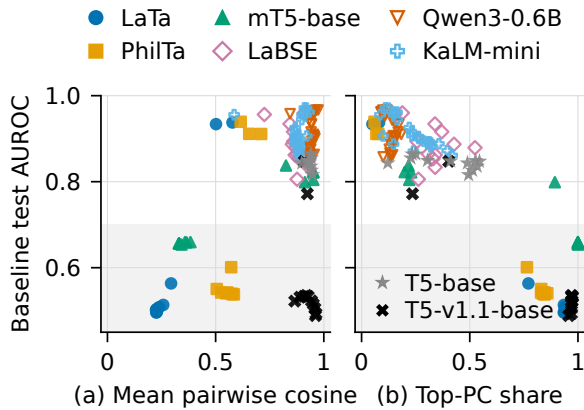


Figure 2: Baseline test AUROC against (a) mean pairwise cosine and (b) top-PC share (share of variance on the first principal component) of the training embeddings, one point per layer, for the six panel models, T5-base and T5-v1.1-base (filled: raw T5-family encoders; open: embedding-trained). Gray band: AUROC below 0.70, the collapsed range. Mean cosine fails in both directions. Every collapsed layer has a top-PC share of at least 0.76, but a high share does not imply collapse (mT5-base layer 4: 0.893, AUROC 0.799).

cosine (0.22 to 0.58) than the embedding-trained models, which retrieve well (0.59 to 0.97). T5-v1.1-base (Section 3), which collapses on the same Latin text (Table 1), sits at a high mean cosine (0.87 to 0.96). So neither a high nor a low mean cosine marks a collapsed layer. Mean cosine is already known to be a poor measure of isotropy (Rudman et al., 2022). Our result adds that it gives no warning at the layers where retrieval fails.

**Top-PC share is high at every collapsed layer.** Top-PC share needs no labels either (Figure 2b). It is at least 0.76 at every collapsed layer of the panel and at every collapsed layer of T5-v1.1-base. In the three embedding-trained panel models it peaks at 0.524 (LaBSE). A high share is necessary for collapse in our models, but it is not sufficient. mT5-base layer 4 has a share of 0.893 and still ranks at 0.799. Sentence-T5 (Ni et al., 2022), an embedding-trained T5 encoder reported in the appendix, combines shares of 0.66 to 0.94 with AUROC 0.800 to 0.891 at layers 2–11 (Table 7). The share marks the collapsed regime. It does not order layers inside it (Appendix C).

## 5 Collapse Across Models

This section asks which models collapse and what they share. Of the ten models in Table 1, four collapse: LaTa, PhilTa, mT5-base and T5-v1.1-base. All four have the T5 v1.1 layout, and none of

the other six collapses at any layer.

**Encoder-only siblings pretrained on the same corpora do not collapse.** LaTa and PhilTa each have an encoder-only sibling with no embedding objective, released by the same authors with the same pretraining data (Section 3). The T5 members collapse, with lowest AUROC 0.496 (LaTa) and 0.538 (PhilTa). The siblings never fall below 0.826 (LaBERTa) and 0.883 (PhilBERTa). Their top-PC share also stays low, peaking at 0.254 and 0.162 against 0.952 and 0.858 in LaTa and PhilTa. Shared pretraining text and a missing embedding objective therefore do not produce the collapse on their own. Architecture, pretraining objective and tokenizer vary together within each pair, so the pairs cannot say which of them matters (Appendix D).

**T5-v1.1-base collapses; the original T5-base does not.** This contrast stays inside the T5 family. The two models share their tokenizer, the span-corruption objective and the C4 pretraining text (Raffel et al., 2020; Google Research, 2020). On the Latin corpus, T5-v1.1-base falls to 0.489 at layer 2, and its layers 2 to 11 collapse. T5-base stays at 0.816 or above at every layer, and its top-PC share peaks at 0.545. The two differ in layout (Section 3): T5-v1.1-base has the T5 v1.1 layout, and T5-base the original ReLU, tied-embedding one. By their checkpoint configurations, LaTa, PhilTa and mT5-base have the T5 v1.1 layout too (Table 7). Across the ten models we test, the collapse tracks the T5 v1.1 layout; we do not isolate which change is responsible. T5-base also saw supervised tasks in pretraining (Raffel et al., 2020), so this contrast does not separate layout from pretraining mix. We did not apply the projection to the controls (Limitations).

**In three T5 v1.1-layout models, the middle layers with high top-PC share appear on English text too.** T5-v1.1-base was pretrained on English alone, so its collapse on Latin could be an out-of-distribution effect. The middle-layer geometry, which needs no labels, says otherwise. We read mT5-base, PhilTa and T5-v1.1-base on 847 English passages of US court opinions (Harvard Law School Library Innovation Lab, 2024; Kandpal et al., 2025), paired one to one with the Latin training witnesses and matched in length. In all three models, layers 4 to 11 keep a top-PC share of at least 0.82 on English, and the first and last layers stay at or below 0.34 (Table 9 and Figure 7).

At layers 2 to 11 of T5-v1.1-base, top-PC share is 0.931 to 0.950 on English and 0.958 to 0.975 on Latin. In these models, then, the high-share middle layers do not depend on Latin input. The check covers geometry only: without labels, it says nothing about retrieval on English.

**In LaTa, contrastive fine-tuning leaves the middle layers collapsed.** We fine-tuned LaTa contrastively on 499 training pairs (Appendix H). Fine-tuning lifts the last layer from 0.938 to 0.984 AUROC. Layers 2 to 11 stay at 0.50 to 0.57. Their geometry barely moves: peak top-PC share is 0.945 after fine-tuning, against 0.952 before (Table 10). This is consistent with Hämmerl et al. (2023), who report that a sentence-tuned multilingual encoder has no large outlier dimension at its output layer but still has one at a middle layer. Output supervision at this scale does not reach these layers.

## 6 Localizing the Collapse

This section asks what masks the retrieval signal in the collapsed layers and where the mask sits. The mask is about three directions, and removing them restores every collapsed layer. Zeroing the ten coordinates that carry most of the dominant direction restores none. In LaTa three token types feed these directions; in mT5-base no set of up to 100 token types does.

We run each test at the 26 collapsed layers, fit every intervention on training embeddings, and read test ranking AUROC; a layer counts as restored at 0.90. We fixed most tests before running them (†, Table 2), either as our expectation or as the decision rule of a rival account (passage length); four of these fail. Figure 3 shows the two geometric tests; Appendix E gives per-layer tables and further controls.

**Zeroing the ten largest coordinates restores no collapsed layer.** In pretrained transformers a few outlier coordinates dominate cosine similarity (Timkey and van Schijndel, 2021). Zeroing them improves pooled similarity (Luo et al., 2021; Hämmerl et al., 2023). We therefore expected that zeroing five or fewer coordinates of largest variance on training embeddings would restore the collapsed layers. Zeroing up to ten, ranked by variance or by mean absolute value, restores none of the 26, and the best layer reaches 0.794 (Figure 3a). The dominant direction does load mostly on these coordinates: the ten of highest variance

| Test                            | Expected                         | Outcome                                                                                   | Result        |
|---------------------------------|----------------------------------|-------------------------------------------------------------------------------------------|---------------|
| <i>Coordinates</i>              |                                  |                                                                                           |               |
| Zero top $k$ coordinates        | $k \leq 5$ restores <sup>†</sup> | 0/26 restored at $k \leq 10$ , either ranking; best 0.794                                 | <b>failed</b> |
| Zero random ones                | no change                        | $\leq 0.0004$ at 10; 0/26 restored up to 400                                              | held          |
| <i>Principal directions</i>     |                                  |                                                                                           |               |
| Center only                     | little gain <sup>†</sup>         | median 0.503 (raw 0.541); 0/26 restored                                                   | held          |
| Remove PC1 only                 | $\geq 80\%$ of gain <sup>†</sup> | median 45%; $\geq 80\%$ at 1/26                                                           | <b>failed</b> |
| Remove the top 3 PCs            | –                                | 26/26 $\geq 0.91$ (lowest 0.914); mT5-base 0.934, 0.978 at 10                             | –             |
| Rank on the 10 removed PCs      | below retained <sup>†</sup>      | 0.489 vs. 0.977; below at 26/26                                                           | held          |
| <i>Tokens</i>                   |                                  |                                                                                           |               |
| LaTa: drop top types            | restores <sup>†</sup>            | comma, period, </s> (11% of tokens): 10/10, median 0.919; mass-matched random types 0.500 | held          |
| mT5-base: drop $\leq 100$ types | restores <sup>†</sup>            | 0/7; best 0.748                                                                           | <b>failed</b> |
| PC1 score vs. log length        | $ \rho  \geq 0.5^{\dagger}$      | 0/26; largest $ \rho $ 0.44                                                               | <b>failed</b> |

Table 2: Localization tests at the 26 collapsed layers (AUROC below 0.70) of LaTa, PhilTa and mT5-base, unless a row names one model. Outcomes are test ranking AUROC or medians over layers; “restores” means AUROC  $\geq 0.90$ . Every intervention is fit on training embeddings only. Coordinates are ranked by variance or mean  $|x|$ . PC: principal component of the centered training embeddings; PC1: the first. Gain: that of removing ten PCs; top types: those that feed the top PCs. †: written down before the run, as our expectation or as the decision rule of a rival account (passage length); the other rows are a control and an observation added afterwards.

carry a median 92% of its squared loading. Yet zeroing them leaves the direction dominant. On the remaining coordinates it still holds a median 72% of the variance, and it still masks retrieval. Zeroing ten random coordinates instead moves AUROC by at most 0.0004. Zeroing far more coordinates does restore LaTa and PhilTa: ranked by variance, the first layers recover at 50 coordinates and all 19 by 300 (by 400 under the mean-absolute-value ranking of Figure 3a); no mT5-base layer recovers even at 400. This extends the outlier-dimension account: the masking direction is concentrated on a few coordinates, yet its effect reaches beyond them. It agrees with Hämmerl et al. (2023), who find that whitening improves a multilingual encoder more than zeroing its outlier dimensions.

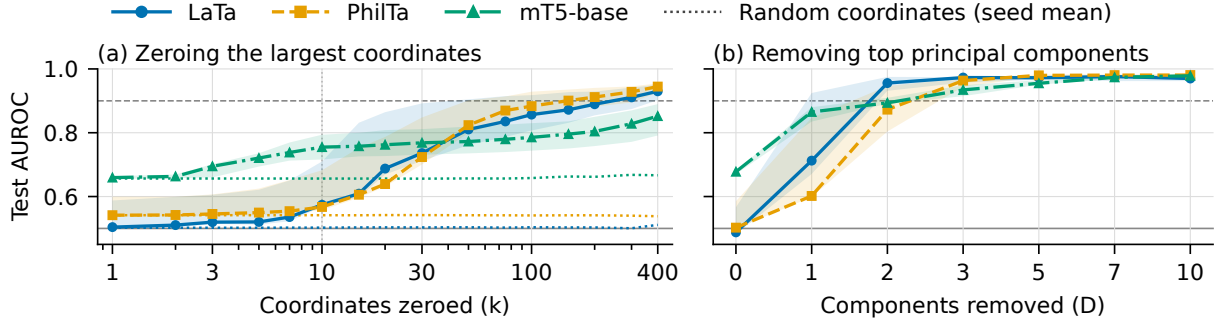


Figure 3: Test ranking AUROC at the 26 collapsed layers of LaTa (10 layers), PhilTa (9) and mT5-base (7); lines: median over a model’s collapsed layers; bands: their range. (a) Zeroing the  $k$  coordinates of largest mean absolute value; dotted:  $k$  random coordinates (mean of five draws), within 0.001 of the baseline up to  $k = 10$  and within 0.03 at 400. Up to  $k = 10$  (vertical line) the best layer reaches 0.794. (b) Removing the top  $D$  principal components ( $D = 0$ : centering only). Dashed line: AUROC 0.90. All fits use training embeddings.

**Removing three directions restores every collapsed layer.** ABTT removes whole directions instead of coordinates, so we vary the number  $D$  of principal components it removes (Figure 3b). Centering alone ( $D = 0$ ) does not help, as expected: the median AUROC over the collapsed layers falls from 0.541 to 0.503. This agrees with reports that centering alone barely changes embedding models (Ren et al., 2026). At these layers the first component holds a median 94% of the centered training variance. We expected that removing it alone would recover at least 80% of the gain from removing ten. It recovers a median 45%, and reaches 80% at only one layer of 26. Three components suffice for 0.91: they bring all 26 layers to 0.914 or above. mT5-base keeps improving up to ten components (median 0.934 at three, 0.978 at ten). At the collapsed layers the removed directions rank near chance. Ranking with the ten removed components alone gives a median AUROC of 0.489, against 0.977 for the retained subspace. At these layers, then, the mask occupies a few directions and the signal lies outside them.

**In LaTa, three token types feed the removed directions.** A pooled vector is the mean of its token vectors. If a few token types add large values along the removed directions, a witness’s position on them follows how often it uses those types. We ranked token types on training witnesses by how much of the variance of the witness scores on the top three components they contribute, then pooled without the top  $m$  types, for  $m$  up to 100. We expected some  $m$  to restore the collapsed layers. In LaTa,  $m = 3$  does. Pooling without the comma, the period and  $\langle /s \rangle$  drops 11% of the test tokens and restores all 10 collapsed layers (median 0.919).

Dropping random token types of the same training mass leaves the median at 0.500. The dominant direction also does not track passage length: its score correlates with log token count at  $|\rho| \leq 0.44$  at every collapsed layer. The tokens are carriers, not a cause: LaBERTa also pools commas, periods and  $\langle /s \rangle$ , and it does not collapse (Section 5). PhilTa needs more types: dropping its top 30, about a third of its tokens, restores all 9 of its collapsed layers.

**In mT5-base, no set of up to 100 token types restores a collapsed layer.** Dropping up to 100 token types, ranked the same way, restores none of its 7 collapsed layers; the best reaches 0.748. mT5-base also resists coordinate zeroing, yet removing three to ten directions repairs it. Its mask is thus confined neither to a few coordinates nor to a few token types, and we have no token-level account of it. Section 7 turns from ranking to routing, the task an editor faces.

## 7 Routing and Limits

This section asks what the projection gains for routing and where it stops. After ABTT the six encoders route about equally well at their train-selected layers. For the embedding-trained encoders, the gain also shows in a score that uses no threshold. No frozen encoder, at its train-selected layer, beats a character n-gram reference, and fine-tuning still adds about two routing points to Qwen3-0.6B and KaLM-mini.

**At train-selected layers, routing after the projection converges across the six encoders.** After ABTT, the six encoders route within 3.3 DirAcc@1 points of each other (95% CI 1.8 to 6.0), from 86.1 to 89.4, at the train-selected layers of Table 3. Without correction they span 39.3 points (34.6 to 43.7),

| Model                                                          | Ranking AUROC |             | Routing DirAcc@1 |             |
|----------------------------------------------------------------|---------------|-------------|------------------|-------------|
|                                                                | Base          | ABTT        | Base             | ABTT        |
| LaTa                                                           | 0.938         | 0.971       | 72.1             | 86.1        |
|                                                                | [.910,.963]   | [.948,.991] | [68.4,75.8]      | [83.2,88.9] |
| PhilTa                                                         | 0.939         | 0.983       | 69.3             | 88.3        |
|                                                                | [.918,.958]   | [.969,.993] | [65.2,73.5]      | [85.6,90.9] |
| mT5-base                                                       | 0.838         | 0.975       | 46.6             | 88.5        |
|                                                                | [.810,.866]   | [.954,.992] | [42.5,51.0]      | [85.7,91.2] |
| LaBSE                                                          | 0.956         | 0.987       | 81.5             | 88.5        |
|                                                                | [.932,.977]   | [.977,.995] | [78.1,84.7]      | [85.8,90.9] |
| Qwen3-0.6B                                                     | 0.966         | 0.973       | 80.3             | 89.4        |
|                                                                | [.950,.981]   | [.948,.990] | [76.9,83.6]      | [86.7,92.0] |
| KaLM-mini                                                      | 0.972         | 0.981       | 85.9             | 89.4        |
|                                                                | [.955,.986]   | [.963,.993] | [82.9,88.7]      | [86.7,91.9] |
| <i>Spread</i>                                                  | 0.134         | 0.016       | 39.3             | 3.3         |
| <i>Fine-tuned; Base = fine-tuned, ABTT = fine-tuned + ABTT</i> |               |             |                  |             |
| LaTa                                                           | 0.984         | 0.970       | 81.6             | 85.2        |
| Qwen3-0.6B                                                     | 0.996         | 0.994       | 90.8             | 91.4        |
| KaLM-mini                                                      | 0.997         | 0.994       | 91.7             | 92.5        |
| Char. n-grams                                                  | 0.987         |             | 89.9             |             |
|                                                                | [.976,.996]   |             | [87.3,92.3]      |             |

Table 3: Ranking (test AUROC) and routing (test DirAcc@1, %) without correction (Base) and after ABTT, each at its train-selected layer (Table 20); ABTT is fit on training embeddings only. Fine-tuned: contrastive fine-tuning on 499 training pairs (intervals in Table 15). Char. n-grams: TF-IDF over character 3–5-grams. Brackets: 95% bootstrap intervals over the 514 test directories. Spread: largest minus smallest printed value across the six encoders.

from 46.6 for mT5-base to 85.9 for KaLM-mini. The bootstrap interval of every encoder’s routing gain excludes zero. The gains run from 3.5 points for KaLM-mini (1.0 to 6.0) to 41.8 for mT5-base (Table 16, Appendix F). At train-selected layers, the choice of encoder then matters little for routing (t-SNE and UMAP maps: Appendix L).

**For the embedding-trained encoders, the routing gain does not depend on the threshold.** LaBSE, Qwen3-0.6B and KaLM-mini already rank well without correction (AUROC 0.956 to 0.972), so their routing gain cannot come from a ranking repair. ABTT still raises their DirAcc@1 by 7.0, 9.1 and 3.5 points. Routing applies a threshold  $\tau$  fit on training pairs, so we also score the existing-versus-new decision without one. The score is the AUROC of each test witness’s best-match cosine, existing against new witnesses. ABTT raises it by 0.037 for LaBSE, 0.055 for Qwen3-0.6B and 0.027 for KaLM-mini, and every paired interval excludes zero (Appendix G; per-cell values in Table 17). The gain in assignment accuracy also holds when both settings use their best test threshold, and a finer threshold grid changes no conclusion (Appendix F).

For these three encoders, centering alone, which removes only the training mean, scores 12 to 26 points lower in assignment accuracy than ABTT at the same layer (Table 16). We do not test what in the removed components produces the routing gain.

**No frozen encoder at its train-selected layer beats a character n-gram reference.** We compare against the character n-gram reference of Section 3 (Table 3). The best projected cells tie it: LaBSE ranks at 0.987, and Qwen3-0.6B and KaLM-mini route at 89.4, with every difference interval including zero. Two ranking cells fall below it with intervals that exclude zero, LaTa and Qwen3-0.6B (by 0.016 and 0.014 AUROC); mT5-base falls 0.012 below, with an interval that touches zero (0.000 to 0.027). LaTa’s routing cell falls 3.7 points below it (Table 16). Schelb et al. (2026) report the same for Latin intertextuality, where lexical methods match or outperform dense retrievers. Our claims therefore concern where the retrieval signal sits inside the encoders and what masks it.

**Fine-tuning Qwen3-0.6B and KaLM-mini adds about two routing points over the projection.** As a supervised reference, we fine-tune LaTa, Qwen3-0.6B and KaLM-mini contrastively on 499 training pairs (Appendix H). These references are flattered: 206 of the 535 existing test witnesses (38.5%) sit in a directory that supplied a training pair. Fine-tuned Qwen3-0.6B and KaLM-mini rank above the character n-gram reference in point estimate (0.996 and 0.997 AUROC; Table 3). After fine-tuning, the projection no longer helps ranking: AUROC falls slightly in all three fine-tuned models (by 0.002 to 0.014), in line with Rajae and Pilehvar (2021). For routing, fine-tuning plus ABTT beats the zero-shot projection by 1.8 points for Qwen3-0.6B (92.3 against 90.5) and by 2.1 for KaLM-mini (93.0 against 90.9); on LaTa the two tie at 87.7. These are means over five reseeds of the query and reference witnesses (Table 18, Appendix H); Table 3 shows the single-split cells, where the gains are 2.0 and 3.1 points and LaTa falls 0.9 points.

## 8 Discussion

Before reading an intermediate layer of a frozen encoder, fit a projection on training embeddings: it is cheap insurance. In the six panel models it scored above the baseline at all 100 layers, in rank-

ing and in routing, with no parameter update (Appendix I); after contrastive fine-tuning it no longer helped ranking (Section 7). It restores ranking at every layer, but routing after it still varies by layer (75.6 to 90.2 DirAcc@1), so the layer still matters. For raw states, check top-PC share, not mean pairwise cosine. In our models mean cosine gave no reliable warning in either direction, while every collapsed layer of the pretrained models had a top-PC share of at least 0.76 (Section 4). The 0.76 is the lowest share we observed at a collapsed layer of a pretrained checkpoint, not a cutoff validated on held-out models. The share is a screen, not a verdict: a high share is necessary for collapse here but not sufficient, and it does not order the layers inside the collapsed regime. Choosing among layers still needs a train-only task signal, such as the training AUROC we use (Section 3).

In decoder language models, gated feed-forward blocks are reported to amplify outlier activations (Fishman et al., 2025; Yang et al., 2025), but the original T5 also carries large outliers (Zhao et al., 2025); why the T5 v1.1 layout collapses remains open. Our localization names carriers in LaTa, not a cause. Concurrently, Mikkelsen (2026) reports a mid-depth retrieval trough with final-layer recovery in retrieval-trained encoders on clinical text. That pattern differs from ours: in our panel the embedding-trained encoders do not collapse, and that study tests no T5 model.

For the editor of a text collection, the projection makes the choice of encoder matter little (Section 7). The existing-versus-new decision is still wrong for about one test witness in ten (assignment accuracy 88.5 to 91.7; Table 15), so the final call stays with the scholar. Because the task is lexically solvable, a practical tool can start from character n-grams; dense encoders are reported to help where reuse is paraphrastic (de la Selle, 2026), a case our testbed does not measure.

## Limitations

Four factors are uncontrolled. (i) We use one labeled corpus in one language and genre, early medieval Latin canon law; its 1,705 witnesses are many for this field but few by current NLP standards. (ii) Within each matched pair, architecture, pretraining objective and tokenizer vary together, so the pairs do not say which of them matters; that LaTa and LaBERTa share authors and a pretraining corpus rests on their model cards (Riemenschneider and Frank, 2023d,c). (iii) The T5 v1.1 changes (gated-GELU feed-forward, untied embeddings) vary together, and T5-base also saw supervised tasks in pretraining, so our models do not separate layout from pretraining mix. (iv) The token ablation accounts for the masking directions in LaTa only; in mT5-base no set of up to 100 token types restores a collapsed layer, so its mask has no token-level account.

The scope of three results is narrow. We claim the projection result for the six-model panel only, since we did not apply the projection to the four controls. The English check measures geometry only; we have no labels for English retrieval. The task is lexically solvable by design, and a character n-gram reference ties the best projected encoders, so we do not claim that encoders beat surface overlap on this corpus.

The fine-tuning test is small (499 pairs, at most eight epochs), and its references are flattered by directory overlap with the training pairs (Section 7). Our intervals resample test directories with the layer, threshold and projection held fixed, so they cover test sampling and not the choice of split. ABTT needs enough text to estimate principal components, so smaller archives need separate study.

The labels are expert-assigned, but by one project. Its director or supervised graduate students assigned the keys, and verification is by collation in the project’s parallel display, not by independent double annotation, so no inter-annotator agreement figure exists for this testbed. Collation compares the witnesses of one key, so it has nothing to compare when a key has a single witness, and 545 of the 840 directories hold one witness in this testbed.

## Ethical Considerations

The testbed holds early medieval legal texts, transcribed from manuscripts and early printings, and contains no personal data. The Carolingian Canon Law project publishes its material under a

Creative Commons Attribution-NonCommercial-ShareAlike 4.0 license (CCL, 2026a). The project has confirmed in writing that the license covers the transcriptions, annotations and markup in the exports we use and that, in the view of the project’s Advisory Board, attribution to the project suffices; the project’s interface credits the transcriber and proofreader of each unit. The project’s corpus keeps growing and being corrected, so the testbed is a dated snapshot (version 1), fixed by a sha256 manifest of its 1,705 labeled files, and every number in this paper comes from it. We will release it under the same license, in coordination with the project’s periodic versioned exports, and later versions will follow those exports. The intended use is research on retrieval and on assisted source identification: the system ranks candidates for an expert to judge and does not identify sources on its own.

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## A Corpus Provenance and Derivation

This appendix records where the testbed text comes from and how our files were derived from it, in the detail that Section 3 leaves out.

**The project.** CCL (Firey, 2009) is an open-access digital edition founded in 2009; Eichbauer (2014) reviews it.<sup>1</sup> Its texts are early

<sup>1</sup>Founding year, pre-2019 workflow, proofreading standard, the one manuscript transcribed by handwritten text recognition (HTR), the design of the identifiers, the assignment and verification of keys, and the extent of unidentified material: unpublished documentation and correspondence supplied by the project, 2026.

medieval canon-law manuscripts, most of them ninth-century and never edited to modern standards. Each manuscript is identified by its siglum (Place-Library-Shelfmark), and a witness is one manuscript’s copy of a source text, transcribed as one unit (one canon or excerpt).

**Transcription.** Transcription is manual, from manuscript images or, for some texts, eighteenth-century printings, with one exception: HTR has been attempted on one manuscript (Paris, BnF, lat. 1454; 142 of the 1,705 witnesses), and its transcription is still in proofreading. Before 2019 transcribers worked from purchased microfilm into word-processed files, which CCL contributors converted to TEI-P5 (the Text Encoding Initiative’s XML standard for scholarly editions). Since 2019 they transcribe in the CCL Transcription Desk beside a IIIF image of the folio (a zoomable web-served scan of the manuscript leaf); an approved proofreader on the Proofreading Desk then checks the text against the image to the letter and adds TEI markup (CCL, 2026b). That proofreading stage belongs to the post-2019 workflow. The interface credits the transcriber and proofreader of each unit (CCL, 2026a).

**Identifiers.** The project has two identifier systems, both designed at its launch in 2009 by its director in consultation with a mathematician, a software developer and a second scholar of early medieval canon law. The first is a location identifier. Every manuscript is a unique object, so every unit has exactly one place in one manuscript, and the identifier records it as the manuscript siglum, the folium, and the unit’s position on that folium. Our file names are these identifiers. The second is the source key. Many units in different manuscripts copy the same prescription issued by the same authority, and the key names that common source. It encodes the type of source (a church council, a papal decretal, a royal capitulary, a theologian, a penitential), the point of origin (a particular council, pope or author), the date of origin, and the item’s reference in modern scholarship, such as a canon, letter or register number. The key is designed so that a researcher can find the modern authoritative edition of the source where one exists.

**Labels.** CCL keys a unit to its original source, such as a fourth-century council or a papal decretal, once that source is identified, and our directories reuse that key as supplied. In a label, a

four-letter code carries the type of source and its point of origin, followed by the year and the canon or chapter number, as in CTOU.567.16; some labels carry a trailing annotation, as in CTOU.567.16 (15) or CARL.501?.18. Counting a directory as key-named when its label begins with a four-letter code, a period, a year and a period, whatever follows, 689 of the 840 directory labels are source keys. The other 151 are modern edition citations, the Apostolic Canons (Can.apost.N), references to other collections such as the Capitula Martini, biblical references, and a handful of unidentified sources; each unidentified source is a singleton directory and contributes no positive pairs.

**Assignment and verification.** Keys were assigned by the project’s director, a specialist in early medieval canon law, or by graduate students working under the director’s supervision, who had enough Latin to check a manuscript version against the modern authoritative version of the source. Verification, and occasional correction, runs through the project’s parallel display, which sets side by side all units that share a key and shows how far their wording agrees. A misidentified unit stands out there as a text that cannot be a version of the others. This is verification by collation (see Limitations). The project supplied two label corrections during this study, and both are logged with the frozen testbed. We assigned no label ourselves and changed only those two. What remains unidentified is large: across the several hundred manuscripts that contain the project’s texts, thousands of units still carry no key, and they are the queries that routing is meant to serve.

**Derivation.** We hold the witnesses as plain text derived from CCL TEI-P5 exports. The rule is one file per unit: the directory is the source key from the unit’s scholSource division, which is stripped from the text so that no file carries its own label; the text is the manuscript rubric followed by the canon text, with inline markup flattened (editorial notes kept, deletions dropped) and layout whitespace collapsed; a unit without a source key receives no directory. The rule is scripted; checked against a current export of one manuscript (Paris, BnF, lat. 2123), it reproduces all 279 units present in both the export and the corpus, 100 percent after whitespace normalization with no character-level disagreement, and places every one of them in the same directory.

## B Testbed and Protocol Details

This appendix gives the parts of the protocol that Section 3 leaves out: how the split treats directories of each size, the further layer statistics, the SIF weighting, and how the lexical reference fares by degree of overlap.

**Split.** Singleton directories (545 of 840) are divided file by file between the two splits. Doubletons go whole to one side. Larger directories are divided internally, so that every size class yields training positives. Pairs form only within a split.

**Further layer statistics.** Let  $p_i = \lambda_i / \sum_j \lambda_j$  for the covariance eigenvalues  $\lambda_1 \geq \lambda_2 \geq \dots$  of the centered training embeddings; the top-PC share is  $p_1$ . The *effective rank* is  $\exp(-\sum_i p_i \log p_i)$ , the entropy effective rank of Roy and Vetterli (2007) applied to the covariance eigenvalues. It is 1 when one direction holds all the variance. The *cosine gap*  $g$  is the mean same-directory test cosine minus the mean different-directory test cosine. The *standardized gap*  $g/s$  divides it by the standard deviation  $s$  of all pairwise test cosines, which puts models with different cosine spreads on one scale. The two gaps need labels; the effective rank, like the top-PC share and the mean pairwise cosine, does not. Our mean pairwise cosine averages over all pairs of pooled passages, an adaptation of the word-level statistic of Ethayarajh (2019), which uses random pairs.

**SIF.** SIF (Arora et al., 2017) reweights each token by  $a/(a + p(w))$  before pooling, with  $a = 10^{-3}$  and  $p(w)$  estimated on training texts. It gives special tokens, such as the end-of-sequence token  $\langle /s \rangle$ , zero weight. We use only this weighting step of SIF, not its common-component removal. SIF+ABTT applies both corrections, and ABTT-only names ABTT when we contrast the two.

**Lexical reference by overlap.** We cut the positive test pairs into terciles of n-gram overlap, using the TF-IDF score itself. In every tercile, the character n-gram reference ranks a witness’s partner first at least as often as every baseline and ABTT encoder configuration at its train-selected layer. Table 19 gives all three lexical scorers.

## C Collapse by Depth: Further Diagnostics

This appendix adds detail to the four findings of Section 4. It gives the geometry of each panel

model over its layers, the correlations behind the statement that top-PC share marks the collapsed regime without ordering layers inside it, two label-based gaps, the spread of cosines at the collapsed layers, the effect of SIF there, and the score distributions before and after ABTT. Every statistic is computed on baseline mean-pooled hidden states unless a paragraph says otherwise.

**Geometry by model.** Table 4 summarizes each panel model over its layers. At the 26 collapsed layers, effective rank lies between 1.0 and 4.6; in the embedding-trained models it ranges from 16 to 150. At mT5-base layers 5–11 one direction holds all the variance to three decimals. The table reads mean cosine on test passages, while Figure 2 reads it on training passages. On test passages the collapsed layers span 0.23 to 0.62, so the PhilTa layers overlap the bottom of the embedding-trained range. ABTT with ten components, fit on training vectors, removes the concentration on held-out passages too: at the collapsed layers, test top-PC share falls to 0.03–0.05 and effective rank rises to 148–173. Table 6 shows this at the most anisotropic layer of each T5 encoder.

**Correlations with ranking.** Table 5 correlates each statistic with baseline AUROC, one observation per model-layer. Across all 100 panel layers, top-PC share and effective rank reach Spearman  $-0.77$  and  $+0.77$ . Both values are inflated by the two clusters of collapsed and healthy layers, and layers within a model are not independent. Within the 26 collapsed layers, top-PC share carries no ordering (Spearman  $+0.24$ ,  $p = 0.23$ ). Within the 74 other layers it orders them only moderately ( $-0.49$ ). Mean pairwise cosine correlates positively with AUROC within the three panel T5 encoders ( $+0.67$  on test passages,  $+0.72$  on training passages). This pattern does not carry over to T5-v1.1-base, whose collapsed layers sit at mean cosines of 0.87 to 0.96 on training passages, around the 0.91 and 0.92 of its two healthy layers; within that model the correlation is  $-0.64$ . The lower block of the table adds no separate evidence. ABTT lifts every layer into a narrow band (0.962 to 0.987), so its gain is nearly a constant minus the baseline, and the gain correlations mirror the baseline ones. The absence of a reliable ordering agrees with reports that geometric statistics do not reliably predict downstream performance (Kulkarni et al., 2026; Nastase and Merlo, 2025).

**Raw and standardized gap.** The raw cosine gap, the margin a routing threshold works with, tracks ranking poorly: across the panel it correlates with AUROC at Spearman  $+0.22$  (Figure 4). The standardized gap reaches  $+0.97$ , but it is a label-based signal-to-noise ratio close to the  $d'$  from which a binormal model computes AUROC, so it tracks AUROC by construction and we do not count it as a predictor. It explains one pattern in mT5-base (Figure 5). There the raw gap rises from about 0.03 to 0.21–0.25 exactly where ranking fails. Over the same layers the standard deviation of all pairwise cosines grows from 0.05–0.07 to about 0.75. Divided by that spread, the gap falls to 0.28–0.33, below the 0.41–0.49 of layers 1–3 and the 0.78 of layer 12. The collapsed layers thus spread all cosines widely, and same-source pairs gain little relative to that spread. mT5-base layer 4 is an exception: its standardized gap is the lowest of the model (0.19), yet it ranks at 0.799.

**Cosine spread at the collapsed layers.** At the collapsed layers, test cosines spread across both signs. At mT5-base’s collapsed layers the 5th percentile lies between  $-0.83$  and  $-0.77$  and the median between 0.84 and 0.90, with a standard deviation of about 0.75. At LaTa’s collapsed layers the 5th percentile lies between  $-0.50$  and  $-0.26$ , and the standard deviation reaches 0.63. In the embedding-trained models the standard deviation stays between 0.007 and 0.10 at every layer. Pooled passages at a collapsed layer therefore lie on both sides of one dominant axis, which Section 6 examines.

**SIF at the collapsed layers.** SIF pooling (Arora et al., 2017) down-weights frequent tokens and drops special tokens (Appendix B). At the collapsed layers it recovers about 80 percent of the ABTT gain in LaTa and PhilTa, and under 10 percent in mT5-base. This split agrees with the token ablation of Section 6: in LaTa, mean pooling without the comma, the period and  $</s>$  restores every collapsed layer, while in mT5-base no set of up to 100 token types, ranked on training passages, does. SIF stays below ABTT at every collapsed layer, and its full sweeps are in Appendix K.

**Score distributions.** Figure 6 restates the collapse and the repair as score distributions rather than as summary metrics. At PhilTa’s collapsed layer 8 the cosines of same-source and different-source pairs overlap almost completely, and ABTT

| Model      | Type         | $L$ | Geometry (baseline) |           |         |          | Baseline test AUROC |             |             |
|------------|--------------|-----|---------------------|-----------|---------|----------|---------------------|-------------|-------------|
|            |              |     | Mean cos.           | Cos. SD   | Max PC1 | Min rank | $\ell^*$            | at $\ell^*$ | Min (layer) |
| LaTa       | T5 raw       | 12  | 0.23–0.58           | 0.10–0.63 | 0.952   | 1.36     | 4                   | 0.500       | 0.496 (6)   |
| PhilTa     | T5 raw       | 12  | 0.53–0.71           | 0.09–0.46 | 0.858   | 1.81     | 6                   | 0.541       | 0.538 (10)  |
| mT5-base   | T5 raw       | 12  | 0.30–0.95           | 0.05–0.76 | 1.000   | 1.00     | 5                   | 0.654       | 0.654 (5)   |
| LaBSE      | BERT emb.    | 12  | 0.73–0.93           | 0.04–0.10 | 0.524   | 16.19    | 8                   | 0.879       | 0.806 (1)   |
| Qwen3-0.6B | Decoder emb. | 28  | 0.85–0.97           | 0.01–0.04 | 0.183   | 71.60    | 16                  | 0.910       | 0.858 (1)   |
| KaLM-mini  | Decoder emb. | 24  | 0.58–0.93           | 0.02–0.07 | 0.419   | 23.83    | 5                   | 0.861       | 0.861 (5)   |

Table 4: Geometry and retrieval regimes of the six encoders under baseline mean pooling (no post-hoc correction). Type: T5 raw is a T5 encoder pretrained without a sentence-embedding objective; emb. marks a BERT encoder or a decoder trained with one.  $L$ : number of transformer layers. Mean cos. and Cos. SD: range over layers of the mean and the standard deviation of the cosine over all pairs of test passages. Max PC1: peak top-PC share over layers, the largest share of variance on the first principal component of the centered training embeddings. Min rank: smallest entropy effective rank of the same centered training embeddings over layers. All four geometry columns are label-free.  $\ell^*$ : layer where the PC1 share peaks. The last two columns give the baseline test pairwise AUROC of cosine at  $\ell^*$  and its minimum over layers, with the layer in parentheses. Layer 1 is the first transformer block.

| Statistic                      | All (100) |        | T5 (36) |        | Non-T5 (64) |        | LOMO $\rho$ |       |
|--------------------------------|-----------|--------|---------|--------|-------------|--------|-------------|-------|
|                                | $r$       | $\rho$ | $r$     | $\rho$ | $r$         | $\rho$ | min         | max   |
| <i>Target: baseline AUROC</i>  |           |        |         |        |             |        |             |       |
| Mean pairwise cosine           | 0.83      | 0.55   | 0.61    | 0.67   | −0.06       | 0.10   | 0.42        | 0.67  |
| Top-PC share                   | −0.92     | −0.77  | −0.86   | −0.50  | −0.47       | −0.43  | −0.83       | −0.70 |
| Effective rank                 | 0.80      | 0.77   | 0.87    | 0.53   | 0.48        | 0.40   | 0.70        | 0.85  |
| Raw cosine gap                 | 0.24      | 0.22   | 0.63    | 0.72   | 0.15        | −0.03  | −0.10       | 0.48  |
| Standardized gap               | 0.86      | 0.97   | 0.87    | 0.96   | 0.89        | 0.93   | 0.96        | 0.98  |
| <i>Target: ABTT AUROC gain</i> |           |        |         |        |             |        |             |       |
| Mean pairwise cosine           | −0.83     | −0.54  | −0.60   | −0.66  | 0.07        | −0.07  | −0.66       | −0.41 |
| Top-PC share                   | 0.92      | 0.76   | 0.86    | 0.50   | 0.45        | 0.41   | 0.69        | 0.83  |
| Effective rank                 | −0.80     | −0.76  | −0.88   | −0.52  | −0.44       | −0.37  | −0.84       | −0.68 |
| Raw cosine gap                 | −0.23     | −0.23  | −0.61   | −0.71  | −0.16       | 0.00   | −0.48       | 0.09  |
| Standardized gap               | −0.85     | −0.97  | −0.87   | −0.95  | −0.88       | −0.92  | −0.98       | −0.96 |

Table 5: Correlation between per-layer geometry and retrieval, one observation per model-layer, baseline mean pooling.  $r$ : Pearson;  $\rho$ : Spearman. Columns pool all 100 model-layers, the 36 layers of the three T5 encoders, or the 64 layers of the three embedding-trained models; LOMO gives the minimum and maximum Spearman over the six leave-one-model-out subsets. Mean pairwise cosine is on test passages; top-PC share and effective rank are on centered training embeddings. These three statistics are label-free. The two gaps are label-based separation indices. Raw cosine gap is the mean test cosine of same-directory pairs minus that of different-directory pairs; standardized gap divides it by the standard deviation of all test pairwise cosines at that layer. Targets are the baseline test AUROC and the ABTT gain (AUROC with ABTT at its train-selected  $D$  minus baseline AUROC). The standardized gap is a signal-to-noise ratio of the kind AUROC summarizes, so it tracks AUROC by construction. ABTT AUROC varies little (0.962–0.987), so the gain correlations mirror the baseline ones. The pooled Pearson and Spearman values are both inflated by the two clusters of collapsed and healthy layers, and layers within a model are not independent.

separates them.

| Model    | Most anisotropic layer | PC1 before | PC1 after | Eff. rank before→after |
|----------|------------------------|------------|-----------|------------------------|
| LaTa     | 8                      | 0.956      | 0.036     | 1.34→168.40            |
| PhilTa   | 6                      | 0.862      | 0.039     | 1.83→155.02            |
| mT5-base | 5                      | 1.000      | 0.044     | 1.00→151.74            |

Table 6: Most anisotropic layers, by top-PC share on the test split, for the three T5 encoders. PC1: top-PC share. PC1 and effective-rank values compare baseline geometry with ABTT-D10 at the most anisotropic layer, on the test split. Table 4 and the main text read top-PC share on the training split, where LaTa peaks at layer 4 instead of 8; LaTa’s training curve is flat, with a top-PC share of 0.93 to 0.95 over layers 3–11.

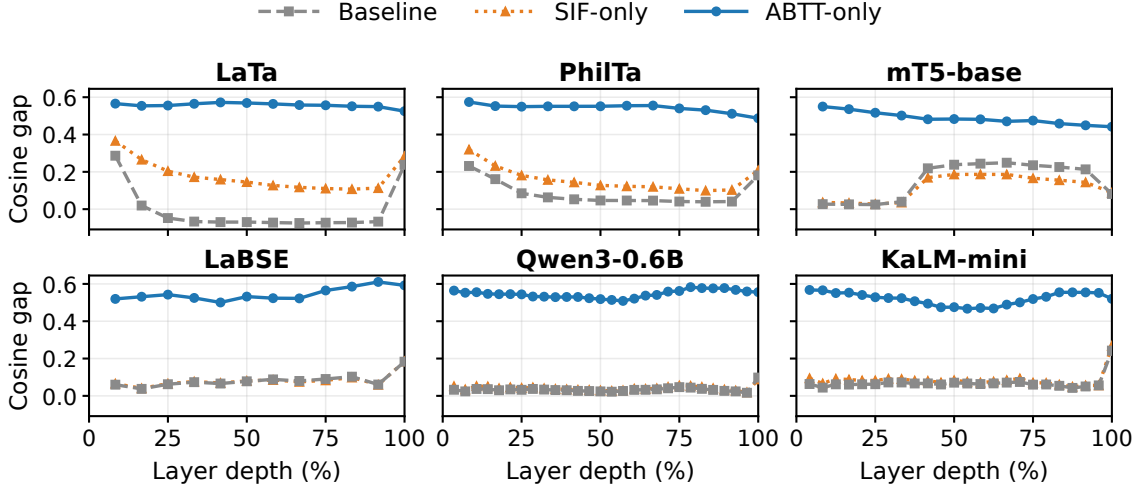


Figure 4: Raw cosine gap against normalized layer depth for the six panel models, test split. The gap is the mean cosine of same-directory pairs minus that of different-directory pairs, the margin a threshold works with. Gray: baseline mean pooling. Orange: SIF-weighted pooling. Blue: ABTT, with  $D$  chosen per layer on training data. Baseline gaps stay below 0.29 at every layer of every model. In the middle layers the LaTa and PhilTa gaps stay within 0.09 of zero, while mT5-base’s gap rises as its ranking fails (compare Figure 1). Figure 5 resolves the mT5-base case by dividing the gap by the spread of all cosines. ABTT raises every model’s gap to between 0.44 and 0.61 at every depth. SIF raises the LaTa and PhilTa gaps and leaves the other four near or below baseline.

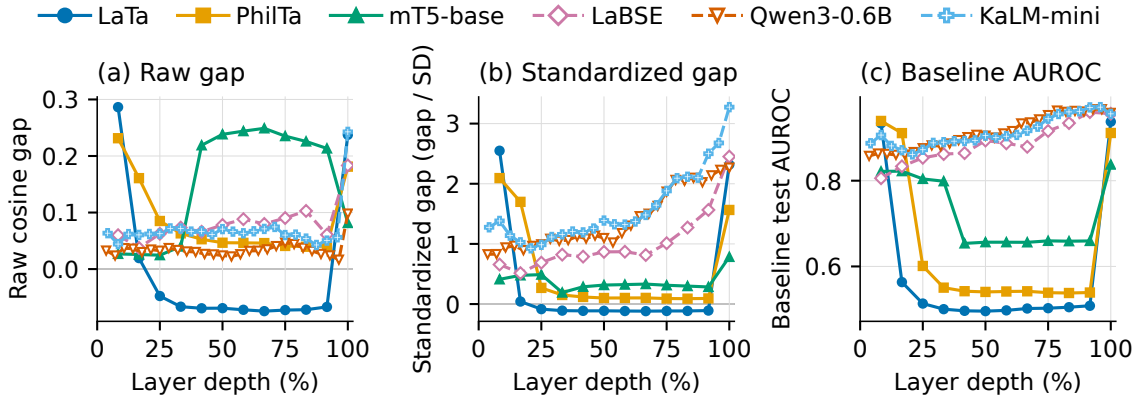


Figure 5: Cosine gap and retrieval against normalized layer depth for the six panel models, baseline mean pooling, test split. Solid lines with filled markers: T5 encoders. Dashed lines with hollow markers: embedding-trained models. (a) Raw gap: mean cosine of same-directory pairs minus that of different-directory pairs. (b) Standardized gap: the raw gap divided by the standard deviation of all pairwise test cosines at that layer. (c) Baseline test ranking AUROC. In the middle layers of mT5-base the raw gap rises to 0.21–0.25 while AUROC falls to 0.654, and the standardized gap falls with AUROC. Across all 100 model-layers the standardized gap tracks AUROC at Spearman 0.97, and the raw gap at 0.22.

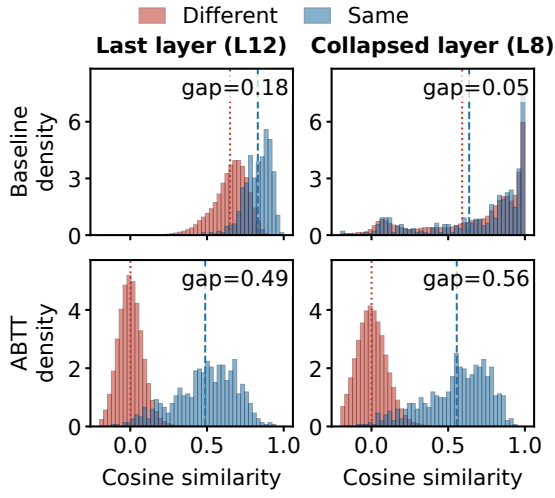


Figure 6: Cosine similarity distributions for PhilTa at its last layer and at the collapsed layer 8 (test AUROC 0.542, close to chance), before and after ABTT. Equivalent pairs share a directory (Same in the legend); non-equivalent pairs do not (Different). Layer 8 is not PhilTa’s worst layer: its lowest baseline test AUROC, 0.538, falls at layer 10. Before ABTT the two distributions sit almost on top of each other, and ABTT separates them.

## D All Twelve Models

This appendix supports Section 5. Table 7 extends Table 1 in two ways. It adds two embedding-trained models that the main text leaves out, Sentence-T5 (Ni et al., 2022) and SPhilBERTa (Riemenschneider and Frank, 2023b). It also gives each checkpoint’s feed-forward type and width, whether its embeddings are tied, its lowest effective rank and its range of mean pairwise cosine. Neither added model collapses. The rest of the appendix gives the per-layer readouts behind the matched pairs, the English-text check and the fine-tuned LaTa model.

**The matched pairs.** Before the run we required each encoder-only sibling to stay above 0.80 AUROC at every layer, and both did (Table 8). LaBERTa clears that floor by 0.026, at layer 1, and rises to 0.910 at layer 12; PhilBERTa stays between 0.883 and 0.923. Effective rank falls to 1.36 in LaTa and 1.81 in PhilTa, and never below 57.87 in LaBERTa or 82.12 in PhilBERTa. Within each pair, several factors differ together. The T5 member is the encoder of an encoder-decoder, pretrained with span corruption; the sibling is an encoder-only RoBERTa-style model, pretrained with masked language modeling (Riemenschneider and Frank, 2023a). The T5 members use a SentencePiece tokenizer and the siblings a byte-level BPE tokenizer, so the tokens that enter the mean also differ. By their checkpoint configurations, the T5 members also use relative position biases, a gated-GELU feed-forward block and untied embeddings, and the siblings learned absolute positions, a GELU feed-forward block and tied embeddings. The pairs therefore rule out the shared pretraining text and the missing embedding objective as sufficient causes, and they cannot say which within-pair difference matters.

**T5-base and T5-v1.1-base.** The two checkpoints share their tokenizer, which splits the Latin corpus identically, the span-corruption objective, relative position biases and pre-layer RMS normalization, at the same depth and width (Raffel et al., 2020; Google Research, 2020). They differ in feed-forward block (ReLU with inner width 3072, against gated GELU with 2048), in embedding tying, and in pretraining. T5-base was pretrained on C4 mixed with supervised tasks, with dropout (Raffel et al., 2020); T5-v1.1-base on C4 alone, without dropout (Google Research, 2020). Dropout does not separate the collapsing models from the

others, because PhilTa was also pretrained with dropout (Riemenschneider and Frank, 2023a). T5-base shows a weak form of the same depth profile: its top-PC share rises to 0.545 and its effective rank falls to 14.85 at layer 8, while its AUROC stays between 0.816 and 0.864. T5-v1.1-base recovers only partly at its last layer, to 0.772.

**Sentence-T5.** Sentence-T5 has the original T5 layout and a contrastive embedding objective. At its layers 2 to 11, top-PC share is 0.659 to 0.943 while AUROC stays between 0.800 and 0.891. A high top-PC share is therefore necessary for collapse in our data but not sufficient: every one of the 36 collapsed layers of the pretrained checkpoints in Table 7 (26 in the panel, 10 in T5-v1.1-base) has a share of at least 0.76, and high-share layers also occur without collapse.

**English text.** Table 9 and Figure 7 give the label-free check of Section 5. The English sample comes from the public-domain US court opinions of the Caselaw Access Project (Harvard Law School Library Innovation Lab, 2024), in the redistribution of the Common Pile (Kandpal et al., 2025); every sampled document is marked public domain. Legal prose is the nearest openly licensed English register to canon law. Each non-empty Latin passage receives one English passage from a distinct opinion (the two empty Latin files receive an empty one): a run of whole words from a random sentence start in the body text, whose mT5 token count is within  $\max(2, 3\%)$  tokens of the Latin passage’s, and the English passage inherits its partner’s split. The two texts then have the same mT5 length distribution (median 98 tokens). The match holds only under the mT5 tokenizer: under PhilTa’s tokenizer the English passages are about 40 percent longer, and under T5-v1.1’s English-only tokenizer the Latin passages are about 55 percent longer. Re-pairing the training passages on each model’s own token lengths (653 to 847 pairs) moves top-PC share by at most 0.010 at the layers that Table 9 lists. We had expected each model to show a top-PC share above 0.8 and an effective rank near 1 at mid-depth on both texts. The share holds in every cell. The effective rank holds for mT5-base and T5-v1.1-base but not for PhilTa, whose effective rank is 1.82 to 2.30 at English layers 4 to 11, and 1.81 to 4.34 at Latin layers 3 to 11. The two texts differ in language, era and genre together, and the English sample covers one register.

| Model                | Type | Emb. obj.    | Checkpoint   |                 |      | Over layers          |                    |                     |           | Layers |                 |
|----------------------|------|--------------|--------------|-----------------|------|----------------------|--------------------|---------------------|-----------|--------|-----------------|
|                      |      |              | Feed-forward | $d_{\text{ff}}$ | Tied | AUROC <sub>min</sub> | PC1 <sub>max</sub> | Rank <sub>min</sub> | Mean cos. | Coll.  | PC1 $\geq 0.76$ |
| <i>Main text</i>     |      |              |              |                 |      |                      |                    |                     |           |        |                 |
| LaTa                 | T5   | none         | gated GELU   | 2048            | no   | 0.496 (6)            | 0.952 (4)          | 1.36 (8)            | 0.22–0.58 | 2–11   | 2–11            |
| PhilTa               | T5   | none         | gated GELU   | 2048            | no   | 0.538 (10)           | 0.858 (6)          | 1.81 (6)            | 0.51–0.71 | 3–11   | 3–11            |
| mT5-base             | T5   | none         | gated GELU   | 2048            | no   | 0.654 (5)            | 1.000 (5)          | 1.00 (5)            | 0.33–0.95 | 5–11   | 4–11            |
| T5-v1.1-base         | T5   | none         | gated GELU   | 2048            | no   | 0.489 (2)            | 0.975 (9)          | 1.14 (9)            | 0.87–0.96 | 2–11   | 2–11            |
| T5-base              | T5   | none         | ReLU         | 3072            | yes  | 0.816 (11)           | 0.545 (8)          | 14.85 (8)           | 0.90–0.95 | none   | none            |
| LaBERTa              | Enc. | none         | GELU         | 3072            | yes  | 0.826 (1)            | 0.254 (9)          | 57.87 (9)           | 0.70–0.91 | none   | none            |
| PhilBERTa            | Enc. | none         | GELU         | 3072            | yes  | 0.883 (6)            | 0.162 (6)          | 82.12 (11)          | 0.84–0.96 | none   | none            |
| LaBSE                | Enc. | contrastive  | GELU         | 3072            | n/a  | 0.806 (1)            | 0.524 (8)          | 16.19 (8)           | 0.73–0.93 | none   | none            |
| Qwen3-0.6B           | Dec. | contrastive  | gated SiLU   | 3072            | yes  | 0.858 (1)            | 0.183 (16)         | 71.60 (16)          | 0.84–0.97 | none   | none            |
| KaLM-mini            | Dec. | contrastive  | gated SiLU   | 4864            | yes  | 0.861 (5)            | 0.419 (5)          | 23.83 (5)           | 0.59–0.93 | none   | none            |
| <i>Appendix only</i> |      |              |              |                 |      |                      |                    |                     |           |        |                 |
| Sentence-T5          | T5   | contrastive  | ReLU         | 3072            | n/a  | 0.800 (11)           | 0.943 (6)          | 1.56 (6)            | 0.72–0.98 | none   | 2–9             |
| SPhilBERTa           | Enc. | distillation | GELU         | 3072            | n/a  | 0.892 (1)            | 0.127 (11)         | 64.67 (11)          | 0.61–0.95 | none   | none            |

Table 7: All twelve models, read on the Latin corpus as mean-pooled hidden states with no post-hoc correction. Type: the encoder of a T5 encoder-decoder (T5), an encoder-only model (Enc.), or a decoder (Dec.). Emb. obj.: the embedding objective trained after pretraining, if any. Checkpoint: the feed-forward activation and inner width  $d_{\text{ff}}$ , and whether input and output embeddings are tied, as the checkpoint’s configuration file states them (T5-base, LaBERTa and PhilBERTa leave tying at the library default, tied; n/a: the checkpoint has no output head). Over layers: the lowest test ranking AUROC, the peak top-PC share and the lowest entropy effective rank, each with its layer, and the range of mean pairwise cosine; top-PC share, effective rank and mean cosine are computed on the training embeddings. Layers: the collapsed layers (Coll.), with AUROC below 0.70, and those with top-PC share of at least 0.76. Every one of the 36 collapsed layers (26 in the panel, 10 in T5-v1.1-base) has a top-PC share of at least 0.76, but a high share also occurs without collapse (mT5-base layer 4, Sentence-T5). Checkpoints: LaTa, bowphs/LaTa; PhilTa, bowphs/PhilTa; mT5-base, google/mt5-base; T5-v1.1-base, google/t5-v1\_1-base; T5-base, google-t5/t5-base; LaBERTa, bowphs/LaBerta; PhilBERTa, bowphs/PhilBerta; LaBSE, sentence-transformers/LaBSE; Qwen3-0.6B, Qwen/Qwen3-Embedding-0.6B; KaLM-mini, KaLM-Embedding/KaLM-embedding-multilingual-mini-instruct-v2.5; Sentence-T5, sentence-transformers/sentence-t5-base; SPhilBERTa, bowphs/SPhilBerta.

**Fine-tuned LaTa.** Table 10 gives fine-tuned LaTa at every layer, with no post-hoc correction, beside the pretrained model. Fine-tuning lifts the last layer from 0.938 to 0.984 AUROC and leaves layers 2 to 11 at 0.499 to 0.569. Their top-PC share stays at 0.759 to 0.945, against 0.771 to 0.952 before fine-tuning, so the collapsed geometry stays in place (fine-tuned layer 2, at 0.759, sits just below the 0.76 screen). The fine-tuning runs at most eight epochs on 499 pairs (Appendix H). It is therefore a weak test of whether an embedding objective repairs the middle layers: small-scale supervision at the output may leave them untouched whatever produces the collapse.

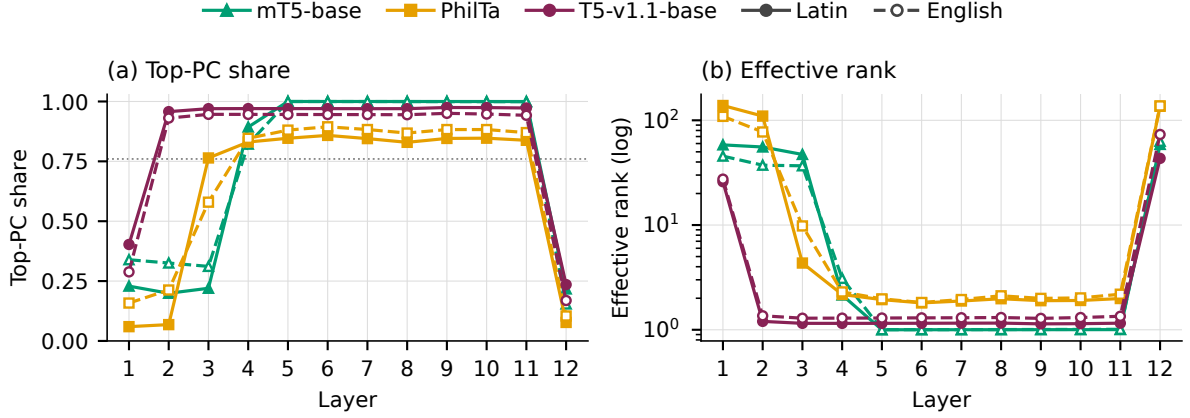


Figure 7: Label-free layer geometry of three T5 encoders on the Latin corpus (solid, filled markers) and on English US court opinions matched to the Latin passages in mT5 token length (dashed, hollow markers), read on 847 passages per cell. (a) Top-PC share, the share of centered variance on the first principal component (dotted line: a share of 0.76, the lowest share of any collapsed layer of the pretrained models, drawn as a visual guide); every collapsed Latin layer of the pretrained models has a share of at least 0.76. (b) Entropy effective rank, log scale. Every model is low-rank from layer 2 to 4 through layer 11 on both texts and recovers at layer 12.

| Layer | LaBERTa |       |       | PhilBERTa |       |        |
|-------|---------|-------|-------|-----------|-------|--------|
|       | AUROC   | PC1   | Rank  | AUROC     | PC1   | Rank   |
| 1     | 0.826   | 0.170 | 87.15 | 0.893     | 0.092 | 134.77 |
| 2     | 0.832   | 0.193 | 75.32 | 0.905     | 0.096 | 109.71 |
| 3     | 0.839   | 0.184 | 78.12 | 0.913     | 0.112 | 97.87  |
| 4     | 0.830   | 0.191 | 74.89 | 0.903     | 0.124 | 98.85  |
| 5     | 0.828   | 0.191 | 72.73 | 0.885     | 0.151 | 95.27  |
| 6     | 0.834   | 0.198 | 68.46 | 0.883     | 0.162 | 97.28  |
| 7     | 0.836   | 0.228 | 61.98 | 0.896     | 0.149 | 97.34  |
| 8     | 0.844   | 0.248 | 58.96 | 0.905     | 0.157 | 94.21  |
| 9     | 0.857   | 0.254 | 57.87 | 0.911     | 0.140 | 92.12  |
| 10    | 0.873   | 0.245 | 61.55 | 0.919     | 0.120 | 89.92  |
| 11    | 0.892   | 0.233 | 65.75 | 0.913     | 0.143 | 82.12  |
| 12    | 0.910   | 0.154 | 85.69 | 0.923     | 0.083 | 105.56 |

Table 8: Per-layer readouts of LaBERTa and PhilBERTa, the encoder-only siblings of LaTa and PhilTa in the matched pairs of Table 1, on the Latin corpus with no post-hoc correction. AUROC: test ranking AUROC of cosine on mean-pooled vectors. PC1: top-PC share, the share of centered variance on the first principal component of the 847 training passages. Rank: entropy effective rank of the same passages. Two facts about the inputs: LaBERTa’s tokenizer encodes every space as its own token, so the indentation of the corpus files becomes tokens, which the pooling filter drops from the mean; PhilBERTa’s published tokenizer adds no sequence-boundary tokens, so the two whitespace-only files of the corpus pool to zero vectors, and scoring without those two files changes no AUROC by more than 0.0003.

| Model        | Text    | $PC1_{\max}(\ell)$ | Layers with $PC1 \geq 0.76$ |             |           |
|--------------|---------|--------------------|-----------------------------|-------------|-----------|
|              |         |                    | Layers                      | PC1         | Eff. rank |
| mT5-base     | Latin   | 1.000 (5)          | 4–11                        | 0.893–1.000 | 1.00–2.15 |
|              | English | 1.000 (5)          | 4–11                        | 0.822–1.000 | 1.00–3.10 |
| PhilTa       | Latin   | 0.858 (6)          | 3–11                        | 0.764–0.858 | 1.81–4.34 |
|              | English | 0.894 (6)          | 4–11                        | 0.846–0.894 | 1.82–2.30 |
| T5-v1.1-base | Latin   | 0.975 (9)          | 2–11                        | 0.958–0.975 | 1.14–1.20 |
|              | English | 0.950 (9)          | 2–11                        | 0.931–0.950 | 1.28–1.36 |

Table 9: Label-free layer geometry crossed by model and text. Each cell reads the mean-pooled hidden states of 847 passages: the Latin training passages, or the English passages paired with them one to one and matched in mT5 token length (US court opinions, Caselaw Access Project).  $PC1_{\max}$ : peak top-PC share over the 12 layers, the share of centered variance on the first principal component, with its layer  $\ell$ . The last three columns cover the layers whose top-PC share is at least 0.76, a share that every collapsed Latin layer of the pretrained models reaches, and give the range of top-PC share and of entropy effective rank over them. No labels are used, so the English rows say nothing about retrieval.

| Layer | AUROC  |       | Top-PC share |       | Eff. rank |        |
|-------|--------|-------|--------------|-------|-----------|--------|
|       | Before | After | Before       | After | Before    | After  |
| 1     | 0.934  | 0.936 | 0.050        | 0.050 | 134.33    | 134.44 |
| 2     | 0.563  | 0.569 | 0.771        | 0.759 | 4.61      | 4.92   |
| 3     | 0.513  | 0.512 | 0.935        | 0.929 | 1.59      | 1.64   |
| 4     | 0.500  | 0.499 | 0.952        | 0.945 | 1.38      | 1.43   |
| 5     | 0.496  | 0.500 | 0.943        | 0.931 | 1.41      | 1.48   |
| 6     | 0.496  | 0.500 | 0.934        | 0.921 | 1.44      | 1.51   |
| 7     | 0.498  | 0.501 | 0.937        | 0.923 | 1.43      | 1.50   |
| 8     | 0.502  | 0.505 | 0.950        | 0.940 | 1.36      | 1.42   |
| 9     | 0.503  | 0.507 | 0.943        | 0.931 | 1.40      | 1.48   |
| 10    | 0.505  | 0.510 | 0.942        | 0.930 | 1.43      | 1.50   |
| 11    | 0.508  | 0.512 | 0.939        | 0.927 | 1.47      | 1.55   |
| 12    | 0.938  | 0.984 | 0.081        | 0.069 | 132.22    | 137.68 |

Table 10: LaTa before and after contrastive fine-tuning on training pairs, per layer, with no post-hoc correction. AUROC: test ranking AUROC of cosine on mean-pooled vectors. Top-PC share and effective rank: label-free geometry of the centered training vectors, which include the passages of the fine-tuning pairs. Fine-tuning lifts the last layer and leaves the middle layers collapsed.

## E Localizing the Collapse: Details

This appendix gives the per-layer tables and the further controls behind Section 6. Collapsed layers are the 26 layers of LaTa, PhilTa and mT5-base with baseline AUROC below 0.70. Every coordinate ranking, component, token ranking and threshold is fit on training passages only, and every AUROC is test ranking AUROC. A layer counts as restored when it reaches 0.90.

**Coordinate ablation.** Table 11 reads each model at its lowest-AUROC baseline layer. The two coordinate rankings pick nearly the same coordinates at the collapsed layers: on average 9.1 of the top ten are shared. Massive coordinates alone do not mark a collapse. At LaTa layer 6, the lowest-AUROC layer of the panel, the three pooled coordinates of largest mean magnitude reach 720, 684 and 676, against a median of 8.5. At mT5-base layer 1, a healthy layer (AUROC 0.822), they reach 135, 131 and 63, against a median of 1.9. Per-coordinate standardization with training statistics (Timkey and van Schijndel, 2021) is a partial repair. It recovers a median 79% of the gain from removing ten components, but it restores only 9 of the 26 layers (1 of 10 in LaTa, 5 of 9 in PhilTa, 3 of 7 in mT5-base). The embedding-trained models barely react to zeroing: at the train-selected layers of LaBSE, Qwen3-0.6B and KaLM-mini, zeroing up to ten coordinates changes AUROC by at most 0.007.

**Zeroing more coordinates.** Figure 8 extends the zeroing to 400 coordinates. We added this sweep after reading the results above, and it carries no prediction. Under the variance ranking, zeroing restores none of the 26 layers up to 30 coordinates, 7 at 100, and 19 at 300 and 400. The 19 are every collapsed layer of LaTa and PhilTa; no mT5-base layer is restored. Under the mean-absolute-value ranking of Figure 3a, the first layers are restored at 50 coordinates and all 19 at 400. Zeroing 400 random coordinates removes a median 55% of the training variance, yet the median AUROC stays at 0.538, against 0.541 uncorrected. Top-PC share falls before AUROC recovers: at 100 zeroed coordinates it is below 0.2 at 19 layers, while only 7 are restored.

**What survives zeroing.** A follow-up, also designed after the results, zeroes the ten highest-variance coordinates and asks which direction then dominates. These ten coordinates carry a median 92% of the first component’s squared loading. The

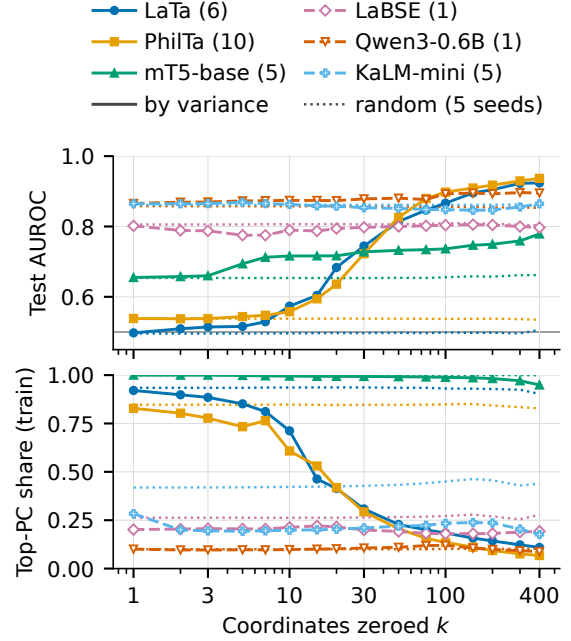


Figure 8: Test ranking AUROC (top) and top-PC share of the training embeddings (bottom) against the number  $k$  of zeroed residual coordinates on a logarithmic axis, one color per model at its lowest-AUROC baseline layer (in parentheses in the legend). Coordinates are ranked by variance across training passages; the dotted line in the same color zeroes  $k$  random coordinates instead (mean over five seeds). Rankings and top-PC share are fit on training embeddings only; the gray horizontal line marks chance (AUROC 0.5). Filled markers and solid lines: T5 encoders; hollow markers and dashed lines: embedding-trained models. LaTa and PhilTa pass AUROC 0.90 only after 200 and 150 coordinates are zeroed (0.924 and 0.937 at  $k = 400$ ). mT5-base reaches only 0.779 at  $k = 400$ , where its top-PC share is still 0.949. At the plotted T5 layers random zeroing moves AUROC by at most 0.010, and zeroing by variance moves the embedding-trained models by at most 0.038. The sweep was added after the first coordinate ablation and carries no prediction.

new dominant direction is the remainder of the first component on the other coordinates (median cosine 0.999 with that remainder). It holds a median 72% of the remaining variance. Removing the original second and third components while keeping the first restores none of the 26 layers. Zeroing followed by ABTT with one, two or three components restores 9, 21 and 26 layers, against 1, 13 and 26 without zeroing. These intervention cells were added after the follow-up’s first run and carry no prediction.

**Number of directions.** Figure 9 plots the sweep over  $D$  at each model’s lowest-AUROC layer, and

| Model                                       | L  | Base  | Zeroed, ranked by mean $ x $ |       |       |       | Zeroed, ranked by variance |       |       |       | Std.  | $D=0$ | ABTT $_{D=1}$ | ABTT $_{D=3}$ |
|---------------------------------------------|----|-------|------------------------------|-------|-------|-------|----------------------------|-------|-------|-------|-------|-------|---------------|---------------|
|                                             |    |       | 1                            | 3     | 5     | 10    | 1                          | 3     | 5     | 10    |       |       |               |               |
| <i>Test ranking AUROC</i>                   |    |       |                              |       |       |       |                            |       |       |       |       |       |               |               |
| LaTa                                        | 6  | 0.496 | 0.498                        | 0.514 | 0.512 | 0.574 | 0.498                      | 0.514 | 0.516 | 0.574 | 0.822 | 0.482 | 0.671         | 0.973         |
| PhilTa                                      | 10 | 0.538 | 0.537                        | 0.540 | 0.540 | 0.557 | 0.538                      | 0.538 | 0.544 | 0.558 | 0.907 | 0.501 | 0.602         | 0.964         |
| mT5-base                                    | 5  | 0.654 | 0.655                        | 0.660 | 0.694 | 0.716 | 0.655                      | 0.660 | 0.694 | 0.716 | 0.806 | 0.670 | 0.835         | 0.914         |
| LaBSE                                       | 1  | 0.806 | 0.804                        | 0.793 | 0.794 | 0.788 | 0.802                      | 0.788 | 0.775 | 0.790 | 0.907 | 0.867 | 0.886         | 0.929         |
| Qwen3-0.6B                                  | 1  | 0.858 | 0.866                        | 0.874 | 0.877 | 0.880 | 0.866                      | 0.870 | 0.873 | 0.874 | 0.963 | 0.954 | 0.959         | 0.970         |
| KaLM-mini                                   | 5  | 0.861 | 0.862                        | 0.857 | 0.857 | 0.862 | 0.865                      | 0.865 | 0.869 | 0.863 | 0.933 | 0.917 | 0.926         | 0.969         |
| <i>Top-PC share of the training vectors</i> |    |       |                              |       |       |       |                            |       |       |       |       |       |               |               |
| LaTa                                        | 6  | 0.934 | 0.921                        | 0.885 | 0.859 | 0.712 | 0.921                      | 0.885 | 0.851 | 0.712 | 0.211 | 0.934 | 0.712         | 0.071         |
| PhilTa                                      | 10 | 0.847 | 0.827                        | 0.796 | 0.760 | 0.580 | 0.828                      | 0.777 | 0.733 | 0.608 | 0.143 | 0.847 | 0.763         | 0.088         |
| mT5-base                                    | 5  | 1.000 | 1.000                        | 1.000 | 0.998 | 0.995 | 1.000                      | 1.000 | 0.998 | 0.995 | 0.910 | 1.000 | 0.458         | 0.138         |
| LaBSE                                       | 1  | 0.263 | 0.263                        | 0.267 | 0.258 | 0.199 | 0.202                      | 0.206 | 0.205 | 0.213 | 0.189 | 0.263 | 0.138         | 0.077         |
| Qwen3-0.6B                                  | 1  | 0.100 | 0.101                        | 0.096 | 0.095 | 0.095 | 0.101                      | 0.096 | 0.097 | 0.098 | 0.102 | 0.100 | 0.077         | 0.049         |
| KaLM-mini                                   | 5  | 0.419 | 0.298                        | 0.301 | 0.306 | 0.192 | 0.284                      | 0.195 | 0.193 | 0.198 | 0.196 | 0.419 | 0.210         | 0.067         |

Table 11: Coordinate ablation at each model’s lowest-AUROC baseline layer (L). Top block: test ranking AUROC for the unmodified mean-pooled vectors (Base); after zeroing the  $k$  residual coordinates with the largest mean absolute training value, or with the largest variance across training passages; after per-coordinate standardization with training statistics (Std.); after centering alone ( $D=0$ ); and after ABTT with one or three components (ABTT $_{D=1}$ , ABTT $_{D=3}$ ), the two reference repairs. Bottom block: top-PC share of the training vectors after the same interventions, the share of their centered variance on the first principal component; centering leaves it unchanged by definition. In the three T5 rows, AUROC stays at or below 0.716 after zeroing ten coordinates under either ranking, while standardization reaches 0.806 to 0.907 and ABTT with three components 0.914 to 0.973. At the collapsed T5 layers (baseline AUROC below 0.70, 26 layers), one component recovers a median 45 percent of the AUROC gain of  $D=10$ , and three recover at least 80 percent at every layer (Table 12). All statistics are fit on training embeddings only. Figure 8 extends the zeroing to  $k = 400$ .

Table 12 gives medians over groups of layers. LaTa reaches the full gain of removing ten components with two (median 98%), and PhilTa with three (95%). mT5-base reaches a median 85% with three and keeps rising up to ten. The failure of a single component is not an estimation artifact. At every collapsed layer the first component fit on training passages and the one fit on test passages agree to  $|\cos| \geq 0.994$ . The selection grid stops at  $D = 10$ ; extending it to  $D = 50$  changes test AUROC by at most 0.023.

**Removed and retained subspaces.** Table 13 ranks passages within the removed and within the retained subspace. At the collapsed layers the removed subspace holds a median 99% of the centered variance of vectors that already rank at chance, so its chance-level ranking there is close to a tautology. The healthy layers are more informative. At the 74 healthy panel layers the removed subspace still ranks at 0.736 to 0.937, so ABTT discards some signal. Yet at all 36 layers of the three pretrained T5 encoders the removed directions rank below the next  $D$  components. At LaTa’s last layer, contrastive fine-tuning moves the top ten directions from below the next ten (0.868 against 0.907) to above them (0.953 against 0.900), consistent with Rajae and Pilehvar (2021).

**Token audit.** Table 14 gives the token audit at each model’s lowest-AUROC layer. In LaTa, a token mix with one mean contribution per token type, fit on training tokens, explains a median 0.985 of the variance of the first component’s score on test passages. For random directions the same statistic is 0.18. At layer 6 the comma alone holds 0.962 of the first component’s score variance over training passages. Ranking token types by the first component alone restores at most 2 of LaTa’s 10 collapsed layers, so the second and third components matter too. Two pooling variants separate the two changes that SIF makes. In LaTa, dropping the 100 most frequent training tokens recovers at least 80% of SIF’s gain at all 10 layers (median AUROC 0.901). Dropping special tokens alone recovers it at none (median 0.498). In PhilTa the token mix explains only a median 0.42 of the first component’s score. Its ablation needs more types: dropping the top 3 restores 1 of 9 layers, the top 10 restore 8, and the top 30 restore all 9 (median 0.925). Dropping random types of the same training mass stays at a median of 0.542. In mT5-base the token mix explains 0.066 of the first component’s score, against 0.160 for random directions. Dropping its top 100 types lifts AUROC to 0.724 to 0.748, short of 0.90. In none of the three models does the first token

| Layers                                           | $n$ | raw   | Removed components $D$ |       |       |       |       |       |       |       |       |       |       |
|--------------------------------------------------|-----|-------|------------------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                  |     |       | 0                      | 1     | 2     | 3     | 5     | 7     | 10    | 15    | 20    | 30    | 50    |
| <i>Lowest-AUROC baseline layer of each model</i> |     |       |                        |       |       |       |       |       |       |       |       |       |       |
| LaTa (6)                                         | 1   | 0.496 | 0.482                  | 0.671 | 0.940 | 0.973 | 0.973 | 0.977 | 0.964 | 0.973 | 0.966 | 0.981 | 0.986 |
| PhilTa (10)                                      | 1   | 0.538 | 0.501                  | 0.602 | 0.888 | 0.964 | 0.977 | 0.982 | 0.982 | 0.975 | 0.977 | 0.973 | 0.969 |
| mT5-base (5)                                     | 1   | 0.654 | 0.670                  | 0.835 | 0.892 | 0.914 | 0.951 | 0.971 | 0.978 | 0.972 | 0.965 | 0.965 | 0.967 |
| LaBSE (1)                                        | 1   | 0.806 | 0.867                  | 0.886 | 0.915 | 0.929 | 0.962 | 0.967 | 0.977 | 0.977 | 0.977 | 0.977 | 0.979 |
| Qwen3-0.6B (1)                                   | 1   | 0.858 | 0.954                  | 0.959 | 0.962 | 0.970 | 0.972 | 0.979 | 0.978 | 0.980 | 0.981 | 0.976 | 0.984 |
| KaLM-mini (5)                                    | 1   | 0.861 | 0.917                  | 0.926 | 0.969 | 0.969 | 0.983 | 0.982 | 0.983 | 0.978 | 0.974 | 0.974 | 0.976 |
| <i>Median over model-layers</i>                  |     |       |                        |       |       |       |       |       |       |       |       |       |       |
| Collapsed T5                                     | 26  | 0.541 | 0.503                  | 0.713 | 0.902 | 0.964 | 0.973 | 0.977 | 0.977 | 0.974 | 0.975 | 0.974 | 0.971 |
| LaTa                                             | 10  | 0.502 | 0.488                  | 0.713 | 0.956 | 0.973 | 0.972 | 0.976 | 0.970 | 0.971 | 0.969 | 0.980 | 0.983 |
| PhilTa                                           | 9   | 0.542 | 0.502                  | 0.602 | 0.872 | 0.963 | 0.980 | 0.981 | 0.981 | 0.975 | 0.978 | 0.973 | 0.970 |
| mT5-base                                         | 7   | 0.657 | 0.678                  | 0.865 | 0.894 | 0.934 | 0.955 | 0.974 | 0.978 | 0.975 | 0.974 | 0.965 | 0.962 |
| Other T5                                         | 10  | 0.874 | 0.936                  | 0.952 | 0.967 | 0.973 | 0.979 | 0.976 | 0.973 | 0.972 | 0.972 | 0.973 | 0.971 |
| Embedding-trained                                | 64  | 0.903 | 0.944                  | 0.966 | 0.971 | 0.976 | 0.979 | 0.980 | 0.978 | 0.977 | 0.975 | 0.974 | 0.973 |
| All                                              | 100 | 0.885 | 0.936                  | 0.958 | 0.968 | 0.973 | 0.978 | 0.979 | 0.977 | 0.975 | 0.974 | 0.974 | 0.972 |

Table 12: Test ranking AUROC against the number  $D$  of top principal components that ABTT removes, fit on training embeddings only. Raw: mean-pooled vectors with no correction;  $D = 0$ : centering on the training mean alone. The top block reads each model at its lowest-AUROC baseline layer (layer in parentheses); the bottom block gives medians over the 26 collapsed T5 layers (baseline AUROC below 0.70), also split by model, the other 10 T5 layers, the 64 layers of the embedding-trained models, and all 100 model-layers. The paper’s selection grid for  $D$  stops at 10.

| Model                                                        | $\ell$ | $D$ | Var. | Full vector |       | Removed subspace |            |            | Retained subspace |           |       |
|--------------------------------------------------------------|--------|-----|------|-------------|-------|------------------|------------|------------|-------------------|-----------|-------|
|                                                              |        |     |      | Raw         | Cent. | PC1              | PCs 2- $D$ | PCs 1- $D$ | Next $D$          | Rand. $D$ | All   |
| <i>Pretrained, train-selected layer and <math>D</math></i>   |        |     |      |             |       |                  |            |            |                   |           |       |
| LaTa                                                         | 12     | 10  | 0.35 | 0.938       | 0.972 | 0.670            | 0.825      | 0.868      | 0.907             | 0.861     | 0.971 |
| PhilTa                                                       | 9      | 10  | 0.99 | 0.539       | 0.501 | 0.462            | 0.545      | 0.487      | 0.922             | 0.873     | 0.983 |
| mT5-base                                                     | 2      | 10  | 0.57 | 0.822       | 0.910 | 0.610            | 0.735      | 0.751      | 0.896             | 0.866     | 0.975 |
| LaBSE                                                        | 11     | 10  | 0.48 | 0.960       | 0.958 | 0.647            | 0.932      | 0.909      | 0.958             | 0.902     | 0.987 |
| Qwen3-0.6B                                                   | 2      | 10  | 0.44 | 0.863       | 0.940 | 0.607            | 0.792      | 0.798      | 0.906             | 0.882     | 0.973 |
| KaLM-mini                                                    | 1      | 10  | 0.41 | 0.887       | 0.958 | 0.518            | 0.846      | 0.819      | 0.910             | 0.880     | 0.981 |
| Collapsed T5, median (26)                                    | –      | –   | 0.99 | 0.541       | 0.503 | 0.463            | 0.601      | 0.489      | 0.919             | 0.874     | 0.977 |
| <i>Fine-tuning contrast, same layer, <math>D = 10</math></i> |        |     |      |             |       |                  |            |            |                   |           |       |
| LaTa, pretrained                                             | 12     | 10  | 0.35 | 0.938       | 0.972 | 0.670            | 0.825      | 0.868      | 0.907             | 0.861     | 0.971 |
| LaTa, fine-tuned                                             | 12     | 10  | 0.33 | 0.984       | 0.985 | 0.697            | 0.935      | 0.953      | 0.900             | 0.874     | 0.970 |
| Qwen3-0.6B, pretrained                                       | 27     | 10  | 0.41 | 0.967       | 0.969 | 0.643            | 0.913      | 0.917      | 0.909             | 0.884     | 0.982 |
| Qwen3-0.6B, fine-tuned                                       | 27     | 10  | 0.31 | 0.993       | 0.996 | 0.684            | 0.980      | 0.983      | 0.969             | 0.927     | 0.992 |
| KaLM-mini, pretrained                                        | 24     | 10  | 0.32 | 0.956       | 0.979 | 0.663            | 0.909      | 0.924      | 0.909             | 0.855     | 0.981 |
| KaLM-mini, fine-tuned                                        | 24     | 10  | 0.23 | 0.997       | 0.996 | 0.699            | 0.977      | 0.978      | 0.967             | 0.918     | 0.994 |

Table 13: Test ranking AUROC of cosine within parts of the pooled vector. ABTT removes the top  $D$  principal components of the centered training embeddings; Var. is their share of the centered training variance. Raw: uncorrected vector; Cent.: centered on the training mean; PC1: projection onto the first component alone, where cosine reduces to a sign match; PCs 2- $D$  and PCs 1- $D$ : projections onto the rest of the removed subspace and onto all of it. Retained subspace, a dimension-matched control: Next  $D$ , PCs  $D+1$  to  $2D$ ; Rand.  $D$ , random  $D$ -dimensional projections of the retained vectors (mean over five seeds); All, the ABTT output. Top block: each pretrained model at its train-selected ABTT layer  $\ell$  with  $D$  chosen on training DirAcc@1, and the median over the 26 collapsed T5 layers (baseline AUROC below 0.70), each at its own layer and  $D$ . Bottom block: each contrastively fine-tuned encoder at its train-selected ABTT layer, next to its pretrained model at the same layer, both with  $D = 10$ . All components are fit on training embeddings only.

carry the dominant direction (share at most 0.005). The same three delimiter roles carry outlier activations in BERT (Bondarenko et al., 2023), and delimiter tokens carry massive activations in large language models (Sun et al., 2024). Removing punctuation and special tokens before pooling also helps BERT (Fuster Baggetto and Fresno, 2022);

here it separates chance from recovery.

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| Model      | L  | AUROC by pooling |          |           |       |          | Token ablation |       | Token-mix EV |       | Share by token group |       |       |          |
|------------|----|------------------|----------|-----------|-------|----------|----------------|-------|--------------|-------|----------------------|-------|-------|----------|
|            |    | Mean             | No spec. | SIF+spec. | SIF   | No freq. | Drop           | Rand. | PC1          | Rand. | Spec.                | Freq. | Other | $ \rho $ |
| LaTa       | 6  | 0.496            | 0.493    | 0.860     | 0.878 | 0.904    | 0.922          | 0.483 | 0.98         | 0.19  | 0.01                 | 0.97  | 0.02  | 0.38     |
| PhilTa     | 10 | 0.538            | 0.496    | 0.677     | 0.879 | 0.692    | 0.917          | 0.540 | 0.44         | 0.14  | 0.23                 | 0.75  | 0.02  | 0.11     |
| mT5-base   | 5  | 0.654            | 0.653    | 0.675     | 0.673 | 0.726    | 0.724          | 0.656 | 0.07         | 0.24  | 0.00                 | 0.53  | 0.47  | 0.20     |
| LaBSE      | 1  | 0.806            | 0.809    | 0.812     | 0.818 | 0.816    | 0.827          | 0.800 | 0.36         | 0.73  | 0.09                 | 0.37  | 0.53  | 0.95     |
| Qwen3-0.6B | 1  | 0.858            | 0.855    | 0.856     | 0.852 | 0.869    | 0.885          | 0.836 | 0.99         | 0.71  | 0.74                 | 0.08  | 0.18  | 0.58     |
| KaLM-mini  | 5  | 0.861            | 0.859    | 0.867     | 0.863 | 0.872    | 0.879          | 0.854 | 0.98         | 0.47  | 0.01                 | 0.02  | 0.97  | 0.51     |

Table 14: Token audit at each model’s lowest-AUROC baseline layer (L). Pooling: test ranking AUROC of mean pooling (Mean); mean pooling without special tokens (No spec.); SIF frequency weights with special tokens kept (SIF+spec.); SIF pooling with training-only token frequencies, which also drops special tokens (SIF); and mean pooling without the 100 most frequent training tokens (No freq.). Token ablation: test AUROC of mean pooling after dropping the  $m$  token types that contribute most to the top principal components (Drop), with  $m$  (1 to 100) and the ranking (one of two) chosen by training AUROC, and the mean test AUROC of five random sets of other token types, sampled in proportion to training count, that hold at least as many training tokens as the dropped types (Rand.). Token-mix EV: the share of the variance of the first principal component’s score across test passages that is explained by which token types a passage contains, with one mean per token type fit on training tokens (PC1), next to the mean of the same quantity over 20 random directions orthogonal to the top ten principal components (Rand.). Share by token group: the covariance share of that variance contributed by special tokens, by the 100 most frequent tokens and by all other tokens; the three sum to one, but a single share can fall below zero or exceed one.  $|\rho|$ : absolute Spearman correlation between the PC1 score and the log token count of test passages (the sign of a principal component is a convention). Components, token frequencies and token rankings are fit on training passages only.

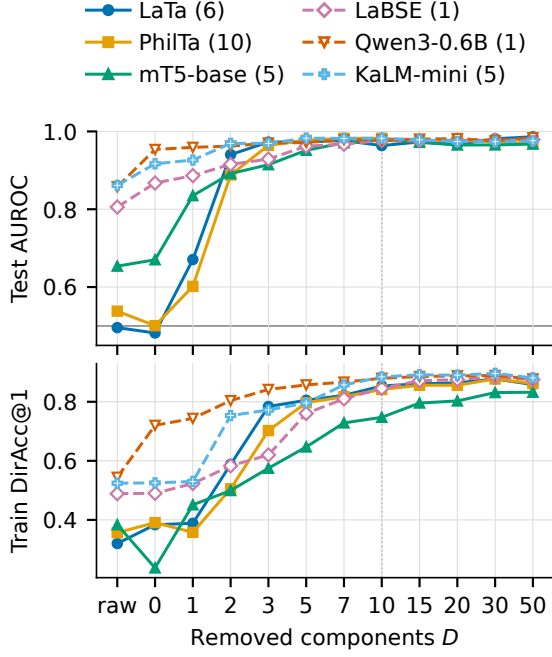


Figure 9: Test ranking AUROC (top) and training DirAcc@1 (bottom) against the number  $D$  of removed principal components, one line per model at its lowest-AUROC baseline layer (in parentheses in the legend). The horizontal axis is categorical: raw is the uncorrected vector,  $D = 0$  is centering on the training mean alone, and the ticks are not evenly spaced in  $D$ . ABTT is fit on training embeddings only; the dotted vertical line marks  $D = 10$ , the top of the selection grid, and the gray horizontal line chance (AUROC 0.5). Filled markers and solid lines: T5 encoders; hollow markers and dashed lines: embedding-trained models. At the plotted T5 layers one component restores part of the ranking. LaTa and PhilTa recover nearly all of it with two or three components; mT5-base keeps improving up to ten.

## F Bootstrap Intervals and Routing Checks

This appendix gives the intervals behind Table 3 and the routing checks that Section 7 quotes.

**Intervals.** A bootstrap replicate draws the 514 test directories with replacement ( $B = 10,000$ , seed 233). A drawn directory keeps every file and pair: a directory drawn  $c$  times weights its files and within-directory pairs by  $c$  and its pairs with a directory drawn  $c'$  times by  $cc'$ , and no pair is formed between two copies of one directory. The layer, threshold, ABTT components,  $D$ , and SIF weights stay fit on the training split, and every score is computed once on the full test set, so a replicate only reweights test pairs and files. Intervals are 95% percentiles; a paired difference recomputes both cells on each replicate. Before resampling, every cell was recomputed with the paper’s evaluation code and matched its printed value. Table 15 gives every cell and Table 16 the contrasts quoted in Section 7.

**Routing checks.** Table 17 reports, at the headline routing cells, the AUROC of each test witness’s maximum cosine for existing against new witnesses (no threshold), the gap between the best test threshold and the training threshold, the skewness of the 10-occurrence distribution (Radovanović et al., 2010), centering alone ( $D = 0$ ), and assignment accuracy with  $\tau$  refit as the exact best-F1 cut over all training pair scores.

**Threshold grid.** We keep the grid of 200 evenly spaced thresholds in  $[0, 1]$  used for every printed number. Every finer grid optimizes the same training objective, the best F1 over same- against different-directory training pairs, and differs only in resolution. With the layer and  $D$  fixed, a grid of 10,001 values moves 45 of the 66 printed routing numbers, by a median of 3 files (0.35 points); ABTT, SIF+ABTT, and fine-tuned cells move by at most 0.8 points, and Base and SIF cells by up to 2.6 (mT5-base’s baseline). The exact best-F1 cut moves 48 of them. Refitting would trade one set of differences of one to a few files for another and changes no conclusion of Section 7.

| Model                   | Setting  | Ranking              |                      | Routing           |                   |
|-------------------------|----------|----------------------|----------------------|-------------------|-------------------|
|                         |          | AUROC                | Cosine gap           | Assignment acc.   | DirAcc@1          |
| LaTa                    | Base     | 0.938 [0.910, 0.963] | 0.237 [0.222, 0.253] | 73.8 [70.2, 77.3] | 72.1 [68.4, 75.8] |
|                         | SIF      | 0.948 [0.915, 0.974] | 0.366 [0.342, 0.390] | 83.2 [80.1, 86.3] | 81.8 [78.5, 85.1] |
|                         | ABTT     | 0.971 [0.948, 0.991] | 0.525 [0.491, 0.559] | 88.5 [85.8, 90.9] | 86.1 [83.2, 88.9] |
|                         | SIF+ABTT | 0.966 [0.938, 0.989] | 0.580 [0.541, 0.617] | 90.2 [87.6, 92.6] | 88.2 [85.4, 90.9] |
| PhilTa                  | Base     | 0.939 [0.918, 0.958] | 0.231 [0.217, 0.247] | 70.9 [66.9, 74.9] | 69.3 [65.2, 73.5] |
|                         | SIF      | 0.955 [0.935, 0.972] | 0.319 [0.300, 0.339] | 79.0 [75.6, 82.4] | 77.7 [74.2, 81.3] |
|                         | ABTT     | 0.983 [0.969, 0.993] | 0.540 [0.508, 0.571] | 91.1 [88.9, 93.2] | 88.3 [85.6, 90.9] |
|                         | SIF+ABTT | 0.982 [0.969, 0.993] | 0.573 [0.538, 0.608] | 91.4 [89.1, 93.6] | 88.9 [86.2, 91.5] |
| mT5-base                | Base     | 0.838 [0.810, 0.866] | 0.081 [0.072, 0.092] | 47.2 [43.1, 51.6] | 46.6 [42.5, 51.0] |
|                         | SIF      | 0.845 [0.816, 0.876] | 0.036 [0.031, 0.042] | 54.5 [50.3, 58.8] | 54.1 [49.8, 58.4] |
|                         | ABTT     | 0.975 [0.954, 0.992] | 0.536 [0.498, 0.574] | 90.3 [87.9, 92.7] | 88.5 [85.7, 91.2] |
|                         | SIF+ABTT | 0.974 [0.950, 0.993] | 0.519 [0.480, 0.559] | 90.3 [88.0, 92.6] | 87.5 [84.8, 90.2] |
| LaBSE                   | Base     | 0.956 [0.932, 0.977] | 0.183 [0.171, 0.195] | 83.3 [80.2, 86.4] | 81.5 [78.1, 84.7] |
|                         | SIF      | 0.956 [0.931, 0.977] | 0.184 [0.172, 0.197] | 83.4 [80.2, 86.5] | 81.9 [78.6, 85.2] |
|                         | ABTT     | 0.987 [0.977, 0.995] | 0.611 [0.575, 0.646] | 90.8 [88.5, 92.9] | 88.5 [85.8, 90.9] |
|                         | SIF+ABTT | 0.976 [0.960, 0.989] | 0.580 [0.543, 0.617] | 89.6 [87.2, 91.9] | 86.9 [84.2, 89.5] |
| Qwen3-0.6B              | Base     | 0.966 [0.950, 0.981] | 0.024 [0.023, 0.026] | 82.3 [79.1, 85.4] | 80.3 [76.9, 83.6] |
|                         | SIF      | 0.942 [0.919, 0.963] | 0.053 [0.048, 0.060] | 76.0 [72.5, 79.4] | 74.2 [70.6, 77.8] |
|                         | ABTT     | 0.973 [0.948, 0.990] | 0.553 [0.513, 0.590] | 91.5 [89.1, 93.7] | 89.4 [86.7, 92.0] |
|                         | SIF+ABTT | 0.972 [0.947, 0.991] | 0.540 [0.499, 0.580] | 90.8 [88.3, 93.1] | 88.6 [85.8, 91.2] |
| KaLM-mini               | Base     | 0.972 [0.955, 0.986] | 0.056 [0.053, 0.060] | 87.5 [84.8, 90.1] | 85.9 [82.9, 88.7] |
|                         | SIF      | 0.964 [0.944, 0.982] | 0.056 [0.051, 0.062] | 85.7 [82.8, 88.5] | 84.0 [80.9, 87.1] |
|                         | ABTT     | 0.981 [0.963, 0.993] | 0.568 [0.529, 0.605] | 91.7 [89.5, 93.8] | 89.4 [86.7, 91.9] |
|                         | SIF+ABTT | 0.981 [0.962, 0.995] | 0.535 [0.496, 0.573] | 90.8 [88.4, 93.0] | 88.0 [85.2, 90.7] |
| LaTa (fine-tuned)       | Base     | 0.984 [0.973, 0.992] | 0.387 [0.368, 0.406] | 83.4 [80.2, 86.5] | 81.6 [78.1, 84.9] |
|                         | ABTT     | 0.970 [0.947, 0.990] | 0.548 [0.511, 0.583] | 87.8 [85.2, 90.2] | 85.2 [82.3, 87.9] |
| Qwen3-0.6B (fine-tuned) | Base     | 0.996 [0.993, 0.999] | 0.684 [0.653, 0.714] | 91.4 [88.7, 93.8] | 90.8 [88.1, 93.3] |
|                         | ABTT     | 0.994 [0.989, 0.998] | 0.716 [0.680, 0.751] | 92.1 [89.6, 94.3] | 91.4 [88.8, 93.7] |
| KaLM-mini (fine-tuned)  | Base     | 0.997 [0.993, 0.999] | 0.636 [0.605, 0.666] | 92.4 [90.1, 94.6] | 91.7 [89.3, 94.0] |
|                         | ABTT     | 0.994 [0.988, 0.998] | 0.651 [0.612, 0.687] | 93.4 [91.2, 95.4] | 92.5 [90.3, 94.7] |
| BM25 (word)             | –        | 0.974 [0.956, 0.987] | 0.243 [0.222, 0.266] | 62.5 [58.2, 66.6] | 56.3 [52.0, 60.5] |
| Char. n-grams           | –        | 0.987 [0.976, 0.996] | 0.555 [0.519, 0.591] | 91.8 [89.7, 93.8] | 89.9 [87.3, 92.3] |
| Levenshtein             | –        | 0.952 [0.927, 0.975] | 0.362 [0.332, 0.393] | 87.8 [85.0, 90.4] | 86.2 [83.2, 89.2] |

Table 15: Every headline cell, those of Table 3 and the SIF and SIF+ABTT cells, with its 95% percentile intervals from a directory-level bootstrap ( $B = 10,000$  resamples of the 514 test directories with replacement, seed 233). A replicate keeps every file and pair of a drawn directory; a directory drawn twice counts twice, and pairs between its two copies are not formed. The threshold  $\tau$ , the ABTT components and  $D$ , the SIF weights and the layer stay fit on the fixed training split; only the test evaluation is resampled, so the intervals cover test-set sampling and not the choice of split. Each cell is read at its train-selected layer (Table 20); routing in percent. The lexical rows are the lexical reference systems of Table 19; Char. n-grams is TF-IDF over character 3–5-grams.

| Contrast                         | Model      | $\Delta$ AUROC          | $\Delta$ Assignment acc. | $\Delta$ DirAcc@1  |
|----------------------------------|------------|-------------------------|--------------------------|--------------------|
| ABTT — Base                      | LaTa       | +0.034 [0.017, 0.054]   | +14.7 [11.1, 18.3]       | +14.0 [10.4, 17.6] |
|                                  | PhilTa     | +0.043 [0.029, 0.060]   | +20.3 [16.2, 24.5]       | +19.0 [15.0, 23.1] |
|                                  | mT5-base   | +0.137 [0.109, 0.166]   | +43.1 [38.6, 47.6]       | +41.8 [37.2, 46.4] |
|                                  | LaBSE      | +0.030 [0.014, 0.049]   | +7.5 [4.6, 10.4]         | +7.0 [4.1, 10.0]   |
|                                  | Qwen3-0.6B | +0.006 [−0.010, 0.019]  | +9.2 [6.3, 12.2]         | +9.1 [6.1, 12.2]   |
|                                  | KaLM-mini  | +0.009 [−0.002, 0.021]  | +4.2 [1.7, 6.7]          | +3.5 [1.0, 6.0]    |
| Fine-tuned: ABTT — Base          | LaTa       | −0.014 [−0.032, 0.001]  | +4.3 [1.7, 7.1]          | +3.6 [1.0, 6.3]    |
|                                  | Qwen3-0.6B | −0.002 [−0.005, 0.001]  | +0.7 [−0.7, 2.1]         | +0.6 [−0.8, 2.0]   |
|                                  | KaLM-mini  | −0.003 [−0.007, 0.001]  | +0.9 [−0.3, 2.2]         | +0.8 [−0.5, 2.1]   |
| Fine-tuned ABTT — zero-shot ABTT | LaTa       | −0.002 [−0.010, 0.005]  | −0.7 [−2.6, 1.2]         | −0.9 [−2.8, 1.0]   |
|                                  | Qwen3-0.6B | +0.022 [0.006, 0.044]   | +0.6 [−1.6, 2.9]         | +2.0 [0.0, 4.0]    |
|                                  | KaLM-mini  | +0.013 [0.002, 0.029]   | +1.6 [−0.3, 3.6]         | +3.1 [1.2, 5.2]    |
| Char. n-grams — ABTT             | LaTa       | +0.016 [0.003, 0.030]   | +3.4 [1.1, 5.7]          | +3.7 [1.4, 6.2]    |
|                                  | PhilTa     | +0.004 [−0.002, 0.011]  | +0.7 [−1.0, 2.4]         | +1.5 [−0.2, 3.3]   |
|                                  | mT5-base   | +0.012 [0.000, 0.027]   | +1.5 [−0.6, 3.7]         | +1.4 [−0.7, 3.6]   |
|                                  | LaBSE      | 0.000 [−0.004, 0.005]   | +1.0 [−1.0, 3.1]         | +1.4 [−0.8, 3.6]   |
|                                  | Qwen3-0.6B | +0.014 [0.002, 0.031]   | +0.3 [−1.6, 2.3]         | +0.5 [−1.4, 2.5]   |
|                                  | KaLM-mini  | +0.006 [−0.002, 0.016]  | +0.1 [−1.6, 1.9]         | +0.5 [−1.3, 2.3]   |
| Centering — Base (ABTT layer)    | LaTa       | +0.034 [0.021, 0.049]   | −2.2 [−5.2, 0.9]         | +6.8 [4.6, 9.1]    |
|                                  | PhilTa     | −0.038 [−0.060, −0.016] | +12.9 [9.0, 16.9]        | +12.4 [8.5, 16.3]  |
|                                  | mT5-base   | +0.088 [0.066, 0.109]   | +20.5 [16.0, 24.8]       | +15.6 [11.3, 19.9] |
|                                  | LaBSE      | −0.002 [−0.009, 0.004]  | −4.7 [−7.4, −2.0]        | −4.7 [−7.3, −2.0]  |
|                                  | Qwen3-0.6B | +0.076 [0.061, 0.092]   | +19.2 [15.4, 23.2]       | +16.7 [12.7, 20.7] |
|                                  | KaLM-mini  | +0.070 [0.052, 0.089]   | +19.3 [15.5, 23.2]       | +16.9 [13.1, 20.7] |
| ABTT — centering (ABTT layer)    | LaTa       | −0.001 [−0.016, 0.012]  | +40.4 [35.8, 44.8]       | +49.2 [44.1, 53.9] |
|                                  | PhilTa     | +0.482 [0.460, 0.504]   | +7.3 [4.9, 10.0]         | +6.6 [4.1, 9.3]    |
|                                  | mT5-base   | +0.065 [0.050, 0.081]   | +24.5 [20.7, 28.3]       | +27.7 [23.7, 31.7] |
|                                  | LaBSE      | +0.029 [0.013, 0.048]   | +12.1 [8.8, 15.5]        | +11.7 [8.4, 15.0]  |
|                                  | Qwen3-0.6B | +0.033 [0.018, 0.050]   | +25.9 [22.0, 29.9]       | +27.7 [23.5, 32.1] |
|                                  | KaLM-mini  | +0.023 [0.008, 0.040]   | +12.0 [9.1, 15.1]        | +13.8 [10.7, 17.0] |
| Spread over six models           | Base       | 0.134 [0.110, 0.158]    | 40.3 [35.7, 44.7]        | 39.3 [34.6, 43.7]  |
|                                  | ABTT       | 0.015 [0.006, 0.034]    | 3.3 [1.8, 5.8]           | 3.3 [1.8, 6.0]     |

Table 16: Paired differences between headline cells, with 95% intervals from the same directory-level bootstrap replicates as Table 15 ( $B = 10,000$ , 514 test directories, seed 233). Both cells of a contrast are recomputed on each replicate, so the interval reflects their correlation. Each cell sits at its own train-selected layer. The two centering contrasts compare baseline, centering alone ( $D = 0$ ) and ABTT at the layer of the ABTT cell, the ranking ABTT layer for  $\Delta$  AUROC and the routing ABTT layer for the two routing columns. Spread: the largest minus the smallest of the six zero-shot models. Routing in points. Spreads and differences are computed before rounding, so they can differ by one unit in the last digit from the difference of the rounded cells. The threshold  $\tau$ , the ABTT components and  $D$ , the SIF weights and the layer stay fit on the fixed training split; only the test evaluation is resampled, so the intervals cover test-set sampling and not the choice of split.

| Model      | Setting            | L  | Cos. SD | Exist./new AUROC     | Assign. | Oracle gap         | $\tau$ | Exact | Skew |
|------------|--------------------|----|---------|----------------------|---------|--------------------|--------|-------|------|
| LaTa       | Base               | 1  | 0.112   | 0.869 [0.840, 0.896] | 73.8    | +5.6 [2.8, 9.4]    | 0.844  | 74.6  | 2.81 |
|            | Base @ ABTT L      | 8  | 0.632   | 0.621 [0.583, 0.658] | 50.2    | +16.4 [11.9, 21.9] | 0.995  | 44.1  | 1.59 |
|            | Centering @ ABTT L | 8  | 0.901   | 0.604 [0.564, 0.644] | 48.0    | +14.3 [9.4, 21.8]  | 0.995  | 42.8  | 1.97 |
|            | ABTT               | 8  | 0.100   | 0.943 [0.925, 0.960] | 88.5    | +0.6 [0.2, 1.7]    | 0.472  | 88.5  | 1.03 |
| PhilTa     | Base               | 1  | 0.111   | 0.843 [0.813, 0.873] | 70.9    | +7.0 [3.7, 11.5]   | 0.894  | 71.0  | 3.15 |
|            | Centering @ ABTT L | 1  | 0.126   | 0.906 [0.882, 0.928] | 83.8    | +1.2 [0.6, 2.5]    | 0.548  | 84.4  | 1.78 |
|            | ABTT               | 1  | 0.086   | 0.960 [0.944, 0.974] | 91.1    | +0.7 [0.2, 1.8]    | 0.417  | 91.1  | 0.72 |
| mT5-base   | Base               | 12 | 0.104   | 0.685 [0.642, 0.726] | 47.2    | +19.6 [14.3, 26.8] | 0.970  | 49.5  | 2.59 |
|            | Base @ ABTT L      | 1  | 0.065   | 0.696 [0.654, 0.737] | 45.3    | +23.1 [16.5, 30.2] | 0.995  | 52.2  | 3.83 |
|            | Centering @ ABTT L | 1  | 0.242   | 0.765 [0.729, 0.801] | 65.9    | +5.6 [2.0, 10.5]   | 0.688  | 66.0  | 3.29 |
|            | ABTT               | 1  | 0.081   | 0.952 [0.935, 0.967] | 90.3    | +0.9 [0.2, 2.4]    | 0.437  | 91.1  | 1.13 |
| LaBSE      | Base               | 11 | 0.039   | 0.922 [0.901, 0.942] | 83.3    | +1.0 [0.1, 3.5]    | 0.975  | 82.5  | 2.00 |
|            | Centering @ ABTT L | 11 | 0.209   | 0.891 [0.866, 0.914] | 78.7    | +2.8 [0.7, 6.3]    | 0.704  | 79.0  | 1.84 |
|            | ABTT               | 11 | 0.102   | 0.960 [0.944, 0.973] | 90.8    | +0.6 [0.1, 1.6]    | 0.452  | 90.8  | 0.58 |
| Qwen3-0.6B | Base               | 28 | 0.043   | 0.900 [0.874, 0.923] | 82.3    | +1.0 [0.2, 2.6]    | 0.945  | 82.6  | 2.51 |
|            | Base @ ABTT L      | 5  | 0.034   | 0.708 [0.666, 0.748] | 46.4    | +23.4 [17.7, 30.4] | 0.990  | 53.0  | 2.49 |
|            | Centering @ ABTT L | 5  | 0.171   | 0.802 [0.767, 0.835] | 65.6    | +9.6 [5.9, 14.5]   | 0.653  | 66.6  | 1.77 |
|            | ABTT               | 5  | 0.086   | 0.954 [0.935, 0.971] | 91.5    | +0.7 [0.2, 1.7]    | 0.417  | 91.5  | 1.07 |
| KaLM-mini  | Base               | 23 | 0.021   | 0.931 [0.910, 0.950] | 87.5    | 0.0 [0.0, 1.1]     | 0.965  | 86.2  | 1.97 |
|            | Base @ ABTT L      | 3  | 0.054   | 0.756 [0.719, 0.792] | 60.4    | +11.3 [6.7, 17.0]  | 0.975  | 58.3  | 3.58 |
|            | Centering @ ABTT L | 3  | 0.135   | 0.881 [0.853, 0.906] | 79.7    | +2.7 [1.2, 4.8]    | 0.558  | 79.5  | 1.75 |
|            | ABTT               | 3  | 0.085   | 0.958 [0.941, 0.973] | 91.7    | +0.2 [0.0, 1.6]    | 0.397  | 91.7  | 0.87 |

Table 17: Routing checks at the headline routing cells (Table 3) of the six zero-shot models: the Base and ABTT cells at their train-selected layers (L), and baseline and centering alone ( $D = 0$ , subtract the training mean) at the ABTT cell’s layer (a second baseline row only where that layer differs). Cos. SD: standard deviation of all test pairwise cosines. Exist./new AUROC: threshold-free AUROC of each test file’s maximum cosine for existing against new files. Assign.: assignment accuracy at the train-fit  $\tau$  (the printed cell). Oracle gap: best assignment accuracy over every test threshold minus Assign. Exact: assignment accuracy with  $\tau$  re-fit as the exact best-F1 cut over all training pair scores instead of the 200-point grid. Skew: skewness of the  $k$ -occurrence distribution  $N_{10}$  over test files (Radovanović et al., 2010). Brackets: 95% directory-bootstrap intervals ( $B = 10,000$ ). Holding layer and  $D$  fixed, the fine grid (step  $10^{-4}$ ) moves 45 and the exact cut 48 of the 66 printed routing numbers; re-selecting layer and  $D$  under the new grid as well moves 42 (fine) and 44 (exact) of 60.

## G Routing Gain Without a Threshold

This appendix gives the paired contrasts behind the second finding of Section 7, for the three embedding-trained encoders. Each contrast compares the ABTT cell with the baseline cell, each at its own train-selected routing layer, and its 95% interval comes from the directory-level bootstrap of Appendix F. Per-cell values are in Table 17.

**No threshold.** The existing-versus-new AUROC of each test witness’s best-match cosine rises by 0.037 (0.021 to 0.055) for LaBSE, 0.055 (0.034 to 0.077) for Qwen3-0.6B, and 0.027 (0.012 to 0.043) for KaLM-mini.

**Best test threshold.** At the baseline cells the train-fit threshold comes within 1.0 point of the best test threshold. When both cells use their best test threshold, ABTT still raises assignment accuracy by 7.0 points (4.2 to 9.3) for LaBSE, 8.9 (6.1

to 11.7) for Qwen3-0.6B, and 4.4 (2.2 to 6.8) for KaLM-mini.

**The training mean.** Centering alone ( $D = 0$ ) at the baseline cell’s layer changes assignment accuracy by  $-4.7$  points ( $-7.4$  to  $-2.0$ ) for LaBSE,  $+2.6$  ( $-0.4$  to  $5.5$ ) for Qwen3-0.6B, and  $-1.9$  ( $-3.7$  to  $-0.1$ ) for KaLM-mini. At the ABTT cell’s layer, the removed components add 12.1, 25.9 and 12.0 points on top of centering (Table 16).

**Hubness.** From the baseline cell to the ABTT cell, the skewness of the 10-occurrence distribution (Radovanović et al., 2010) falls from 2.0–2.5 to 0.6–1.1, while centering alone at the ABTT layer leaves it near 1.8. Across representations, low hubness does not track good routing. The character n-gram reference has a skewness of 10.6 and routes as well as the best projected encoder, and SIF raises skewness in LaTa and PhilTa while it raises their routing. We therefore report the drop in hubness as an observation and do not test it as a cause.

## H Reference Systems

Table 3 carries four reference rows below the six encoders: one supervised fine-tuning ceiling for each of the three fine-tuned models, and the character 3–5-gram TF-IDF reference. This appendix describes them. Table 18 gives the three fine-tuning ceilings with the zero-shot rows they are compared against. Some routing numbers are means over *five reseedings*: each reseeding keeps the training/test split and redraws, within each directory’s test witnesses, which serve as queries and which as references; queries are routed against the reference directories. Every model we fine-tuned is reported, on whichever side of its own zero-shot row it lands.

Three encoders are fine-tuned under one recipe, LaTa, Qwen3-0.6B and KaLM-mini. Fine-tuning only the Latin-pretrained model would leave the ceiling confounded with Latin pretraining, so the second model is a multilingual decoder-only encoder with no Latin-specific pretraining, Qwen3-0.6B. The third answers the remaining question, whether supervision can clear the best cell the correction produces anywhere in the paper: KaLM-mini holds it, at 91.7 assignment accuracy and at 89.4 DirAcc@1 tied with Qwen3-0.6B. All three are trained with symmetric InfoNCE over positive pairs and in-batch negatives (no two pairs from one directory in a batch), temperature 0.05, AdamW at learning rate  $2 \times 10^{-5}$  with weight decay 0.01,

linear warmup over the first 10 percent of steps then linear decay, gradient clipping at 1.0, 16 pairs per batch, up to 8 epochs with early stopping after 3 epochs without dev improvement, and seed 42. All three are mean-pooled with the same token filter as their zero-shot rows, so the fine-tuned vectors live in the space those rows are measured in; Qwen3-0.6B and KaLM-mini add gradient checkpointing, which recomputes activations rather than storing them and leaves the gradients unchanged. The dev slice holds out 28 of the 190 train directories with at least two files (71 files), the same directories for every model, so each trains on 499 of the 565 positive train pairs, and the checkpoint is chosen by dev DirAcc@1 with dev AUROC as the tie-break. Accuracy differences at or below one dev file are treated as ties, because one file out of 71 is the pool’s whole resolution; without that, a model whose zero-shot encoder already routes all 71 dev files can never select a trained checkpoint. A 71-file pool is the weak link in this protocol: one borderline file flipped between two otherwise identical Qwen3-0.6B runs at a training loss of  $6 \times 10^{-4}$ , which is GPU non-determinism rather than a difference in the model, and it is worth 1.4 points of the selection criterion. LaTa selects epoch 7, the last epoch run before early stopping fired, and its ABTT sweep selects  $D = 10$ , the top of the grid, at every layer, so the tuned and fixed ABTT variants coincide there. Qwen3-0.6B selects epoch 3 of 6 run, and its sweep is a genuine one, selecting the top of the grid at 12 of 28 layers and  $D = 2$  at the layer its routing row reports. KaLM-mini selects epoch 1 of 4 run, the only model of the three whose zero-shot encoder leaves the dev pool room to improve, and its sweep selects the top of the grid at 19 of 24 layers including the layer its routing row reports. All three rows are ceilings at this training budget rather than asymptotes: training loss reaches  $4 \times 10^{-5}$  on Qwen3-0.6B by epoch 4 and  $3 \times 10^{-5}$  on KaLM-mini, so the 499 pairs are memorized and more epochs buy nothing.

**Fine-tuning ties the projection on LaTa routing and adds about two points on Qwen3-0.6B and KaLM-mini.** Fine-tuning LaTa on the positive training pairs gives an uncorrected ranking AUROC of 0.984, above every zero-shot baseline cell (0.838 to 0.972), and 83.4 assignment accuracy. With ABTT on top it reaches 87.8 assignment accuracy and 85.2 DirAcc@1. In point estimate that is below every zero-shot ABTT cell (88.5 to 91.7

and 86.1 to 89.4). Against LaTa’s own zero-shot ABTT cell, however, the DirAcc@1 difference is  $-0.9$  points ( $-2.8$  to  $1.0$ ; Table 16), and over five reseeds both reach 87.7. ABTT lowers the fine-tuned row’s AUROC from 0.984 to 0.970 in point estimate ( $-0.014$ , interval  $-0.032$  to  $0.001$ ) while it widens the cosine gap from 0.387 to 0.548. On the other two encoders the same recipe goes further. With ABTT, fine-tuned Qwen3-0.6B reaches 92.1 assignment accuracy and 91.4 DirAcc@1, and fine-tuned KaLM-mini 93.4 and 92.5, both above every zero-shot ABTT cell in point estimate. On the single split only KaLM-mini’s DirAcc@1 lead over its zero-shot ABTT cell excludes zero ( $+3.1$ ,  $1.2$  to  $5.2$ ). Over five reseeds the fine-tuned encoder with ABTT beats its zero-shot ABTT row by 1.8 points on Qwen3-0.6B (92.3 against 90.5) and by 2.1 points on KaLM-mini (93.0 against 90.9). The tie with the projection is therefore a statement about LaTa, not about post-processing in general. All three ceilings are also flattered: 206 of the 535 test witnesses whose directory already exists (38.5 percent) sit in a directory that supplied a training pair, and witnesses inside a directory are near-duplicates, so each margin is an upper bound.

**Lexical reference systems.** Table 19 gives the three lexical reference systems (BM25, character 3–5-gram TF-IDF, and Levenshtein similarity) on both tasks; the character n-gram row is the reference quoted in Section 7.

| System                         | Ranking |            | Routing      |          |                          |
|--------------------------------|---------|------------|--------------|----------|--------------------------|
|                                | AUROC   | Cosine gap | Assign. acc. | DirAcc@1 | DirAcc@1<br>5 reseedings |
| LaTa (zero-shot)               | 0.938   | 0.237      | 73.8         | 72.1     | 73.1 $\pm$ 1.0           |
| LaTa (zero-shot) + ABTT        | 0.971   | 0.525      | 88.5         | 86.1     | 87.7 $\pm$ 0.4           |
| LaTa (fine-tuned)              | 0.984   | 0.387      | 83.4         | 81.6     | 83.4 $\pm$ 0.9           |
| LaTa (fine-tuned) + ABTT       | 0.970   | 0.548      | 87.8         | 85.2     | 87.7 $\pm$ 0.8           |
| Qwen3-0.6B (zero-shot)         | 0.966   | 0.024      | 82.3         | 80.3     | 82.4 $\pm$ 1.0           |
| Qwen3-0.6B (zero-shot) + ABTT  | 0.973   | 0.553      | 91.5         | 89.4     | 90.5 $\pm$ 0.4           |
| Qwen3-0.6B (fine-tuned)        | 0.996   | 0.684      | 91.4         | 90.8     | 91.7 $\pm$ 0.6           |
| Qwen3-0.6B (fine-tuned) + ABTT | 0.994   | 0.716      | 92.1         | 91.4     | 92.3 $\pm$ 0.2           |
| KaLM-mini (zero-shot)          | 0.972   | 0.056      | 87.5         | 85.9     | 87.3 $\pm$ 0.7           |
| KaLM-mini (zero-shot) + ABTT   | 0.981   | 0.568      | 91.7         | 89.4     | 90.9 $\pm$ 0.6           |
| KaLM-mini (fine-tuned)         | 0.997   | 0.636      | 92.4         | 91.7     | 92.6 $\pm$ 0.8           |
| KaLM-mini (fine-tuned) + ABTT  | 0.994   | 0.651      | 93.4         | 92.5     | 93.0 $\pm$ 0.5           |

Table 18: Supervised fine-tuning reference ceiling on LaTa, Qwen3-0.6B and KaLM-mini. Fine-tuning raises ranking AUROC for every model, and ABTT on top of it lowers AUROC again in point estimate; on routing over the five reseedings, the fine-tuned encoder with ABTT is level with its zero-shot ABTT row for LaTa and above it for Qwen3-0.6B and KaLM-mini. The fine-tuned encoders are trained contrastively on 499 of the 565 positive training pairs (in-batch negatives, symmetric InfoNCE), with a dev slice carved out of the training split by directory for model selection; the test split is untouched until this evaluation. Ranking columns are test AUROC and cosine gap at the layer chosen by training AUROC; routing columns are test assignment accuracy and DirAcc@1, in percent, at the layer chosen by training DirAcc@1, with  $\tau$  learned on the training split. The selected layers are listed in Table 20. The last column repeats DirAcc@1 at the same layer as a mean  $\pm$  standard deviation over five reseedings of the query and reference files, with the training/test split fixed. ABTT rows for LaTa sweep  $D$  per layer on the training split and select  $D = 10$  everywhere, which is the top of the grid  $\{1, 2, 3, 5, 7, 10\}$ , so the tuned and fixed variants coincide. The selected checkpoint for LaTa is epoch 7, the terminal epoch of the sweep under the 8-epoch budget, so this is a ceiling at this training budget rather than an asymptote. ABTT rows for Qwen3-0.6B sweep  $D$  per layer on the training split over the grid  $\{1, 2, 3, 5, 7, 10\}$ ; 12 of 28 layer rows select the top of the grid. The selected checkpoint for Qwen3-0.6B is epoch 3 of 6 run under the 8-epoch budget, so this is a ceiling at this training budget rather than an asymptote. ABTT rows for KaLM-mini sweep  $D$  per layer on the training split over the grid  $\{1, 2, 3, 5, 7, 10\}$ ; 19 of 24 layer rows select the top of the grid. The selected checkpoint for KaLM-mini is epoch 1 of 4 run under the 8-epoch budget, so this is a ceiling at this training budget rather than an asymptote. This is a reference ceiling, not a proposed method: it consumes supervision the zero-shot pipeline never sees.

| System          | Ranking |       | Routing |          |
|-----------------|---------|-------|---------|----------|
|                 | AUROC   | Gap   | Assign. | DirAcc@1 |
| BM25 (word)     | 0.974   | 0.243 | 62.5    | 56.3     |
| TF-IDF char 3–5 | 0.987   | 0.555 | 91.8    | 89.9     |
| Levenshtein     | 0.952   | 0.362 | 87.8    | 86.2     |

Table 19: Lexical reference systems on the same 50/50 split, the same held-out test set and the same evaluation code as the embedding models, so the numbers are directly comparable to the headline ranking and routing tables. Ranking is measured over the test  $n \times n$  score matrix: AUROC treats same-directory test pairs as positives, and Gap is the mean same-directory minus the mean different-directory score. Routing is reported in percent: Assign. is the existing-versus-new decision against a threshold  $\tau$  fitted on the training split, and DirAcc@1 additionally requires the correct directory. TF-IDF char 3–5 is the strongest of the three at 89.9 percent DirAcc@1, which puts plain surface-form overlap within reach of the best embedding configurations on this corpus of hand-copied witnesses. BM25 separates pairs well but routes poorly: a raw BM25 score grows with query length, so each file’s maximum score tracks document length as much as relatedness, and the  $\tau$  fitted on the training split transfers badly. BM25 uses  $k_1 = 1.5$ ,  $b = 0.75$  with document frequencies, vocabulary and average document length taken from training files only; the character TF-IDF vectorizer is likewise fitted on training files only, and normalized Levenshtein has nothing to fit. Every score matrix is min-max rescaled using the training block; the map is monotone, so AUROC is unchanged, while the threshold  $\tau$  is fitted on a fixed grid afterwards, which matters for BM25 (unrescaled,  $\tau$  pins at the grid ceiling) and is the identity for char TF-IDF, whose training scores already span  $[0, 1]$ . Under five reseeds of the query and reference files, in which only the query/reference partition of the test half is redrawn and the training/test split stays fixed, DirAcc@1 is BM25 (word)  $68.3 \pm 7.1$ ; TF-IDF char 3–5  $91.1 \pm 0.5$ ; Levenshtein  $87.2 \pm 0.3$ .

## I Per-Layer Results

This appendix gives the per-layer numbers behind Figure 1 and the routing results at every layer. Table 20 lists the train-selected layer behind every headline cell. Tables 21, 22 and 23 give ranking AUROC, routing assignment accuracy and routing DirAcc@1 at every layer of LaTa, PhilTa and mT5-base. Tables 24, 25 and 26 give the same for LaBSE, Qwen3-0.6B and KaLM-mini. Each compares baseline mean pooling with ABTT, its  $D$  tuned per layer on the training split. Fixing the number of removed components at ten gives the same range (AUROC 0.962 to 0.987 at all 100 panel layers), so the result does not rest on how that number is chosen.

| Model                   | Ranking layer |     |      |          | Routing layer |     |      |          |
|-------------------------|---------------|-----|------|----------|---------------|-----|------|----------|
|                         | Base          | SIF | ABTT | SIF+ABTT | Base          | SIF | ABTT | SIF+ABTT |
| LaTa                    | 12            | 1   | 12   | 7        | 1             | 1   | 8    | 12       |
| PhilTa                  | 1             | 1   | 9    | 8        | 1             | 1   | 1    | 1        |
| mT5-base                | 12            | 1   | 2    | 1        | 12            | 1   | 1    | 2        |
| LaBSE                   | 12            | 12  | 11   | 10       | 11            | 12  | 11   | 11       |
| Qwen3-0.6B              | 26            | 22  | 2    | 2        | 28            | 27  | 5    | 5        |
| KaLM-mini               | 23            | 23  | 1    | 2        | 23            | 22  | 3    | 1        |
| LaTa (fine-tuned)       | 12            | –   | 12   | –        | 12            | –   | 12   | –        |
| Qwen3-0.6B (fine-tuned) | 28            | –   | 27   | –        | 28            | –   | 27   | –        |
| KaLM-mini (fine-tuned)  | 24            | –   | 24   | –        | 24            | –   | 24   | –        |

Table 20: Train-selected layer behind every headline cell (Tables 3 and 15). For each model and post-processing setting, the ranking layer has the highest training-set AUROC and the routing layer the highest training-set DirAcc@1; the test scores in those tables are read at these layers, so no layer is chosen on test. Layer 1 is the first transformer block. The fine-tuned rows below the rule have no SIF setting. Table 18 reads its zero-shot rows at the Base and ABTT layers above the rule and its fine-tuned rows at the layers below it.

| Model    | Layer     | Ranking (T5 encoders) |                 |                      |                      |
|----------|-----------|-----------------------|-----------------|----------------------|----------------------|
|          |           | AUROC<br>(base)       | AUROC<br>(ABTT) | Cosine gap<br>(base) | Cosine gap<br>(ABTT) |
| LaTa     | 1         | 0.934                 | 0.970           | 0.286                | 0.566                |
|          | 2         | 0.563                 | 0.969           | 0.019                | 0.554                |
|          | 3         | 0.513                 | 0.973           | -0.048               | 0.555                |
|          | 4         | 0.500                 | 0.971           | -0.067               | 0.565                |
|          | 5         | 0.496                 | 0.972           | -0.069               | 0.572                |
|          | 6         | 0.496                 | 0.964           | -0.069               | 0.569                |
|          | 7         | 0.498                 | 0.962           | -0.072               | 0.564                |
|          | 8         | 0.502                 | 0.965           | -0.074               | 0.558                |
|          | 9         | 0.503                 | 0.965           | -0.073               | 0.557                |
|          | 10        | 0.505                 | 0.971           | -0.072               | 0.551                |
|          | 11        | 0.508                 | 0.973           | -0.067               | 0.550                |
|          | <b>12</b> | <b>0.938</b>          | <b>0.971</b>    | <b>0.237</b>         | <b>0.525</b>         |
| PhilTa   | 1         | 0.939                 | 0.977           | 0.231                | 0.574                |
|          | 2         | 0.911                 | 0.973           | 0.161                | 0.553                |
|          | 3         | 0.601                 | 0.973           | 0.085                | 0.549                |
|          | 4         | 0.551                 | 0.978           | 0.063                | 0.551                |
|          | 5         | 0.542                 | 0.979           | 0.053                | 0.551                |
|          | 6         | 0.541                 | 0.981           | 0.047                | 0.551                |
|          | 7         | 0.542                 | 0.979           | 0.046                | 0.554                |
|          | 8         | 0.542                 | 0.984           | 0.046                | 0.555                |
|          | <b>9</b>  | <b>0.539</b>          | <b>0.983</b>    | <b>0.041</b>         | <b>0.540</b>         |
|          | 10        | 0.538                 | 0.982           | 0.040                | 0.531                |
|          | 11        | 0.539                 | 0.982           | 0.041                | 0.511                |
|          | 12        | 0.911                 | 0.982           | 0.181                | 0.487                |
| mT5-base | 1         | 0.822                 | 0.979           | 0.027                | 0.550                |
|          | <b>2</b>  | <b>0.822</b>          | <b>0.975</b>    | <b>0.026</b>         | <b>0.536</b>         |
|          | 3         | 0.804                 | 0.968           | 0.025                | 0.517                |
|          | 4         | 0.799                 | 0.974           | 0.040                | 0.502                |
|          | 5         | 0.654                 | 0.978           | 0.219                | 0.482                |
|          | 6         | 0.657                 | 0.978           | 0.238                | 0.483                |
|          | 7         | 0.656                 | 0.980           | 0.244                | 0.482                |
|          | 8         | 0.656                 | 0.982           | 0.249                | 0.471                |
|          | 9         | 0.660                 | 0.982           | 0.235                | 0.475                |
|          | 10        | 0.659                 | 0.977           | 0.226                | 0.459                |
|          | 11        | 0.659                 | 0.974           | 0.213                | 0.449                |
|          | 12        | 0.838                 | 0.972           | 0.081                | 0.442                |

Table 21: Per-layer ranking metrics for the three T5 encoders (LaTa, PhilTa, mT5-base). baseline is mean pooling without correction; abtt\_optimal applies ABTT (no SIF weighting) with  $D$  tuned per layer on the training split. Both AUROC and cosine gap are computed over the test  $n \times n$  cosine matrix with no threshold and no directory routing: AUROC treats same-directory test pairs as positives, and cosine gap is the difference between mean same-directory and mean different-directory cosine. Rows in bold mark the layer chosen on the training split, the layer with the highest training-set AUROC under ABTT. It is the layer behind the ABTT cell of Table 15 (listed in Table 20), and not always the layer with the highest test score in this table.

| Routing (T5 encoders, file-level, $\tau$ -thresholded) |          |                    |                    |               |               |                   |                   |
|--------------------------------------------------------|----------|--------------------|--------------------|---------------|---------------|-------------------|-------------------|
| Model                                                  | Layer    | Existing<br>(base) | Existing<br>(ABTT) | New<br>(base) | New<br>(ABTT) | Overall<br>(base) | Overall<br>(ABTT) |
| LaTa                                                   | 1        | 0.630              | 0.910              | 0.916         | 0.907         | 0.738             | 0.909             |
|                                                        | 2        | 0.153              | 0.905              | 0.913         | 0.892         | 0.439             | 0.900             |
|                                                        | 3        | 0.256              | 0.882              | 0.842         | 0.950         | 0.477             | 0.908             |
|                                                        | 4        | 0.389              | 0.893              | 0.718         | 0.920         | 0.513             | 0.903             |
|                                                        | 5        | 0.374              | 0.892              | 0.715         | 0.913         | 0.502             | 0.900             |
|                                                        | 6        | 0.378              | 0.884              | 0.712         | 0.913         | 0.503             | 0.895             |
|                                                        | 7        | 0.372              | 0.875              | 0.715         | 0.916         | 0.501             | 0.890             |
|                                                        | <b>8</b> | <b>0.379</b>       | <b>0.884</b>       | <b>0.706</b>  | <b>0.885</b>  | <b>0.502</b>      | <b>0.885</b>      |
|                                                        | 9        | 0.366              | 0.886              | 0.724         | 0.873         | 0.501             | 0.881             |
|                                                        | 10       | 0.344              | 0.873              | 0.752         | 0.882         | 0.498             | 0.876             |
|                                                        | 11       | 0.310              | 0.867              | 0.786         | 0.901         | 0.490             | 0.880             |
|                                                        | 12       | 0.576              | 0.890              | 0.932         | 0.833         | 0.710             | 0.868             |
| PhilTa                                                 | <b>1</b> | <b>0.591</b>       | <b>0.918</b>       | <b>0.904</b>  | <b>0.901</b>  | <b>0.709</b>      | <b>0.911</b>      |
|                                                        | 2        | 0.479              | 0.895              | 0.867         | 0.885         | 0.625             | 0.892             |
|                                                        | 3        | 0.193              | 0.897              | 0.889         | 0.879         | 0.455             | 0.890             |
|                                                        | 4        | 0.148              | 0.886              | 0.923         | 0.873         | 0.439             | 0.881             |
|                                                        | 5        | 0.252              | 0.879              | 0.796         | 0.873         | 0.457             | 0.876             |
|                                                        | 6        | 0.338              | 0.905              | 0.700         | 0.817         | 0.474             | 0.872             |
|                                                        | 7        | 0.301              | 0.901              | 0.746         | 0.830         | 0.469             | 0.874             |
|                                                        | 8        | 0.262              | 0.875              | 0.808         | 0.885         | 0.467             | 0.879             |
|                                                        | 9        | 0.303              | 0.890              | 0.762         | 0.870         | 0.476             | 0.882             |
|                                                        | 10       | 0.299              | 0.908              | 0.780         | 0.793         | 0.480             | 0.865             |
|                                                        | 11       | 0.256              | 0.869              | 0.824         | 0.836         | 0.470             | 0.857             |
|                                                        | 12       | 0.338              | 0.880              | 0.898         | 0.876         | 0.549             | 0.879             |
| mT5-base                                               | <b>1</b> | <b>0.135</b>       | <b>0.877</b>       | <b>0.981</b>  | <b>0.947</b>  | <b>0.453</b>      | <b>0.903</b>      |
|                                                        | 2        | 0.110              | 0.890              | 0.975         | 0.910         | 0.436             | 0.897             |
|                                                        | 3        | 0.161              | 0.794              | 0.926         | 0.941         | 0.449             | 0.850             |
|                                                        | 4        | 0.269              | 0.787              | 0.858         | 0.882         | 0.491             | 0.823             |
|                                                        | 5        | 0.422              | 0.746              | 0.659         | 0.864         | 0.512             | 0.790             |
|                                                        | 6        | 0.636              | 0.766              | 0.489         | 0.876         | 0.580             | 0.808             |
|                                                        | 7        | 0.594              | 0.793              | 0.517         | 0.851         | 0.565             | 0.815             |
|                                                        | 8        | 0.557              | 0.809              | 0.560         | 0.842         | 0.558             | 0.822             |
|                                                        | 9        | 0.364              | 0.841              | 0.746         | 0.814         | 0.508             | 0.831             |
|                                                        | 10       | 0.314              | 0.794              | 0.783         | 0.876         | 0.491             | 0.825             |
|                                                        | 11       | 0.297              | 0.763              | 0.780         | 0.854         | 0.479             | 0.797             |
|                                                        | 12       | 0.198              | 0.834              | 0.926         | 0.799         | 0.472             | 0.821             |

Table 22: Per-layer routing accuracy for the three T5 encoders. baseline is mean pooling without correction; abtt\_optimal applies ABTT (no SIF weighting) with  $D$  tuned per layer on the training split. For every test file we compute  $s_i = \max_{j \neq i} \cos(e_i, e_j)$  against the rest of the test set and compare to the learned threshold  $\tau$  (fit on the training split via an  $F_1$ -optimal cut on same-directory vs. different-directory pairs). The decision is binary first (existing if  $s_i \geq \tau$ , else new); only files predicted existing are then routed to their top-1 neighbor’s directory. We report **existing accuracy** (restricted to files whose true partner is in the test set), **new accuracy** (files that should be flagged as novel), and **overall assignment accuracy**. Rows in bold mark the layer chosen on the training split, the layer with the highest training-set DirAcc@1 under ABTT. It is the layer behind the ABTT cell of Table 15 (listed in Table 20), and not always the layer with the highest test score in this table.

| Routing: DirAcc@1 (T5 encoders, single seed) |          |                    |                    |                    |                    |               |               |
|----------------------------------------------|----------|--------------------|--------------------|--------------------|--------------------|---------------|---------------|
| Model                                        | Layer    | DirAcc@1<br>(base) | DirAcc@1<br>(ABTT) | Existing<br>(base) | Existing<br>(ABTT) | New<br>(base) | New<br>(ABTT) |
| LaTa                                         | 1        | 0.721              | 0.880              | 0.630              | 0.910              | 0.916         | 0.907         |
|                                              | 2        | 0.383              | 0.869              | 0.153              | 0.905              | 0.913         | 0.892         |
|                                              | 3        | 0.350              | 0.882              | 0.256              | 0.882              | 0.842         | 0.950         |
|                                              | 4        | 0.303              | 0.881              | 0.389              | 0.893              | 0.718         | 0.920         |
|                                              | 5        | 0.297              | 0.875              | 0.374              | 0.892              | 0.715         | 0.913         |
|                                              | 6        | 0.297              | 0.874              | 0.378              | 0.884              | 0.712         | 0.913         |
|                                              | 7        | 0.302              | 0.868              | 0.372              | 0.875              | 0.715         | 0.916         |
|                                              | <b>8</b> | <b>0.302</b>       | <b>0.861</b>       | <b>0.379</b>       | <b>0.884</b>       | <b>0.706</b>  | <b>0.885</b>  |
|                                              | 9        | 0.309              | 0.858              | 0.366              | 0.886              | 0.724         | 0.873         |
|                                              | 10       | 0.317              | 0.852              | 0.344              | 0.873              | 0.752         | 0.882         |
|                                              | 11       | 0.331              | 0.857              | 0.310              | 0.867              | 0.786         | 0.901         |
|                                              | 12       | 0.689              | 0.841              | 0.576              | 0.890              | 0.932         | 0.833         |
| PhilTa                                       | <b>1</b> | <b>0.693</b>       | <b>0.883</b>       | <b>0.591</b>       | <b>0.918</b>       | <b>0.904</b>  | <b>0.901</b>  |
|                                              | 2        | 0.603              | 0.869              | 0.479              | 0.895              | 0.867         | 0.885         |
|                                              | 3        | 0.389              | 0.865              | 0.193              | 0.897              | 0.889         | 0.879         |
|                                              | 4        | 0.369              | 0.859              | 0.148              | 0.886              | 0.923         | 0.873         |
|                                              | 5        | 0.334              | 0.854              | 0.252              | 0.879              | 0.796         | 0.873         |
|                                              | 6        | 0.310              | 0.845              | 0.338              | 0.905              | 0.700         | 0.817         |
|                                              | 7        | 0.325              | 0.847              | 0.301              | 0.901              | 0.746         | 0.830         |
|                                              | 8        | 0.350              | 0.857              | 0.262              | 0.875              | 0.808         | 0.885         |
|                                              | 9        | 0.337              | 0.857              | 0.303              | 0.890              | 0.762         | 0.870         |
|                                              | 10       | 0.344              | 0.833              | 0.299              | 0.908              | 0.780         | 0.793         |
|                                              | 11       | 0.355              | 0.829              | 0.256              | 0.869              | 0.824         | 0.836         |
|                                              | 12       | 0.528              | 0.854              | 0.338              | 0.880              | 0.898         | 0.876         |
| mT5-base                                     | <b>1</b> | <b>0.451</b>       | <b>0.885</b>       | <b>0.135</b>       | <b>0.877</b>       | <b>0.981</b>  | <b>0.947</b>  |
|                                              | 2        | 0.435              | 0.874              | 0.110              | 0.890              | 0.975         | 0.910         |
|                                              | 3        | 0.445              | 0.830              | 0.161              | 0.794              | 0.926         | 0.941         |
|                                              | 4        | 0.467              | 0.798              | 0.269              | 0.787              | 0.858         | 0.882         |
|                                              | 5        | 0.352              | 0.756              | 0.422              | 0.746              | 0.659         | 0.864         |
|                                              | 6        | 0.346              | 0.775              | 0.636              | 0.766              | 0.489         | 0.876         |
|                                              | 7        | 0.352              | 0.775              | 0.594              | 0.793              | 0.517         | 0.851         |
|                                              | 8        | 0.366              | 0.784              | 0.557              | 0.809              | 0.560         | 0.842         |
|                                              | 9        | 0.394              | 0.787              | 0.364              | 0.841              | 0.746         | 0.814         |
|                                              | 10       | 0.387              | 0.788              | 0.314              | 0.794              | 0.783         | 0.876         |
|                                              | 11       | 0.378              | 0.765              | 0.297              | 0.763              | 0.780         | 0.854         |
|                                              | 12       | 0.466              | 0.779              | 0.198              | 0.834              | 0.926         | 0.799         |

Table 23: Per-layer routing DirAcc@1, overall and for existing and new witnesses, for the three T5 encoders. baseline is mean pooling without correction; abtt\_optimal applies ABTT (no SIF weighting) with  $D$  tuned per layer on the training split. This table is computed once, without the five reseeds of the query and reference files, which were run only for the SIF-conditioned variants (see Table 32 for the five-reseeding SIF+ABTT view). We report DirAcc@1 with an existing/new decomposition. Rows in bold mark the layer chosen on the training split, the layer with the highest training-set DirAcc@1 under ABTT. It is the layer behind the ABTT cell of Table 15 (listed in Table 20), and not always the layer with the highest test score in this table.

Table 24: Per-layer ranking metrics for the embedding-trained models (LaBSE, Qwen3-0.6B, KaLM-mini), under the same two-method comparison as Table 21. baseline is mean pooling without correction; abtt\_optimal applies ABTT (no SIF weighting) with  $D$  tuned per layer on the training split. Both AUROC and cosine gap are computed over the test  $n \times n$  cosine matrix with no threshold and no directory routing: AUROC treats same-directory test pairs as positives, and cosine gap is the difference between mean same-directory and mean different-directory cosine. Rows in bold mark the layer chosen on the training split, the layer with the highest training-set AUROC under ABTT. It is the layer behind the ABTT cell of Table 15 (listed in Table 20), and not always the layer with the highest test score in this table.

| Model      | Layer     | Ranking (embedding-trained models) |              |                   |                   |
|------------|-----------|------------------------------------|--------------|-------------------|-------------------|
|            |           | AUROC (base)                       | AUROC (ABTT) | Cosine gap (base) | Cosine gap (ABTT) |
| LaBSE      | 1         | 0.806                              | 0.977        | 0.060             | 0.520             |
|            | 2         | 0.833                              | 0.979        | 0.037             | 0.531             |
|            | 3         | 0.853                              | 0.978        | 0.062             | 0.543             |
|            | 4         | 0.862                              | 0.977        | 0.073             | 0.525             |
|            | 5         | 0.864                              | 0.968        | 0.066             | 0.501             |
|            | 6         | 0.894                              | 0.976        | 0.078             | 0.532             |
|            | 7         | 0.887                              | 0.975        | 0.088             | 0.524             |
|            | 8         | 0.879                              | 0.975        | 0.080             | 0.522             |
|            | 9         | 0.916                              | 0.979        | 0.090             | 0.565             |
|            | 10        | 0.934                              | 0.982        | 0.103             | 0.586             |
|            | <b>11</b> | <b>0.960</b>                       | <b>0.987</b> | <b>0.061</b>      | <b>0.611</b>      |
|            | 12        | 0.956                              | 0.981        | 0.183             | 0.592             |
| Qwen3-0.6B | 1         | 0.858                              | 0.978        | 0.033             | 0.564             |
|            | <b>2</b>  | <b>0.863</b>                       | <b>0.973</b> | <b>0.024</b>      | <b>0.553</b>      |
|            | 3         | 0.862                              | 0.980        | 0.037             | 0.556             |
|            | 4         | 0.860                              | 0.986        | 0.036             | 0.547             |
|            | 5         | 0.861                              | 0.981        | 0.030             | 0.545             |
|            | 6         | 0.867                              | 0.983        | 0.035             | 0.545             |
|            | 7         | 0.876                              | 0.980        | 0.033             | 0.544             |
|            | 8         | 0.882                              | 0.972        | 0.037             | 0.532             |
|            | 9         | 0.881                              | 0.977        | 0.034             | 0.532             |
|            | 10        | 0.887                              | 0.976        | 0.031             | 0.529             |
|            | 11        | 0.895                              | 0.977        | 0.030             | 0.530             |
|            | 12        | 0.901                              | 0.973        | 0.027             | 0.530             |
|            | 13        | 0.907                              | 0.974        | 0.026             | 0.523             |
|            | 14        | 0.905                              | 0.974        | 0.024             | 0.519             |
|            | 15        | 0.901                              | 0.971        | 0.022             | 0.514             |
|            | 16        | 0.910                              | 0.975        | 0.026             | 0.509             |
|            | 17        | 0.916                              | 0.979        | 0.032             | 0.521             |
|            | 18        | 0.933                              | 0.983        | 0.032             | 0.537             |
|            | 19        | 0.936                              | 0.980        | 0.035             | 0.541             |
|            | 20        | 0.943                              | 0.982        | 0.041             | 0.559             |
|            | 21        | 0.953                              | 0.982        | 0.047             | 0.562             |
|            | 22        | 0.963                              | 0.983        | 0.044             | 0.583             |
|            | 23        | 0.961                              | 0.984        | 0.038             | 0.578             |
|            | 24        | 0.963                              | 0.984        | 0.032             | 0.576             |
|            | 25        | 0.961                              | 0.983        | 0.027             | 0.578             |
|            | 26        | 0.966                              | 0.985        | 0.024             | 0.568             |
|            | 27        | 0.967                              | 0.982        | 0.017             | 0.559             |
|            | 28        | 0.958                              | 0.975        | 0.098             | 0.556             |
| KaLM-mini  | <b>1</b>  | <b>0.887</b>                       | <b>0.981</b> | <b>0.063</b>      | <b>0.568</b>      |
|            | 2         | 0.906                              | 0.983        | 0.044             | 0.566             |
|            | 3         | 0.881                              | 0.981        | 0.062             | 0.551             |
|            | 4         | 0.871                              | 0.983        | 0.060             | 0.553             |
|            | 5         | 0.861                              | 0.983        | 0.062             | 0.541             |
|            | 6         | 0.872                              | 0.983        | 0.063             | 0.529             |
|            | 7         | 0.888                              | 0.979        | 0.072             | 0.524             |
|            | 8         | 0.890                              | 0.976        | 0.072             | 0.524             |
|            | 9         | 0.892                              | 0.974        | 0.066             | 0.507             |
|            | 10        | 0.893                              | 0.971        | 0.067             | 0.494             |
|            | 11        | 0.894                              | 0.970        | 0.061             | 0.475             |
|            | 12        | 0.905                              | 0.970        | 0.071             | 0.475             |
|            | 13        | 0.900                              | 0.968        | 0.066             | 0.467             |

(continued on next page)

| Model | Layer | Ranking (embedding-trained models) |                 |                      |                      |
|-------|-------|------------------------------------|-----------------|----------------------|----------------------|
|       |       | AUROC<br>(base)                    | AUROC<br>(ABTT) | Cosine gap<br>(base) | Cosine gap<br>(ABTT) |
|       | 14    | 0.904                              | 0.971           | 0.063                | 0.471                |
|       | 15    | 0.907                              | 0.973           | 0.067                | 0.468                |
|       | 16    | 0.917                              | 0.972           | 0.071                | 0.489                |
|       | 17    | 0.926                              | 0.975           | 0.075                | 0.501                |
|       | 18    | 0.945                              | 0.977           | 0.060                | 0.519                |
|       | 19    | 0.956                              | 0.979           | 0.061                | 0.531                |
|       | 20    | 0.960                              | 0.980           | 0.054                | 0.555                |
|       | 21    | 0.962                              | 0.981           | 0.043                | 0.555                |
|       | 22    | 0.971                              | 0.985           | 0.050                | 0.555                |
|       | 23    | 0.972                              | 0.986           | 0.056                | 0.552                |
|       | 24    | 0.956                              | 0.981           | 0.243                | 0.520                |

Table 25: Per-layer routing accuracy for the embedding-trained models. `baseline` is mean pooling without correction; `abtt_optimal` applies ABTT (no SIF weighting) with  $D$  tuned per layer on the training split. For every test file we compute  $s_i = \max_{j \neq i} \cos(e_i, e_j)$  against the rest of the test set and compare to the learned threshold  $\tau$  (fit on the training split via an  $F_1$ -optimal cut on same-directory vs. different-directory pairs). The decision is binary first (existing if  $s_i \geq \tau$ , else new); only files predicted existing are then routed to their top-1 neighbor’s directory. We report **existing accuracy** (restricted to files whose true partner is in the test set), **new accuracy** (files that should be flagged as novel), and **overall assignment accuracy**. Rows in bold mark the layer chosen on the training split, the layer with the highest training-set DirAcc@1 under ABTT. It is the layer behind the ABTT cell of Table 15 (listed in Table 20), and not always the layer with the highest test score in this table.

| Model      | Layer     | Routing (embedding-trained models) |                    |               |               |                   |                   |
|------------|-----------|------------------------------------|--------------------|---------------|---------------|-------------------|-------------------|
|            |           | Existing<br>(base)                 | Existing<br>(ABTT) | New<br>(base) | New<br>(ABTT) | Overall<br>(base) | Overall<br>(ABTT) |
| LaBSE      | 1         | 0.222                              | 0.879              | 0.898         | 0.873         | 0.477             | 0.876             |
|            | 2         | 0.381                              | 0.875              | 0.811         | 0.827         | 0.543             | 0.857             |
|            | 3         | 0.277                              | 0.886              | 0.907         | 0.867         | 0.514             | 0.879             |
|            | 4         | 0.372                              | 0.897              | 0.885         | 0.836         | 0.565             | 0.874             |
|            | 5         | 0.327                              | 0.845              | 0.901         | 0.851         | 0.543             | 0.847             |
|            | 6         | 0.548                              | 0.860              | 0.882         | 0.851         | 0.674             | 0.857             |
|            | 7         | 0.398                              | 0.849              | 0.892         | 0.830         | 0.584             | 0.841             |
|            | 8         | 0.385                              | 0.869              | 0.870         | 0.743         | 0.568             | 0.822             |
|            | 9         | 0.421                              | 0.871              | 0.929         | 0.817         | 0.612             | 0.851             |
|            | 10        | 0.606                              | 0.892              | 0.892         | 0.805         | 0.713             | 0.859             |
|            | <b>11</b> | <b>0.785</b>                       | <b>0.923</b>       | <b>0.913</b>  | <b>0.882</b>  | <b>0.833</b>      | <b>0.908</b>      |
|            | 12        | 0.729                              | 0.914              | 0.938         | 0.864         | 0.808             | 0.895             |
| Qwen3-0.6B | 1         | 0.264                              | 0.921              | 0.960         | 0.885         | 0.526             | 0.908             |
|            | 2         | 0.437                              | 0.869              | 0.814         | 0.916         | 0.579             | 0.887             |
|            | 3         | 0.250                              | 0.914              | 0.947         | 0.907         | 0.513             | 0.911             |
|            | 4         | 0.411                              | 0.936              | 0.848         | 0.842         | 0.576             | 0.901             |
|            | <b>5</b>  | <b>0.181</b>                       | <b>0.910</b>       | <b>0.932</b>  | <b>0.923</b>  | <b>0.464</b>      | <b>0.915</b>      |
|            | 6         | 0.422                              | 0.935              | 0.830         | 0.796         | 0.576             | 0.882             |
|            | 7         | 0.250                              | 0.890              | 0.947         | 0.876         | 0.513             | 0.885             |
|            | 8         | 0.409                              | 0.931              | 0.895         | 0.799         | 0.592             | 0.881             |
|            | 9         | 0.460                              | 0.901              | 0.851         | 0.851         | 0.607             | 0.882             |
|            | 10        | 0.533                              | 0.892              | 0.836         | 0.864         | 0.647             | 0.881             |
|            | 11        | 0.293                              | 0.903              | 0.978         | 0.808         | 0.551             | 0.867             |
|            | 12        | 0.407                              | 0.912              | 0.938         | 0.817         | 0.607             | 0.876             |
|            | 13        | 0.432                              | 0.908              | 0.944         | 0.836         | 0.625             | 0.881             |
|            | 14        | 0.456                              | 0.907              | 0.926         | 0.811         | 0.633             | 0.871             |
|            | 15        | 0.469                              | 0.901              | 0.920         | 0.845         | 0.639             | 0.880             |
|            | 16        | 0.639                              | 0.897              | 0.830         | 0.808         | 0.711             | 0.864             |
|            | 17        | 0.480                              | 0.886              | 0.935         | 0.851         | 0.652             | 0.873             |
|            | 18        | 0.553                              | 0.882              | 0.944         | 0.839         | 0.700             | 0.866             |
|            | 19        | 0.514                              | 0.871              | 0.966         | 0.858         | 0.684             | 0.866             |
|            | 20        | 0.654                              | 0.865              | 0.898         | 0.892         | 0.746             | 0.875             |
|            | 21        | 0.647                              | 0.912              | 0.950         | 0.898         | 0.761             | 0.907             |
|            | 22        | 0.731                              | 0.888              | 0.938         | 0.916         | 0.809             | 0.899             |
|            | 23        | 0.622                              | 0.914              | 0.966         | 0.895         | 0.752             | 0.907             |

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| Model     | Layer    | Routing (embedding-trained models) |                 |              |              |                |                |
|-----------|----------|------------------------------------|-----------------|--------------|--------------|----------------|----------------|
|           |          | Existing (base)                    | Existing (ABTT) | New (base)   | New (ABTT)   | Overall (base) | Overall (ABTT) |
|           | 24       | 0.748                              | 0.901           | 0.935        | 0.935        | 0.818          | 0.914          |
|           | 25       | 0.574                              | 0.893           | 0.975        | 0.923        | 0.725          | 0.904          |
|           | 26       | 0.613                              | 0.899           | 0.975        | 0.895        | 0.749          | 0.897          |
|           | 27       | 0.793                              | 0.877           | 0.833        | 0.923        | 0.808          | 0.894          |
|           | 28       | 0.796                              | 0.877           | 0.867        | 0.913        | 0.823          | 0.890          |
| KaLM-mini | 1        | 0.391                              | 0.921           | 0.907        | 0.920        | 0.585          | 0.921          |
|           | 2        | 0.493                              | 0.916           | 0.848        | 0.879        | 0.627          | 0.902          |
|           | <b>3</b> | <b>0.443</b>                       | <b>0.923</b>    | <b>0.870</b> | <b>0.907</b> | <b>0.604</b>   | <b>0.917</b>   |
|           | 4        | 0.308                              | 0.931           | 0.935        | 0.895        | 0.544          | 0.917          |
|           | 5        | 0.338                              | 0.905           | 0.882        | 0.923        | 0.543          | 0.911          |
|           | 6        | 0.256                              | 0.895           | 0.910        | 0.920        | 0.502          | 0.904          |
|           | 7        | 0.475                              | 0.864           | 0.854        | 0.944        | 0.618          | 0.894          |
|           | 8        | 0.467                              | 0.862           | 0.870        | 0.913        | 0.619          | 0.881          |
|           | 9        | 0.469                              | 0.873           | 0.885        | 0.870        | 0.626          | 0.872          |
|           | 10       | 0.449                              | 0.875           | 0.885        | 0.870        | 0.613          | 0.873          |
|           | 11       | 0.458                              | 0.862           | 0.885        | 0.864        | 0.619          | 0.862          |
|           | 12       | 0.507                              | 0.862           | 0.895        | 0.904        | 0.653          | 0.878          |
|           | 13       | 0.540                              | 0.860           | 0.895        | 0.870        | 0.674          | 0.864          |
|           | 14       | 0.593                              | 0.836           | 0.845        | 0.885        | 0.688          | 0.854          |
|           | 15       | 0.566                              | 0.864           | 0.898        | 0.827        | 0.691          | 0.850          |
|           | 16       | 0.550                              | 0.849           | 0.935        | 0.830        | 0.695          | 0.841          |
|           | 17       | 0.675                              | 0.864           | 0.920        | 0.889        | 0.767          | 0.873          |
|           | 18       | 0.697                              | 0.875           | 0.960        | 0.898        | 0.796          | 0.883          |
|           | 19       | 0.721                              | 0.873           | 0.966        | 0.892        | 0.814          | 0.880          |
|           | 20       | 0.705                              | 0.791           | 0.966        | 0.913        | 0.803          | 0.837          |
|           | 21       | 0.727                              | 0.811           | 0.966        | 0.926        | 0.817          | 0.854          |
|           | 22       | 0.850                              | 0.879           | 0.916        | 0.923        | 0.875          | 0.895          |
|           | 23       | 0.849                              | 0.884           | 0.920        | 0.898        | 0.875          | 0.889          |
|           | 24       | 0.787                              | 0.892           | 0.944        | 0.867        | 0.846          | 0.882          |

Table 26: Per-layer routing DirAcc@1, overall and for existing and new witnesses, for the embedding-trained models. baseline is mean pooling without correction; abtt\_optimal applies ABTT (no SIF weighting) with  $D$  tuned per layer on the training split. This table is computed once, without the five reseeds of the query and reference files, which were run only for the SIF-conditioned variants (see Table 32 for the five-reseeding SIF+ABTT view). We report DirAcc@1 with an existing/new decomposition. Rows in bold mark the layer chosen on the training split, the layer with the highest training-set DirAcc@1 under ABTT. It is the layer behind the ABTT cell of Table 15 (listed in Table 20), and not always the layer with the highest test score in this table.

| Model      | Layer     | Routing: DirAcc@1 (embedding-trained models, single seed) |                 |                 |                 |              |              |
|------------|-----------|-----------------------------------------------------------|-----------------|-----------------|-----------------|--------------|--------------|
|            |           | DirAcc@1 (base)                                           | DirAcc@1 (ABTT) | Existing (base) | Existing (ABTT) | New (base)   | New (ABTT)   |
| LaBSE      | 1         | 0.465                                                     | 0.851           | 0.222           | 0.879           | 0.898        | 0.873        |
|            | 2         | 0.494                                                     | 0.830           | 0.381           | 0.875           | 0.811        | 0.827        |
|            | 3         | 0.500                                                     | 0.848           | 0.277           | 0.886           | 0.907        | 0.867        |
|            | 4         | 0.542                                                     | 0.841           | 0.372           | 0.897           | 0.885        | 0.836        |
|            | 5         | 0.521                                                     | 0.811           | 0.327           | 0.845           | 0.901        | 0.851        |
|            | 6         | 0.640                                                     | 0.831           | 0.548           | 0.860           | 0.882        | 0.851        |
|            | 7         | 0.561                                                     | 0.810           | 0.398           | 0.849           | 0.892        | 0.830        |
|            | 8         | 0.544                                                     | 0.779           | 0.385           | 0.869           | 0.870        | 0.743        |
|            | 9         | 0.608                                                     | 0.823           | 0.421           | 0.871           | 0.929        | 0.817        |
|            | 10        | 0.693                                                     | 0.832           | 0.606           | 0.892           | 0.892        | 0.805        |
|            | <b>11</b> | <b>0.815</b>                                              | <b>0.885</b>    | <b>0.785</b>    | <b>0.923</b>    | <b>0.913</b> | <b>0.882</b> |
|            | 12        | 0.791                                                     | 0.868           | 0.729           | 0.914           | 0.938        | 0.864        |
| Qwen3-0.6B | 1         | 0.521                                                     | 0.875           | 0.264           | 0.921           | 0.960        | 0.885        |
|            | 2         | 0.558                                                     | 0.871           | 0.437           | 0.869           | 0.814        | 0.916        |
|            | 3         | 0.508                                                     | 0.890           | 0.250           | 0.914           | 0.947        | 0.907        |
|            | 4         | 0.559                                                     | 0.874           | 0.411           | 0.936           | 0.848        | 0.842        |
|            | <b>5</b>  | <b>0.450</b>                                              | <b>0.894</b>    | <b>0.181</b>    | <b>0.910</b>    | <b>0.932</b> | <b>0.923</b> |
|            | 6         | 0.551                                                     | 0.858           | 0.422           | 0.935           | 0.830        | 0.796        |
|            | 7         | 0.507                                                     | 0.859           | 0.250           | 0.890           | 0.947        | 0.876        |

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| Routing: DirAcc@1 (embedding-trained models, single seed) |       |                    |                    |                    |                    |               |               |
|-----------------------------------------------------------|-------|--------------------|--------------------|--------------------|--------------------|---------------|---------------|
| Model                                                     | Layer | DirAcc@1<br>(base) | DirAcc@1<br>(ABTT) | Existing<br>(base) | Existing<br>(ABTT) | New<br>(base) | New<br>(ABTT) |
|                                                           | 8     | 0.580              | 0.852              | 0.409              | 0.931              | 0.895         | 0.799         |
|                                                           | 9     | 0.587              | 0.852              | 0.460              | 0.901              | 0.851         | 0.851         |
|                                                           | 10    | 0.622              | 0.848              | 0.533              | 0.892              | 0.836         | 0.864         |
|                                                           | 11    | 0.548              | 0.830              | 0.293              | 0.903              | 0.978         | 0.808         |
|                                                           | 12    | 0.603              | 0.848              | 0.407              | 0.912              | 0.938         | 0.817         |
|                                                           | 13    | 0.618              | 0.851              | 0.432              | 0.908              | 0.944         | 0.836         |
|                                                           | 14    | 0.621              | 0.840              | 0.456              | 0.907              | 0.926         | 0.811         |
|                                                           | 15    | 0.627              | 0.848              | 0.469              | 0.901              | 0.920         | 0.845         |
|                                                           | 16    | 0.682              | 0.830              | 0.639              | 0.897              | 0.830         | 0.808         |
|                                                           | 17    | 0.641              | 0.840              | 0.480              | 0.886              | 0.935         | 0.851         |
|                                                           | 18    | 0.691              | 0.837              | 0.553              | 0.882              | 0.944         | 0.839         |
|                                                           | 19    | 0.676              | 0.837              | 0.514              | 0.871              | 0.966         | 0.858         |
|                                                           | 20    | 0.733              | 0.852              | 0.654              | 0.865              | 0.898         | 0.892         |
|                                                           | 21    | 0.752              | 0.881              | 0.647              | 0.912              | 0.950         | 0.898         |
|                                                           | 22    | 0.795              | 0.876              | 0.731              | 0.888              | 0.938         | 0.916         |
|                                                           | 23    | 0.742              | 0.886              | 0.622              | 0.914              | 0.966         | 0.895         |
|                                                           | 24    | 0.808              | 0.896              | 0.748              | 0.901              | 0.935         | 0.935         |
|                                                           | 25    | 0.718              | 0.886              | 0.574              | 0.893              | 0.975         | 0.923         |
|                                                           | 26    | 0.740              | 0.881              | 0.613              | 0.899              | 0.975         | 0.895         |
|                                                           | 27    | 0.787              | 0.876              | 0.793              | 0.877              | 0.833         | 0.923         |
|                                                           | 28    | 0.803              | 0.872              | 0.796              | 0.877              | 0.867         | 0.913         |
| KaLM-mini                                                 | 1     | 0.577              | 0.902              | 0.391              | 0.921              | 0.907         | 0.920         |
|                                                           | 2     | 0.607              | 0.879              | 0.493              | 0.916              | 0.848         | 0.879         |
|                                                           | 3     | <b>0.587</b>       | <b>0.894</b>       | <b>0.443</b>       | <b>0.923</b>       | <b>0.870</b>  | <b>0.907</b>  |
|                                                           | 4     | 0.538              | 0.887              | 0.308              | 0.931              | 0.935         | 0.895         |
|                                                           | 5     | 0.524              | 0.882              | 0.338              | 0.905              | 0.882         | 0.923         |
|                                                           | 6     | 0.490              | 0.876              | 0.256              | 0.895              | 0.910         | 0.920         |
|                                                           | 7     | 0.592              | 0.873              | 0.475              | 0.864              | 0.854         | 0.944         |
|                                                           | 8     | 0.597              | 0.860              | 0.467              | 0.862              | 0.870         | 0.913         |
|                                                           | 9     | 0.601              | 0.843              | 0.469              | 0.873              | 0.885         | 0.870         |
|                                                           | 10    | 0.584              | 0.844              | 0.449              | 0.875              | 0.885         | 0.870         |
|                                                           | 11    | 0.586              | 0.836              | 0.458              | 0.862              | 0.885         | 0.864         |
|                                                           | 12    | 0.622              | 0.850              | 0.507              | 0.862              | 0.895         | 0.904         |
|                                                           | 13    | 0.633              | 0.840              | 0.540              | 0.860              | 0.895         | 0.870         |
|                                                           | 14    | 0.636              | 0.829              | 0.593              | 0.836              | 0.845         | 0.885         |
|                                                           | 15    | 0.650              | 0.817              | 0.566              | 0.864              | 0.898         | 0.827         |
|                                                           | 16    | 0.676              | 0.819              | 0.550              | 0.849              | 0.935         | 0.830         |
|                                                           | 17    | 0.747              | 0.852              | 0.675              | 0.864              | 0.920         | 0.889         |
|                                                           | 18    | 0.777              | 0.865              | 0.697              | 0.875              | 0.960         | 0.898         |
|                                                           | 19    | 0.801              | 0.861              | 0.721              | 0.873              | 0.966         | 0.892         |
|                                                           | 20    | 0.789              | 0.817              | 0.705              | 0.791              | 0.966         | 0.913         |
|                                                           | 21    | 0.802              | 0.831              | 0.727              | 0.811              | 0.966         | 0.926         |
|                                                           | 22    | 0.857              | 0.875              | 0.850              | 0.879              | 0.916         | 0.923         |
|                                                           | 23    | 0.859              | 0.872              | 0.849              | 0.884              | 0.920         | 0.898         |
|                                                           | 24    | 0.833              | 0.858              | 0.787              | 0.892              | 0.944         | 0.867         |

## J Last-Token versus Mean Pooling

Decoder-only embedding models often use last-token pooling, so this appendix checks that mean pooling, which every main-text result uses, does not handicap them. We compare the two poolings under the same ABTT pipeline, training and test split, and ABTT choices. Table 27 reports, for each model and pooling, the layer chosen on the training split by DirAcc@1, the rule behind the routing cells of Table 3, with test assignment accuracy, DirAcc@1 and cosine gap at that layer. Mean pooling routes better for each of the five models in this run, the two decoders included; mT5-base was not part of it.

| Model      | Pooling    | Layer | $D$ | Assignment acc. | DirAcc@1 | Cosine gap |
|------------|------------|-------|-----|-----------------|----------|------------|
| LaTa       | Mean       | 8     | 10  | 0.885           | 0.861    | 0.558      |
|            | Last token | 7     | 10  | 0.664           | 0.466    | 0.308      |
| PhilTa     | Mean       | 1     | 10  | 0.911           | 0.883    | 0.573      |
|            | Last token | 11    | 10  | 0.589           | 0.406    | 0.205      |
| LaBSE      | Mean       | 11    | 10  | 0.908           | 0.885    | 0.610      |
|            | Last token | 12    | 10  | 0.895           | 0.871    | 0.586      |
| Qwen3-0.6B | Mean       | 5     | 10  | 0.915           | 0.894    | 0.545      |
|            | Last token | 26    | 5   | 0.881           | 0.864    | 0.580      |
| KaLM-mini  | Mean       | 3     | 10  | 0.917           | 0.894    | 0.552      |
|            | Last token | 24    | 10  | 0.875           | 0.847    | 0.483      |

Table 27: Mean versus last-token pooling under ABTT with  $D$  tuned per layer on the training split, for the five models whose last-token embeddings were extracted (mT5-base was not part of this run). For each model and pooling the layer is the one with the highest training-set DirAcc@1, the rule behind the routing cells of Table 3 (their layers are listed in Table 20);  $D$  is the number of principal components removed at that layer. The three score columns are test-set routing assignment accuracy, routing DirAcc@1, and ranking cosine gap at that layer.

## K SIF and Whitening Baselines

Section 3 names SIF frequency weighting and reduced-rank whitening as baselines to ABTT, and this appendix gives both. Tables 30, 31 and 32 report SIF alone, SIF followed by ABTT with  $D = 10$ , and SIF followed by ABTT with  $D$  tuned per layer, at every layer of the six panel models, with one primary metric per task.

**SIF.** SIF acts before pooling. It can down-weight the frequent tokens that feed the dominant direction (Appendix C), but it removes no direction from the pooled vector. At every collapsed layer it stays below ABTT.

**Shortlist accuracy.** In the shortlist form of routing, each directory receives its best witness match score, a pseudo-option NEW receives score  $\tau$ , and the simulated expert succeeds when the labeled directory appears in the shortlist. Shortlist accuracy,  $\text{DirAcc}@K$ , is the fraction of test queries whose labeled directory is among the top  $K$  options. Table 28 gives this shortlist view under SIF+ABTT, over five reseeds (Appendix H).  $\text{DirAcc}@2$  ranges from 95.1 to 96.1 percent across the six models, and little is gained after  $\text{DirAcc}@3$ .

**Whitening.** Full-rank PCA whitening (Su et al., 2021) is ill-conditioned here, since it estimates 768 to 1,024 dimensions from 847 training vectors (condition numbers  $3 \times 10^5$  to  $10^{16}$ ). It ranks below ABTT at every layer: its test AUROC is 0.820–0.876 for the T5 encoders and LaBSE, and 0.552–0.643 for the two decoders. It also degenerates the routing threshold. Full-rank whitening leaves the different-directory cosine mean at or below 0.015, so across all 100 whitening cells every file is routed to an existing directory, existing accuracy is 1.000, and new accuracy never exceeds one file in 323. Assignment accuracy then sits at the 62.4 percent class prior. That prior beats the baseline in 60 of 100 cells while  $\text{DirAcc}@1$  never exceeds 41.5 percent, so the apparent gain is a thresholding artifact. Whitening to a reduced dimension, as Su et al. (2021) also do, drops the near-null directions and is the fairer comparison (Table 29). Fit on training embeddings with  $r = 128$  or 256 components, it ranks on par with ABTT. At each model’s train-selected layer,  $r = 128$  reaches test AUROC 0.973–0.984, against 0.971–0.987 for ABTT. Over all 100 panel layers, collapsed layers included, both  $r = 128$  and  $r = 256$  stay within 0.021 of ABTT;  $r = 64$  falls slightly below. On ranking, the repair

therefore does not depend on ABTT in particular. On routing it does. At each model’s train-selected routing layer,  $r = 128$  reaches  $\text{DirAcc}@1$  1.9 to 4.5 points below ABTT, and  $r = 256$  up to 15 points below. We therefore keep ABTT as the reference correction.

| Model      | DirAcc@1       | DirAcc@2       | DirAcc@3       | DirAcc@4       | DirAcc@5       |
|------------|----------------|----------------|----------------|----------------|----------------|
| LaTa       | 88.9 $\pm$ 0.5 | 96.1 $\pm$ 0.5 | 97.4 $\pm$ 0.6 | 97.9 $\pm$ 0.3 | 98.2 $\pm$ 0.2 |
| PhilTa     | 89.8 $\pm$ 0.8 | 95.8 $\pm$ 0.6 | 97.4 $\pm$ 0.3 | 97.7 $\pm$ 0.3 | 97.8 $\pm$ 0.4 |
| mT5-base   | 89.3 $\pm$ 0.5 | 95.1 $\pm$ 0.2 | 96.7 $\pm$ 0.7 | 97.1 $\pm$ 0.4 | 97.2 $\pm$ 0.4 |
| LaBSE      | 90.6 $\pm$ 1.0 | 95.5 $\pm$ 0.5 | 96.4 $\pm$ 0.5 | 97.2 $\pm$ 0.6 | 97.8 $\pm$ 0.5 |
| Qwen3-0.6B | 89.3 $\pm$ 0.7 | 95.7 $\pm$ 0.4 | 97.1 $\pm$ 0.3 | 97.5 $\pm$ 0.4 | 97.6 $\pm$ 0.4 |
| KaLM-mini  | 89.3 $\pm$ 0.5 | 95.1 $\pm$ 0.4 | 96.6 $\pm$ 0.7 | 97.2 $\pm$ 0.7 | 97.6 $\pm$ 0.5 |

Table 28: Routing shortlist accuracy under SIF+ABTT: DirAcc@ $K$ , as mean  $\pm$  standard deviation over five reseeds of the query and reference files, with the training/test split fixed. For each model the reported layer is chosen on the training split: the layer with the highest training-set DirAcc@1 for SIF+ABTT in the single-seed run, the layer behind the SIF+ABTT cell of Table 15, listed in Table 20.  $D$  is tuned per layer on the training split. This is not always the layer with the highest mean over the five reseeds. Each row is the bold row of Table 32. Each cell is the percentage of test queries whose labeled directory appears within the top  $K$  options. Per-layer shortlist accuracy: Tables 23, 26, and 32.

| Model      | Ranking AUROC |       |                |       |       |       | Routing DirAcc@1 |                |      |      |      | Routing assignment |                |      |      |      |
|------------|---------------|-------|----------------|-------|-------|-------|------------------|----------------|------|------|------|--------------------|----------------|------|------|------|
|            | Base          | ABTT  | whitening, $r$ |       |       |       | ABTT             | whitening, $r$ |      |      |      | ABTT               | whitening, $r$ |      |      |      |
|            |               |       | 64             | 128   | 256   | full  |                  | 64             | 128  | 256  | full |                    | 64             | 128  | 256  | full |
| LaTa       | 0.938         | 0.971 | 0.969          | 0.973 | 0.983 | 0.872 | 86.1             | 81.2           | 83.1 | 77.0 | 41.1 | 88.5               | 84.7           | 86.2 | 80.7 | 62.5 |
| PhilTa     | 0.939         | 0.983 | 0.974          | 0.977 | 0.979 | 0.854 | 88.3             | 83.0           | 86.4 | 83.7 | 41.0 | 91.1               | 86.5           | 88.8 | 86.8 | 62.5 |
| mT5-base   | 0.838         | 0.975 | 0.971          | 0.979 | 0.974 | 0.845 | 88.5             | 78.9           | 85.8 | 83.8 | 41.0 | 90.3               | 82.6           | 88.6 | 87.2 | 62.5 |
| LaBSE      | 0.956         | 0.987 | 0.979          | 0.984 | 0.976 | 0.866 | 88.5             | 86.1           | 84.0 | 73.5 | 40.3 | 90.8               | 88.8           | 86.7 | 78.0 | 62.5 |
| Qwen3-0.6B | 0.966         | 0.973 | 0.973          | 0.978 | 0.983 | 0.594 | 89.4             | 84.8           | 85.0 | 75.9 | 20.5 | 91.5               | 87.2           | 88.0 | 79.3 | 62.4 |
| KaLM-mini  | 0.972         | 0.981 | 0.976          | 0.981 | 0.981 | 0.562 | 89.4             | 82.6           | 86.7 | 84.5 | 16.8 | 91.7               | 85.4           | 89.2 | 87.8 | 62.4 |

Table 29: PCA whitening to  $r$  components against the baseline and ABTT. Whitening keeps the top  $r$  principal components of the centered training embeddings and rescales each to unit variance. Full keeps every component: 768 for the T5 encoders and LaBSE, and 847, the number of training passages, for Qwen3-0.6B and KaLM-mini, whose last component then has numerically zero variance. Ranking cells are read at each setting’s train-selected layer (highest training AUROC), routing cells at the layer with the highest training DirAcc@1. Routing values are single-split test DirAcc@1 and assignment accuracy in percent, with the threshold learned on training pairs. Every transform is fit on training embeddings only. Whitening ranks on par with ABTT, but routes below it.

Table 30: Per-layer ranking AUROC across the SIF-conditioned post-processing suite for all six models. `sif_only` replaces mean pooling with SIF-weighted pooling; `sif_abtt_fixed` adds ABTT with fixed  $D=10$ ; `sif_abtt_optimal` tunes  $D$  per layer on the training split. Gap is omitted to keep the table narrow; the pure-ABTT comparison (not SIF-conditioned) is in Tables 21 and 24. Rows in bold mark the layer chosen on the training split, the layer with the highest training-set AUROC under SIF+ABTT. It is the layer behind the SIF+ABTT cell of Table 15 (listed in Table 20), and not always the layer with the highest test score in this table.

| Model  | Layer    | Ranking: SIF-suite AUROC (six models) |                |                                    |                     |
|--------|----------|---------------------------------------|----------------|------------------------------------|---------------------|
|        |          | AUROC<br>(base)                       | AUROC<br>(SIF) | AUROC<br>(SIF+ABTT <sub>10</sub> ) | AUROC<br>(SIF+ABTT) |
| LaTa   | 1        | 0.934                                 | 0.948          | 0.972                              | 0.976               |
|        | 2        | 0.563                                 | 0.938          | 0.964                              | 0.971               |
|        | 3        | 0.513                                 | 0.921          | 0.967                              | 0.966               |
|        | 4        | 0.500                                 | 0.890          | 0.970                              | 0.968               |
|        | 5        | 0.496                                 | 0.878          | 0.970                              | 0.970               |
|        | 6        | 0.496                                 | 0.865          | 0.965                              | 0.965               |
|        | <b>7</b> | <b>0.498</b>                          | <b>0.858</b>   | <b>0.966</b>                       | <b>0.966</b>        |
|        | 8        | 0.502                                 | 0.842          | 0.964                              | 0.964               |
|        | 9        | 0.503                                 | 0.854          | 0.964                              | 0.964               |
|        | 10       | 0.505                                 | 0.859          | 0.966                              | 0.966               |
|        | 11       | 0.508                                 | 0.867          | 0.965                              | 0.965               |
|        | 12       | 0.938                                 | 0.948          | 0.963                              | 0.963               |
| PhilTa | 1        | 0.939                                 | 0.955          | 0.975                              | 0.975               |
|        | 2        | 0.911                                 | 0.941          | 0.973                              | 0.973               |
|        | 3        | 0.601                                 | 0.928          | 0.968                              | 0.968               |
|        | 4        | 0.551                                 | 0.916          | 0.972                              | 0.972               |

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| Ranking: SIF-suite AUROC (six models) |           |                 |                |                                    |                     |
|---------------------------------------|-----------|-----------------|----------------|------------------------------------|---------------------|
| Model                                 | Layer     | AUROC<br>(base) | AUROC<br>(SIF) | AUROC<br>(SIF+ABTT <sub>10</sub> ) | AUROC<br>(SIF+ABTT) |
|                                       | 5         | 0.542           | 0.904          | 0.972                              | 0.972               |
|                                       | 6         | 0.541           | 0.884          | 0.981                              | 0.981               |
|                                       | 7         | 0.542           | 0.891          | 0.981                              | 0.981               |
|                                       | <b>8</b>  | <b>0.542</b>    | <b>0.895</b>   | <b>0.982</b>                       | <b>0.982</b>        |
|                                       | 9         | 0.539           | 0.881          | 0.977                              | 0.977               |
|                                       | 10        | 0.538           | 0.874          | 0.977                              | 0.977               |
|                                       | 11        | 0.539           | 0.872          | 0.979                              | 0.979               |
|                                       | 12        | 0.911           | 0.929          | 0.969                              | 0.969               |
| mT5-base                              | <b>1</b>  | <b>0.822</b>    | <b>0.845</b>   | <b>0.974</b>                       | <b>0.974</b>        |
|                                       | 2         | 0.822           | 0.835          | 0.966                              | 0.966               |
|                                       | 3         | 0.804           | 0.808          | 0.958                              | 0.958               |
|                                       | 4         | 0.799           | 0.809          | 0.966                              | 0.966               |
|                                       | 5         | 0.654           | 0.672          | 0.974                              | 0.974               |
|                                       | 6         | 0.657           | 0.678          | 0.971                              | 0.971               |
|                                       | 7         | 0.656           | 0.680          | 0.967                              | 0.967               |
|                                       | 8         | 0.656           | 0.681          | 0.968                              | 0.968               |
|                                       | 9         | 0.660           | 0.685          | 0.967                              | 0.967               |
|                                       | 10        | 0.659           | 0.684          | 0.966                              | 0.966               |
|                                       | 11        | 0.659           | 0.689          | 0.966                              | 0.966               |
|                                       | 12        | 0.838           | 0.833          | 0.970                              | 0.970               |
| LaBSE                                 | 1         | 0.806           | 0.818          | 0.974                              | 0.974               |
|                                       | 2         | 0.833           | 0.840          | 0.975                              | 0.975               |
|                                       | 3         | 0.853           | 0.862          | 0.968                              | 0.968               |
|                                       | 4         | 0.862           | 0.869          | 0.966                              | 0.966               |
|                                       | 5         | 0.864           | 0.869          | 0.964                              | 0.964               |
|                                       | 6         | 0.894           | 0.895          | 0.965                              | 0.965               |
|                                       | 7         | 0.887           | 0.892          | 0.964                              | 0.964               |
|                                       | 8         | 0.879           | 0.883          | 0.968                              | 0.968               |
|                                       | 9         | 0.916           | 0.915          | 0.973                              | 0.973               |
|                                       | <b>10</b> | <b>0.934</b>    | <b>0.934</b>   | <b>0.976</b>                       | <b>0.976</b>        |
|                                       | 11        | 0.960           | 0.959          | 0.983                              | 0.983               |
|                                       | 12        | 0.956           | 0.956          | 0.980                              | 0.984               |
| Qwen3-0.6B                            | 1         | 0.858           | 0.852          | 0.982                              | 0.981               |
|                                       | <b>2</b>  | <b>0.863</b>    | <b>0.853</b>   | <b>0.972</b>                       | <b>0.972</b>        |
|                                       | 3         | 0.862           | 0.858          | 0.977                              | 0.975               |
|                                       | 4         | 0.860           | 0.860          | 0.976                              | 0.977               |
|                                       | 5         | 0.861           | 0.858          | 0.971                              | 0.971               |
|                                       | 6         | 0.867           | 0.861          | 0.974                              | 0.974               |
|                                       | 7         | 0.876           | 0.866          | 0.971                              | 0.971               |
|                                       | 8         | 0.882           | 0.868          | 0.969                              | 0.969               |
|                                       | 9         | 0.881           | 0.866          | 0.968                              | 0.968               |
|                                       | 10        | 0.887           | 0.870          | 0.966                              | 0.966               |
|                                       | 11        | 0.895           | 0.875          | 0.964                              | 0.964               |
|                                       | 12        | 0.901           | 0.881          | 0.966                              | 0.966               |
|                                       | 13        | 0.907           | 0.887          | 0.967                              | 0.967               |
|                                       | 14        | 0.905           | 0.885          | 0.966                              | 0.966               |
|                                       | 15        | 0.901           | 0.883          | 0.968                              | 0.968               |
|                                       | 16        | 0.910           | 0.894          | 0.965                              | 0.965               |
|                                       | 17        | 0.916           | 0.899          | 0.967                              | 0.974               |
|                                       | 18        | 0.933           | 0.914          | 0.966                              | 0.973               |
|                                       | 19        | 0.936           | 0.915          | 0.966                              | 0.976               |
|                                       | 20        | 0.943           | 0.918          | 0.971                              | 0.977               |
|                                       | 21        | 0.953           | 0.931          | 0.971                              | 0.971               |
|                                       | 22        | 0.963           | 0.942          | 0.973                              | 0.973               |
|                                       | 23        | 0.961           | 0.941          | 0.973                              | 0.973               |
|                                       | 24        | 0.963           | 0.943          | 0.976                              | 0.974               |
|                                       | 25        | 0.961           | 0.938          | 0.978                              | 0.975               |
|                                       | 26        | 0.966           | 0.942          | 0.980                              | 0.980               |
|                                       | 27        | 0.967           | 0.936          | 0.978                              | 0.978               |
|                                       | 28        | 0.958           | 0.925          | 0.968                              | 0.968               |
| KaLM-mini                             | 1         | 0.887           | 0.880          | 0.982                              | 0.980               |
|                                       | <b>2</b>  | <b>0.906</b>    | <b>0.899</b>   | <b>0.981</b>                       | <b>0.981</b>        |

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| Model | Layer | Ranking: SIF-suite AUROC (six models) |                |                                    |                     |
|-------|-------|---------------------------------------|----------------|------------------------------------|---------------------|
|       |       | AUROC<br>(base)                       | AUROC<br>(SIF) | AUROC<br>(SIF+ABTT <sub>10</sub> ) | AUROC<br>(SIF+ABTT) |
|       | 3     | 0.881                                 | 0.882          | 0.977                              | 0.977               |
|       | 4     | 0.871                                 | 0.873          | 0.974                              | 0.974               |
|       | 5     | 0.861                                 | 0.862          | 0.971                              | 0.971               |
|       | 6     | 0.872                                 | 0.867          | 0.967                              | 0.967               |
|       | 7     | 0.888                                 | 0.879          | 0.972                              | 0.972               |
|       | 8     | 0.890                                 | 0.880          | 0.973                              | 0.975               |
|       | 9     | 0.892                                 | 0.882          | 0.968                              | 0.968               |
|       | 10    | 0.893                                 | 0.882          | 0.964                              | 0.964               |
|       | 11    | 0.894                                 | 0.885          | 0.969                              | 0.969               |
|       | 12    | 0.905                                 | 0.896          | 0.966                              | 0.966               |
|       | 13    | 0.900                                 | 0.894          | 0.964                              | 0.964               |
|       | 14    | 0.904                                 | 0.898          | 0.970                              | 0.970               |
|       | 15    | 0.907                                 | 0.901          | 0.970                              | 0.970               |
|       | 16    | 0.917                                 | 0.913          | 0.967                              | 0.967               |
|       | 17    | 0.926                                 | 0.923          | 0.968                              | 0.968               |
|       | 18    | 0.945                                 | 0.939          | 0.977                              | 0.977               |
|       | 19    | 0.956                                 | 0.950          | 0.978                              | 0.978               |
|       | 20    | 0.960                                 | 0.956          | 0.978                              | 0.978               |
|       | 21    | 0.962                                 | 0.958          | 0.980                              | 0.983               |
|       | 22    | 0.971                                 | 0.967          | 0.983                              | 0.984               |
|       | 23    | 0.972                                 | 0.964          | 0.986                              | 0.985               |
|       | 24    | 0.956                                 | 0.956          | 0.976                              | 0.976               |

Table 31: Per-layer routing assignment accuracy across the SIF-conditioned suite for all six models. Restricted to overall routing accuracy (existing/new decomposition omitted) to keep the table compact; see Tables 22 and 25 for the pure-ABTT pairwise comparison with existing/new broken out. Rows in bold mark the layer chosen on the training split, the layer with the highest training-set DirAcc@1 under SIF+ABTT. It is the layer behind the SIF+ABTT cell of Table 15 (listed in Table 20), and not always the layer with the highest test score in this table.

| Model    | Layer     | Routing: SIF-suite assignment accuracy (six models) |                  |                                      |                       |
|----------|-----------|-----------------------------------------------------|------------------|--------------------------------------|-----------------------|
|          |           | Overall<br>(base)                                   | Overall<br>(SIF) | Overall<br>(SIF+ABTT <sub>10</sub> ) | Overall<br>(SIF+ABTT) |
| LaTa     | 1         | 0.738                                               | 0.832            | 0.920                                | 0.899                 |
|          | 2         | 0.439                                               | 0.791            | 0.904                                | 0.900                 |
|          | 3         | 0.477                                               | 0.720            | 0.910                                | 0.907                 |
|          | 4         | 0.513                                               | 0.668            | 0.900                                | 0.902                 |
|          | 5         | 0.502                                               | 0.648            | 0.906                                | 0.906                 |
|          | 6         | 0.503                                               | 0.668            | 0.895                                | 0.895                 |
|          | 7         | 0.501                                               | 0.642            | 0.903                                | 0.903                 |
|          | 8         | 0.502                                               | 0.614            | 0.892                                | 0.892                 |
|          | 9         | 0.501                                               | 0.618            | 0.889                                | 0.889                 |
|          | 10        | 0.498                                               | 0.655            | 0.890                                | 0.890                 |
|          | 11        | 0.490                                               | 0.621            | 0.880                                | 0.880                 |
|          | <b>12</b> | <b>0.710</b>                                        | <b>0.804</b>     | <b>0.902</b>                         | <b>0.902</b>          |
| PhilTa   | <b>1</b>  | <b>0.709</b>                                        | <b>0.790</b>     | <b>0.914</b>                         | <b>0.914</b>          |
|          | 2         | 0.625                                               | 0.738            | 0.892                                | 0.892                 |
|          | 3         | 0.455                                               | 0.713            | 0.892                                | 0.892                 |
|          | 4         | 0.439                                               | 0.663            | 0.886                                | 0.886                 |
|          | 5         | 0.457                                               | 0.626            | 0.899                                | 0.899                 |
|          | 6         | 0.474                                               | 0.612            | 0.906                                | 0.906                 |
|          | 7         | 0.469                                               | 0.625            | 0.906                                | 0.906                 |
|          | 8         | 0.467                                               | 0.614            | 0.886                                | 0.886                 |
|          | 9         | 0.476                                               | 0.590            | 0.895                                | 0.895                 |
|          | 10        | 0.480                                               | 0.580            | 0.888                                | 0.888                 |
|          | 11        | 0.470                                               | 0.565            | 0.886                                | 0.886                 |
|          | 12        | 0.549                                               | 0.667            | 0.899                                | 0.899                 |
| mT5-base | 1         | 0.453                                               | 0.545            | 0.907                                | 0.907                 |
|          | <b>2</b>  | <b>0.436</b>                                        | <b>0.537</b>     | <b>0.903</b>                         | <b>0.903</b>          |
|          | 3         | 0.449                                               | 0.568            | 0.885                                | 0.885                 |
|          | 4         | 0.491                                               | 0.488            | 0.857                                | 0.857                 |

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| Routing: SIF-suite assignment accuracy (six models) |           |                   |                  |                                      |                       |
|-----------------------------------------------------|-----------|-------------------|------------------|--------------------------------------|-----------------------|
| Model                                               | Layer     | Overall<br>(base) | Overall<br>(SIF) | Overall<br>(SIF+ABTT <sub>10</sub> ) | Overall<br>(SIF+ABTT) |
|                                                     | 5         | 0.512             | 0.483            | 0.826                                | 0.826                 |
|                                                     | 6         | 0.580             | 0.558            | 0.817                                | 0.817                 |
|                                                     | 7         | 0.565             | 0.554            | 0.832                                | 0.832                 |
|                                                     | 8         | 0.558             | 0.548            | 0.834                                | 0.834                 |
|                                                     | 9         | 0.508             | 0.495            | 0.845                                | 0.845                 |
|                                                     | 10        | 0.491             | 0.495            | 0.832                                | 0.832                 |
|                                                     | 11        | 0.479             | 0.484            | 0.816                                | 0.816                 |
|                                                     | 12        | 0.472             | 0.541            | 0.858                                | 0.858                 |
| LaBSE                                               | 1         | 0.477             | 0.521            | 0.886                                | 0.886                 |
|                                                     | 2         | 0.543             | 0.508            | 0.875                                | 0.875                 |
|                                                     | 3         | 0.514             | 0.577            | 0.860                                | 0.860                 |
|                                                     | 4         | 0.565             | 0.587            | 0.869                                | 0.869                 |
|                                                     | 5         | 0.543             | 0.598            | 0.857                                | 0.857                 |
|                                                     | 6         | 0.674             | 0.648            | 0.872                                | 0.872                 |
|                                                     | 7         | 0.584             | 0.634            | 0.858                                | 0.858                 |
|                                                     | 8         | 0.568             | 0.571            | 0.854                                | 0.854                 |
|                                                     | 9         | 0.612             | 0.613            | 0.859                                | 0.859                 |
|                                                     | 10        | 0.713             | 0.698            | 0.864                                | 0.864                 |
|                                                     | <b>11</b> | <b>0.833</b>      | <b>0.807</b>     | <b>0.896</b>                         | <b>0.896</b>          |
|                                                     | 12        | 0.808             | 0.834            | 0.890                                | 0.880                 |
| Qwen3-0.6B                                          | 1         | 0.526             | 0.566            | 0.914                                | 0.909                 |
|                                                     | 2         | 0.579             | 0.524            | 0.894                                | 0.894                 |
|                                                     | 3         | 0.513             | 0.537            | 0.904                                | 0.901                 |
|                                                     | 4         | 0.576             | 0.531            | 0.899                                | 0.904                 |
|                                                     | <b>5</b>  | <b>0.464</b>      | <b>0.537</b>     | <b>0.908</b>                         | <b>0.908</b>          |
|                                                     | 6         | 0.576             | 0.526            | 0.906                                | 0.906                 |
|                                                     | 7         | 0.513             | 0.573            | 0.911                                | 0.911                 |
|                                                     | 8         | 0.592             | 0.627            | 0.911                                | 0.911                 |
|                                                     | 9         | 0.607             | 0.576            | 0.910                                | 0.910                 |
|                                                     | 10        | 0.647             | 0.599            | 0.900                                | 0.900                 |
|                                                     | 11        | 0.551             | 0.606            | 0.893                                | 0.893                 |
|                                                     | 12        | 0.607             | 0.663            | 0.900                                | 0.900                 |
|                                                     | 13        | 0.625             | 0.675            | 0.897                                | 0.897                 |
|                                                     | 14        | 0.633             | 0.571            | 0.893                                | 0.893                 |
|                                                     | 15        | 0.639             | 0.600            | 0.893                                | 0.893                 |
|                                                     | 16        | 0.711             | 0.679            | 0.880                                | 0.880                 |
|                                                     | 17        | 0.652             | 0.686            | 0.900                                | 0.876                 |
|                                                     | 18        | 0.700             | 0.731            | 0.899                                | 0.882                 |
|                                                     | 19        | 0.684             | 0.700            | 0.886                                | 0.850                 |
|                                                     | 20        | 0.746             | 0.676            | 0.882                                | 0.866                 |
|                                                     | 21        | 0.761             | 0.739            | 0.892                                | 0.892                 |
|                                                     | 22        | 0.809             | 0.719            | 0.893                                | 0.893                 |
|                                                     | 23        | 0.752             | 0.759            | 0.893                                | 0.893                 |
|                                                     | 24        | 0.818             | 0.706            | 0.900                                | 0.899                 |
|                                                     | 25        | 0.725             | 0.748            | 0.897                                | 0.893                 |
|                                                     | 26        | 0.749             | 0.772            | 0.903                                | 0.903                 |
|                                                     | 27        | 0.808             | 0.760            | 0.899                                | 0.899                 |
|                                                     | 28        | 0.823             | 0.774            | 0.890                                | 0.890                 |
| KaLM-mini                                           | <b>1</b>  | <b>0.585</b>      | <b>0.604</b>     | <b>0.906</b>                         | <b>0.908</b>          |
|                                                     | 2         | 0.627             | 0.594            | 0.908                                | 0.908                 |
|                                                     | 3         | 0.604             | 0.576            | 0.907                                | 0.907                 |
|                                                     | 4         | 0.544             | 0.577            | 0.901                                | 0.901                 |
|                                                     | 5         | 0.543             | 0.533            | 0.897                                | 0.897                 |
|                                                     | 6         | 0.502             | 0.558            | 0.900                                | 0.900                 |
|                                                     | 7         | 0.618             | 0.593            | 0.901                                | 0.901                 |
|                                                     | 8         | 0.619             | 0.607            | 0.892                                | 0.878                 |
|                                                     | 9         | 0.626             | 0.663            | 0.876                                | 0.876                 |
|                                                     | 10        | 0.613             | 0.603            | 0.866                                | 0.866                 |
|                                                     | 11        | 0.619             | 0.627            | 0.874                                | 0.874                 |
|                                                     | 12        | 0.653             | 0.642            | 0.874                                | 0.874                 |
|                                                     | 13        | 0.674             | 0.666            | 0.864                                | 0.864                 |
|                                                     | 14        | 0.688             | 0.648            | 0.861                                | 0.861                 |
|                                                     | 15        | 0.691             | 0.698            | 0.866                                | 0.866                 |

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| Routing: SIF-suite assignment accuracy (six models) |       |                   |                  |                                      |                       |
|-----------------------------------------------------|-------|-------------------|------------------|--------------------------------------|-----------------------|
| Model                                               | Layer | Overall<br>(base) | Overall<br>(SIF) | Overall<br>(SIF+ABTT <sub>10</sub> ) | Overall<br>(SIF+ABTT) |
|                                                     | 16    | 0.695             | 0.738            | 0.859                                | 0.859                 |
|                                                     | 17    | 0.767             | 0.763            | 0.882                                | 0.882                 |
|                                                     | 18    | 0.796             | 0.796            | 0.879                                | 0.879                 |
|                                                     | 19    | 0.814             | 0.804            | 0.878                                | 0.878                 |
|                                                     | 20    | 0.803             | 0.821            | 0.876                                | 0.876                 |
|                                                     | 21    | 0.817             | 0.832            | 0.874                                | 0.876                 |
|                                                     | 22    | 0.875             | 0.857            | 0.876                                | 0.876                 |
|                                                     | 23    | 0.875             | 0.862            | 0.880                                | 0.881                 |
|                                                     | 24    | 0.846             | 0.847            | 0.869                                | 0.869                 |

Table 32: Per-layer routing DirAcc@1, overall and for existing and new witnesses, across all six models, averaged over five reseeds of the query and reference files, with the training/test split fixed (mean  $\pm$  std). `sif_abtt_optimal` applies SIF weighting plus ABTT with  $D$  tuned per layer on the training split. This table covers SIF+ABTT; five-reseeding ABTT-only values for LaTa, Qwen3-0.6B and KaLM-mini, at their train-selected layers, are in Table 18 (Appendix H). Overall assignment accuracy coincides with DirAcc@1 under the five reseeds, so we report the existing/new decomposition instead. Rows in bold mark the layer reported per model, the row printed in Table 28. For each model the reported layer is chosen on the training split: the layer with the highest training-set DirAcc@1 for SIF+ABTT in the single-seed run, the layer behind the SIF+ABTT cell of Table 15, listed in Table 20.  $D$  is tuned per layer on the training split. This is not always the layer with the highest mean over the five reseeds.

| Routing: DirAcc@1 (five reseeds, mean $\pm$ std, SIF+ABTT) |           |                                     |                                     |                                     |                                     |                                     |                                     |
|------------------------------------------------------------|-----------|-------------------------------------|-------------------------------------|-------------------------------------|-------------------------------------|-------------------------------------|-------------------------------------|
| Model                                                      | Layer     | DirAcc@1<br>(base)                  | DirAcc@1<br>(SIF+ABTT)              | Existing<br>(base)                  | Existing<br>(SIF+ABTT)              | New<br>(base)                       | New<br>(SIF+ABTT)                   |
| LaTa                                                       | 1         | 0.731 $\pm$ 0.010                   | 0.889 $\pm$ 0.010                   | 0.563 $\pm$ 0.007                   | 0.844 $\pm$ 0.011                   | 0.949 $\pm$ 0.011                   | 0.948 $\pm$ 0.010                   |
|                                                            | 2         | 0.429 $\pm$ 0.011                   | 0.888 $\pm$ 0.007                   | 0.049 $\pm$ 0.006                   | 0.841 $\pm$ 0.007                   | 0.920 $\pm$ 0.008                   | 0.948 $\pm$ 0.007                   |
|                                                            | 3         | 0.393 $\pm$ 0.011                   | 0.891 $\pm$ 0.010                   | 0.047 $\pm$ 0.006                   | 0.847 $\pm$ 0.010                   | 0.840 $\pm$ 0.020                   | 0.947 $\pm$ 0.012                   |
|                                                            | 4         | 0.331 $\pm$ 0.019                   | 0.889 $\pm$ 0.009                   | 0.048 $\pm$ 0.009                   | 0.860 $\pm$ 0.009                   | 0.696 $\pm$ 0.030                   | 0.925 $\pm$ 0.013                   |
|                                                            | 5         | 0.326 $\pm$ 0.021                   | 0.891 $\pm$ 0.009                   | 0.038 $\pm$ 0.005                   | 0.858 $\pm$ 0.008                   | 0.698 $\pm$ 0.028                   | 0.934 $\pm$ 0.014                   |
|                                                            | 6         | 0.322 $\pm$ 0.018                   | 0.891 $\pm$ 0.009                   | 0.038 $\pm$ 0.007                   | 0.863 $\pm$ 0.010                   | 0.688 $\pm$ 0.026                   | 0.927 $\pm$ 0.013                   |
|                                                            | 7         | 0.328 $\pm$ 0.020                   | 0.896 $\pm$ 0.008                   | 0.045 $\pm$ 0.007                   | 0.848 $\pm$ 0.009                   | 0.693 $\pm$ 0.028                   | 0.957 $\pm$ 0.010                   |
|                                                            | 8         | 0.325 $\pm$ 0.022                   | 0.884 $\pm$ 0.008                   | 0.046 $\pm$ 0.010                   | 0.834 $\pm$ 0.008                   | 0.685 $\pm$ 0.032                   | 0.949 $\pm$ 0.011                   |
|                                                            | 9         | 0.332 $\pm$ 0.019                   | 0.886 $\pm$ 0.011                   | 0.045 $\pm$ 0.010                   | 0.848 $\pm$ 0.009                   | 0.701 $\pm$ 0.024                   | 0.935 $\pm$ 0.015                   |
|                                                            | 10        | 0.346 $\pm$ 0.017                   | 0.889 $\pm$ 0.004                   | 0.045 $\pm$ 0.010                   | 0.846 $\pm$ 0.008                   | 0.735 $\pm$ 0.021                   | 0.944 $\pm$ 0.012                   |
|                                                            | 11        | 0.361 $\pm$ 0.018                   | 0.882 $\pm$ 0.009                   | 0.045 $\pm$ 0.009                   | 0.862 $\pm$ 0.010                   | 0.767 $\pm$ 0.029                   | 0.907 $\pm$ 0.014                   |
|                                                            | <b>12</b> | <b>0.707 <math>\pm</math> 0.014</b> | <b>0.889 <math>\pm</math> 0.005</b> | <b>0.511 <math>\pm</math> 0.016</b> | <b>0.834 <math>\pm</math> 0.005</b> | <b>0.961 <math>\pm</math> 0.008</b> | <b>0.961 <math>\pm</math> 0.010</b> |
| PhilTa                                                     | <b>1</b>  | <b>0.705 <math>\pm</math> 0.015</b> | <b>0.898 <math>\pm</math> 0.008</b> | <b>0.531 <math>\pm</math> 0.016</b> | <b>0.858 <math>\pm</math> 0.010</b> | <b>0.931 <math>\pm</math> 0.012</b> | <b>0.951 <math>\pm</math> 0.011</b> |
|                                                            | 2         | 0.626 $\pm$ 0.019                   | 0.888 $\pm$ 0.010                   | 0.410 $\pm$ 0.017                   | 0.866 $\pm$ 0.010                   | 0.906 $\pm$ 0.018                   | 0.917 $\pm$ 0.017                   |
|                                                            | 3         | 0.440 $\pm$ 0.013                   | 0.893 $\pm$ 0.009                   | 0.071 $\pm$ 0.014                   | 0.858 $\pm$ 0.010                   | 0.917 $\pm$ 0.013                   | 0.939 $\pm$ 0.011                   |
|                                                            | 4         | 0.421 $\pm$ 0.016                   | 0.891 $\pm$ 0.009                   | 0.027 $\pm$ 0.007                   | 0.858 $\pm$ 0.008                   | 0.928 $\pm$ 0.015                   | 0.935 $\pm$ 0.013                   |
|                                                            | 5         | 0.375 $\pm$ 0.016                   | 0.895 $\pm$ 0.010                   | 0.042 $\pm$ 0.013                   | 0.852 $\pm$ 0.010                   | 0.804 $\pm$ 0.018                   | 0.950 $\pm$ 0.014                   |
|                                                            | 6         | 0.337 $\pm$ 0.015                   | 0.897 $\pm$ 0.009                   | 0.060 $\pm$ 0.012                   | 0.861 $\pm$ 0.010                   | 0.696 $\pm$ 0.026                   | 0.944 $\pm$ 0.013                   |
|                                                            | 7         | 0.361 $\pm$ 0.017                   | 0.900 $\pm$ 0.009                   | 0.056 $\pm$ 0.012                   | 0.860 $\pm$ 0.008                   | 0.755 $\pm$ 0.020                   | 0.952 $\pm$ 0.013                   |
|                                                            | 8         | 0.389 $\pm$ 0.016                   | 0.891 $\pm$ 0.010                   | 0.058 $\pm$ 0.014                   | 0.859 $\pm$ 0.009                   | 0.818 $\pm$ 0.018                   | 0.932 $\pm$ 0.015                   |
|                                                            | 9         | 0.371 $\pm$ 0.016                   | 0.893 $\pm$ 0.008                   | 0.066 $\pm$ 0.014                   | 0.861 $\pm$ 0.009                   | 0.766 $\pm$ 0.027                   | 0.934 $\pm$ 0.016                   |
|                                                            | 10        | 0.377 $\pm$ 0.014                   | 0.892 $\pm$ 0.009                   | 0.066 $\pm$ 0.014                   | 0.855 $\pm$ 0.005                   | 0.778 $\pm$ 0.023                   | 0.940 $\pm$ 0.017                   |
|                                                            | 11        | 0.394 $\pm$ 0.014                   | 0.888 $\pm$ 0.008                   | 0.059 $\pm$ 0.010                   | 0.851 $\pm$ 0.010                   | 0.827 $\pm$ 0.021                   | 0.935 $\pm$ 0.013                   |
|                                                            | 12        | 0.566 $\pm$ 0.013                   | 0.897 $\pm$ 0.010                   | 0.282 $\pm$ 0.014                   | 0.863 $\pm$ 0.010                   | 0.934 $\pm$ 0.007                   | 0.941 $\pm$ 0.018                   |
| mT5-base                                                   | 1         | 0.487 $\pm$ 0.012                   | 0.899 $\pm$ 0.005                   | 0.100 $\pm$ 0.009                   | 0.856 $\pm$ 0.007                   | 0.987 $\pm$ 0.007                   | 0.954 $\pm$ 0.011                   |
|                                                            | <b>2</b>  | <b>0.476 <math>\pm</math> 0.012</b> | <b>0.893 <math>\pm</math> 0.005</b> | <b>0.083 <math>\pm</math> 0.008</b> | <b>0.851 <math>\pm</math> 0.012</b> | <b>0.984 <math>\pm</math> 0.008</b> | <b>0.947 <math>\pm</math> 0.013</b> |
|                                                            | 3         | 0.483 $\pm$ 0.014                   | 0.883 $\pm$ 0.011                   | 0.124 $\pm$ 0.009                   | 0.817 $\pm$ 0.014                   | 0.947 $\pm$ 0.008                   | 0.970 $\pm$ 0.007                   |
|                                                            | 4         | 0.500 $\pm$ 0.015                   | 0.852 $\pm$ 0.007                   | 0.206 $\pm$ 0.006                   | 0.772 $\pm$ 0.010                   | 0.879 $\pm$ 0.020                   | 0.956 $\pm$ 0.010                   |
|                                                            | 5         | 0.371 $\pm$ 0.014                   | 0.825 $\pm$ 0.008                   | 0.155 $\pm$ 0.006                   | 0.724 $\pm$ 0.007                   | 0.651 $\pm$ 0.023                   | 0.956 $\pm$ 0.006                   |
|                                                            | 6         | 0.361 $\pm$ 0.012                   | 0.810 $\pm$ 0.005                   | 0.263 $\pm$ 0.021                   | 0.765 $\pm$ 0.012                   | 0.488 $\pm$ 0.018                   | 0.868 $\pm$ 0.016                   |
|                                                            | 7         | 0.370 $\pm$ 0.010                   | 0.827 $\pm$ 0.008                   | 0.255 $\pm$ 0.014                   | 0.760 $\pm$ 0.011                   | 0.519 $\pm$ 0.017                   | 0.914 $\pm$ 0.013                   |
|                                                            | 8         | 0.382 $\pm$ 0.008                   | 0.824 $\pm$ 0.012                   | 0.244 $\pm$ 0.007                   | 0.773 $\pm$ 0.012                   | 0.561 $\pm$ 0.017                   | 0.891 $\pm$ 0.016                   |
|                                                            | 9         | 0.428 $\pm$ 0.014                   | 0.838 $\pm$ 0.010                   | 0.171 $\pm$ 0.008                   | 0.783 $\pm$ 0.013                   | 0.759 $\pm$ 0.024                   | 0.909 $\pm$ 0.006                   |
|                                                            | 10        | 0.421 $\pm$ 0.010                   | 0.821 $\pm$ 0.017                   | 0.140 $\pm$ 0.011                   | 0.767 $\pm$ 0.015                   | 0.783 $\pm$ 0.021                   | 0.892 $\pm$ 0.019                   |
|                                                            | 11        | 0.420 $\pm$ 0.015                   | 0.803 $\pm$ 0.012                   | 0.131 $\pm$ 0.012                   | 0.721 $\pm$ 0.013                   | 0.792 $\pm$ 0.016                   | 0.910 $\pm$ 0.015                   |
|                                                            | 12        | 0.507 $\pm$ 0.009                   | 0.841 $\pm$ 0.009                   | 0.169 $\pm$ 0.007                   | 0.772 $\pm$ 0.016                   | 0.942 $\pm$ 0.011                   | 0.931 $\pm$ 0.011                   |

(continued on next page)

| Routing: DirAcc@1 (five reseeds, mean $\pm$ std, SIF+ABTT) |           |                                     |                                     |                                     |                                     |                                     |                                     |
|------------------------------------------------------------|-----------|-------------------------------------|-------------------------------------|-------------------------------------|-------------------------------------|-------------------------------------|-------------------------------------|
| Model                                                      | Layer     | DirAcc@1<br>(base)                  | DirAcc@1<br>(SIF+ABTT)              | Existing<br>(base)                  | Existing<br>(SIF+ABTT)              | New<br>(base)                       | New<br>(SIF+ABTT)                   |
| LaBSE                                                      | 1         | 0.502 $\pm$ 0.012                   | 0.858 $\pm$ 0.007                   | 0.170 $\pm$ 0.003                   | 0.838 $\pm$ 0.007                   | 0.929 $\pm$ 0.016                   | 0.885 $\pm$ 0.009                   |
|                                                            | 2         | 0.523 $\pm$ 0.011                   | 0.860 $\pm$ 0.004                   | 0.274 $\pm$ 0.005                   | 0.817 $\pm$ 0.003                   | 0.844 $\pm$ 0.022                   | 0.915 $\pm$ 0.008                   |
|                                                            | 3         | 0.535 $\pm$ 0.010                   | 0.847 $\pm$ 0.006                   | 0.218 $\pm$ 0.009                   | 0.831 $\pm$ 0.009                   | 0.945 $\pm$ 0.008                   | 0.867 $\pm$ 0.007                   |
|                                                            | 4         | 0.569 $\pm$ 0.008                   | 0.850 $\pm$ 0.006                   | 0.299 $\pm$ 0.004                   | 0.837 $\pm$ 0.006                   | 0.917 $\pm$ 0.014                   | 0.866 $\pm$ 0.011                   |
|                                                            | 5         | 0.551 $\pm$ 0.012                   | 0.845 $\pm$ 0.005                   | 0.255 $\pm$ 0.012                   | 0.794 $\pm$ 0.005                   | 0.933 $\pm$ 0.013                   | 0.911 $\pm$ 0.010                   |
|                                                            | 6         | 0.661 $\pm$ 0.011                   | 0.859 $\pm$ 0.004                   | 0.461 $\pm$ 0.019                   | 0.817 $\pm$ 0.008                   | 0.919 $\pm$ 0.018                   | 0.914 $\pm$ 0.015                   |
|                                                            | 7         | 0.596 $\pm$ 0.010                   | 0.849 $\pm$ 0.004                   | 0.333 $\pm$ 0.016                   | 0.808 $\pm$ 0.006                   | 0.935 $\pm$ 0.016                   | 0.902 $\pm$ 0.010                   |
|                                                            | 8         | 0.572 $\pm$ 0.012                   | 0.847 $\pm$ 0.006                   | 0.314 $\pm$ 0.019                   | 0.802 $\pm$ 0.009                   | 0.904 $\pm$ 0.016                   | 0.905 $\pm$ 0.008                   |
|                                                            | 9         | 0.631 $\pm$ 0.008                   | 0.854 $\pm$ 0.007                   | 0.381 $\pm$ 0.015                   | 0.828 $\pm$ 0.002                   | 0.954 $\pm$ 0.008                   | 0.887 $\pm$ 0.012                   |
|                                                            | 10        | 0.712 $\pm$ 0.016                   | 0.858 $\pm$ 0.005                   | 0.551 $\pm$ 0.019                   | 0.836 $\pm$ 0.005                   | 0.920 $\pm$ 0.015                   | 0.887 $\pm$ 0.010                   |
|                                                            | <b>11</b> | <b>0.820 <math>\pm</math> 0.010</b> | <b>0.906 <math>\pm</math> 0.010</b> | <b>0.731 <math>\pm</math> 0.012</b> | <b>0.875 <math>\pm</math> 0.013</b> | <b>0.935 <math>\pm</math> 0.012</b> | <b>0.944 <math>\pm</math> 0.017</b> |
|                                                            | 12        | 0.793 $\pm$ 0.015                   | 0.884 $\pm$ 0.006                   | 0.670 $\pm$ 0.019                   | 0.850 $\pm$ 0.012                   | 0.952 $\pm$ 0.013                   | 0.927 $\pm$ 0.015                   |
| Qwen3-0.6B                                                 | 1         | 0.550 $\pm$ 0.008                   | 0.894 $\pm$ 0.005                   | 0.223 $\pm$ 0.006                   | 0.840 $\pm$ 0.010                   | 0.972 $\pm$ 0.003                   | 0.965 $\pm$ 0.011                   |
|                                                            | 2         | 0.572 $\pm$ 0.015                   | 0.882 $\pm$ 0.004                   | 0.363 $\pm$ 0.014                   | 0.828 $\pm$ 0.005                   | 0.841 $\pm$ 0.032                   | 0.952 $\pm$ 0.007                   |
|                                                            | 3         | 0.541 $\pm$ 0.007                   | 0.889 $\pm$ 0.006                   | 0.216 $\pm$ 0.007                   | 0.827 $\pm$ 0.008                   | 0.962 $\pm$ 0.007                   | 0.968 $\pm$ 0.011                   |
|                                                            | 4         | 0.580 $\pm$ 0.012                   | 0.891 $\pm$ 0.004                   | 0.353 $\pm$ 0.008                   | 0.845 $\pm$ 0.007                   | 0.874 $\pm$ 0.028                   | 0.950 $\pm$ 0.006                   |
|                                                            | <b>5</b>  | <b>0.487 <math>\pm</math> 0.017</b> | <b>0.893 <math>\pm</math> 0.007</b> | <b>0.133 <math>\pm</math> 0.012</b> | <b>0.839 <math>\pm</math> 0.011</b> | <b>0.944 <math>\pm</math> 0.014</b> | <b>0.964 <math>\pm</math> 0.009</b> |
|                                                            | 6         | 0.568 $\pm$ 0.012                   | 0.897 $\pm$ 0.005                   | 0.350 $\pm$ 0.010                   | 0.857 $\pm$ 0.008                   | 0.850 $\pm$ 0.023                   | 0.948 $\pm$ 0.012                   |
|                                                            | 7         | 0.541 $\pm$ 0.007                   | 0.902 $\pm$ 0.006                   | 0.211 $\pm$ 0.009                   | 0.862 $\pm$ 0.008                   | 0.967 $\pm$ 0.012                   | 0.953 $\pm$ 0.012                   |
|                                                            | 8         | 0.609 $\pm$ 0.009                   | 0.903 $\pm$ 0.006                   | 0.362 $\pm$ 0.013                   | 0.861 $\pm$ 0.010                   | 0.929 $\pm$ 0.017                   | 0.956 $\pm$ 0.005                   |
|                                                            | 9         | 0.612 $\pm$ 0.013                   | 0.900 $\pm$ 0.004                   | 0.393 $\pm$ 0.016                   | 0.868 $\pm$ 0.008                   | 0.894 $\pm$ 0.021                   | 0.943 $\pm$ 0.006                   |
|                                                            | 10        | 0.638 $\pm$ 0.013                   | 0.890 $\pm$ 0.009                   | 0.452 $\pm$ 0.009                   | 0.847 $\pm$ 0.014                   | 0.878 $\pm$ 0.022                   | 0.944 $\pm$ 0.010                   |
|                                                            | 11        | 0.578 $\pm$ 0.006                   | 0.885 $\pm$ 0.005                   | 0.257 $\pm$ 0.010                   | 0.838 $\pm$ 0.014                   | 0.991 $\pm$ 0.005                   | 0.947 $\pm$ 0.012                   |
|                                                            | 12        | 0.627 $\pm$ 0.009                   | 0.892 $\pm$ 0.005                   | 0.366 $\pm$ 0.016                   | 0.844 $\pm$ 0.010                   | 0.964 $\pm$ 0.013                   | 0.955 $\pm$ 0.007                   |
|                                                            | 13        | 0.639 $\pm$ 0.010                   | 0.887 $\pm$ 0.006                   | 0.383 $\pm$ 0.014                   | 0.849 $\pm$ 0.011                   | 0.970 $\pm$ 0.015                   | 0.937 $\pm$ 0.011                   |
|                                                            | 14        | 0.646 $\pm$ 0.009                   | 0.885 $\pm$ 0.006                   | 0.402 $\pm$ 0.017                   | 0.852 $\pm$ 0.009                   | 0.960 $\pm$ 0.016                   | 0.927 $\pm$ 0.007                   |
|                                                            | 15        | 0.648 $\pm$ 0.012                   | 0.885 $\pm$ 0.007                   | 0.410 $\pm$ 0.017                   | 0.852 $\pm$ 0.010                   | 0.954 $\pm$ 0.017                   | 0.929 $\pm$ 0.018                   |
|                                                            | 16        | 0.691 $\pm$ 0.016                   | 0.873 $\pm$ 0.005                   | 0.546 $\pm$ 0.017                   | 0.837 $\pm$ 0.009                   | 0.878 $\pm$ 0.017                   | 0.920 $\pm$ 0.016                   |
|                                                            | 17        | 0.658 $\pm$ 0.010                   | 0.862 $\pm$ 0.006                   | 0.420 $\pm$ 0.018                   | 0.807 $\pm$ 0.012                   | 0.966 $\pm$ 0.013                   | 0.933 $\pm$ 0.013                   |
|                                                            | 18        | 0.709 $\pm$ 0.012                   | 0.868 $\pm$ 0.002                   | 0.503 $\pm$ 0.018                   | 0.833 $\pm$ 0.013                   | 0.975 $\pm$ 0.006                   | 0.914 $\pm$ 0.017                   |
|                                                            | 19        | 0.691 $\pm$ 0.010                   | 0.874 $\pm$ 0.002                   | 0.462 $\pm$ 0.018                   | 0.860 $\pm$ 0.007                   | 0.986 $\pm$ 0.006                   | 0.893 $\pm$ 0.012                   |
|                                                            | 20        | 0.743 $\pm$ 0.020                   | 0.860 $\pm$ 0.004                   | 0.588 $\pm$ 0.023                   | 0.797 $\pm$ 0.011                   | 0.943 $\pm$ 0.018                   | 0.941 $\pm$ 0.010                   |
|                                                            | 21        | 0.758 $\pm$ 0.017                   | 0.879 $\pm$ 0.006                   | 0.585 $\pm$ 0.021                   | 0.836 $\pm$ 0.008                   | 0.981 $\pm$ 0.010                   | 0.935 $\pm$ 0.010                   |
|                                                            | 22        | 0.799 $\pm$ 0.014                   | 0.882 $\pm$ 0.006                   | 0.671 $\pm$ 0.019                   | 0.855 $\pm$ 0.002                   | 0.965 $\pm$ 0.006                   | 0.918 $\pm$ 0.015                   |
|                                                            | 23        | 0.757 $\pm$ 0.012                   | 0.887 $\pm$ 0.006                   | 0.577 $\pm$ 0.014                   | 0.857 $\pm$ 0.007                   | 0.989 $\pm$ 0.004                   | 0.927 $\pm$ 0.011                   |
|                                                            | 24        | 0.810 $\pm$ 0.014                   | 0.893 $\pm$ 0.005                   | 0.695 $\pm$ 0.021                   | 0.861 $\pm$ 0.004                   | 0.959 $\pm$ 0.003                   | 0.935 $\pm$ 0.011                   |
|                                                            | 25        | 0.732 $\pm$ 0.011                   | 0.890 $\pm$ 0.009                   | 0.527 $\pm$ 0.016                   | 0.866 $\pm$ 0.006                   | 0.997 $\pm$ 0.003                   | 0.920 $\pm$ 0.014                   |
|                                                            | 26        | 0.749 $\pm$ 0.013                   | 0.895 $\pm$ 0.008                   | 0.562 $\pm$ 0.017                   | 0.858 $\pm$ 0.004                   | 0.991 $\pm$ 0.003                   | 0.943 $\pm$ 0.015                   |
|                                                            | 27        | 0.812 $\pm$ 0.004                   | 0.881 $\pm$ 0.005                   | 0.741 $\pm$ 0.008                   | 0.824 $\pm$ 0.006                   | 0.903 $\pm$ 0.011                   | 0.954 $\pm$ 0.011                   |
|                                                            | 28        | 0.824 $\pm$ 0.010                   | 0.871 $\pm$ 0.006                   | 0.736 $\pm$ 0.015                   | 0.802 $\pm$ 0.010                   | 0.939 $\pm$ 0.005                   | 0.961 $\pm$ 0.007                   |
| KaLM-mini                                                  | <b>1</b>  | <b>0.596 <math>\pm</math> 0.012</b> | <b>0.893 <math>\pm</math> 0.005</b> | <b>0.344 <math>\pm</math> 0.003</b> | <b>0.846 <math>\pm</math> 0.005</b> | <b>0.922 <math>\pm</math> 0.022</b> | <b>0.953 <math>\pm</math> 0.006</b> |
|                                                            | 2         | 0.620 $\pm$ 0.018                   | 0.893 $\pm$ 0.003                   | 0.422 $\pm$ 0.014                   | 0.845 $\pm$ 0.005                   | 0.876 $\pm$ 0.023                   | 0.955 $\pm$ 0.008                   |
|                                                            | 3         | 0.605 $\pm$ 0.011                   | 0.893 $\pm$ 0.004                   | 0.379 $\pm$ 0.008                   | 0.843 $\pm$ 0.000                   | 0.896 $\pm$ 0.018                   | 0.959 $\pm$ 0.010                   |
|                                                            | 4         | 0.562 $\pm$ 0.015                   | 0.889 $\pm$ 0.005                   | 0.265 $\pm$ 0.009                   | 0.842 $\pm$ 0.008                   | 0.945 $\pm$ 0.016                   | 0.949 $\pm$ 0.008                   |
|                                                            | 5         | 0.550 $\pm$ 0.013                   | 0.890 $\pm$ 0.004                   | 0.270 $\pm$ 0.005                   | 0.841 $\pm$ 0.004                   | 0.910 $\pm$ 0.019                   | 0.952 $\pm$ 0.009                   |
|                                                            | 6         | 0.522 $\pm$ 0.016                   | 0.893 $\pm$ 0.004                   | 0.203 $\pm$ 0.012                   | 0.850 $\pm$ 0.005                   | 0.934 $\pm$ 0.014                   | 0.949 $\pm$ 0.014                   |
|                                                            | 7         | 0.604 $\pm$ 0.018                   | 0.893 $\pm$ 0.003                   | 0.393 $\pm$ 0.018                   | 0.840 $\pm$ 0.008                   | 0.876 $\pm$ 0.023                   | 0.962 $\pm$ 0.008                   |
|                                                            | 8         | 0.607 $\pm$ 0.014                   | 0.866 $\pm$ 0.003                   | 0.388 $\pm$ 0.011                   | 0.798 $\pm$ 0.008                   | 0.891 $\pm$ 0.017                   | 0.954 $\pm$ 0.007                   |
|                                                            | 9         | 0.614 $\pm$ 0.011                   | 0.874 $\pm$ 0.005                   | 0.387 $\pm$ 0.008                   | 0.821 $\pm$ 0.008                   | 0.908 $\pm$ 0.013                   | 0.943 $\pm$ 0.008                   |
|                                                            | 10        | 0.602 $\pm$ 0.013                   | 0.859 $\pm$ 0.008                   | 0.360 $\pm$ 0.005                   | 0.774 $\pm$ 0.010                   | 0.916 $\pm$ 0.017                   | 0.969 $\pm$ 0.010                   |
|                                                            | 11        | 0.601 $\pm$ 0.012                   | 0.864 $\pm$ 0.008                   | 0.361 $\pm$ 0.007                   | 0.792 $\pm$ 0.010                   | 0.911 $\pm$ 0.020                   | 0.956 $\pm$ 0.008                   |
|                                                            | 12        | 0.640 $\pm$ 0.010                   | 0.868 $\pm$ 0.006                   | 0.416 $\pm$ 0.006                   | 0.801 $\pm$ 0.007                   | 0.929 $\pm$ 0.019                   | 0.954 $\pm$ 0.006                   |
|                                                            | 13        | 0.645 $\pm$ 0.011                   | 0.862 $\pm$ 0.005                   | 0.436 $\pm$ 0.005                   | 0.793 $\pm$ 0.004                   | 0.915 $\pm$ 0.018                   | 0.951 $\pm$ 0.012                   |
|                                                            | 14        | 0.654 $\pm$ 0.010                   | 0.859 $\pm$ 0.006                   | 0.479 $\pm$ 0.004                   | 0.795 $\pm$ 0.008                   | 0.880 $\pm$ 0.017                   | 0.941 $\pm$ 0.011                   |
|                                                            | 15        | 0.661 $\pm$ 0.009                   | 0.870 $\pm$ 0.006                   | 0.460 $\pm$ 0.009                   | 0.814 $\pm$ 0.009                   | 0.921 $\pm$ 0.014                   | 0.941 $\pm$ 0.006                   |
|                                                            | 16        | 0.685 $\pm$ 0.008                   | 0.870 $\pm$ 0.007                   | 0.477 $\pm$ 0.009                   | 0.824 $\pm$ 0.008                   | 0.953 $\pm$ 0.015                   | 0.930 $\pm$ 0.008                   |
|                                                            | 17        | 0.757 $\pm$ 0.006                   | 0.886 $\pm$ 0.010                   | 0.612 $\pm$ 0.016                   | 0.841 $\pm$ 0.005                   | 0.944 $\pm$ 0.010                   | 0.943 $\pm$ 0.016                   |
|                                                            | 18        | 0.781 $\pm$ 0.008                   | 0.880 $\pm$ 0.008                   | 0.636 $\pm$ 0.013                   | 0.840 $\pm$ 0.002                   | 0.967 $\pm$ 0.007                   | 0.931 $\pm$ 0.016                   |
|                                                            | 19        | 0.805 $\pm$ 0.012                   | 0.877 $\pm$ 0.008                   | 0.673 $\pm$ 0.020                   | 0.831 $\pm$ 0.007                   | 0.976 $\pm$ 0.003                   | 0.935 $\pm$ 0.011                   |
|                                                            | 20        | 0.794 $\pm$ 0.010                   | 0.878 $\pm$ 0.006                   | 0.650 $\pm$ 0.018                   | 0.831 $\pm$ 0.003                   | 0.979 $\pm$ 0.004                   | 0.937 $\pm$ 0.011                   |
|                                                            | 21        | 0.805 $\pm$ 0.010                   | 0.871 $\pm$ 0.007                   | 0.673 $\pm$ 0.016                   | 0.831 $\pm$ 0.004                   | 0.975 $\pm$ 0.006                   | 0.924 $\pm$ 0.012                   |
|                                                            | 22        | 0.871 $\pm$ 0.006                   | 0.873 $\pm$ 0.006                   | 0.811 $\pm$ 0.005                   | 0.813 $\pm$ 0.007                   | 0.949 $\pm$ 0.006                   | 0.951 $\pm$ 0.007                   |
|                                                            | 23        | 0.873 $\pm$ 0.007                   | 0.885 $\pm$ 0.005                   | 0.811 $\pm$ 0.006                   | 0.835 $\pm$ 0.007                   | 0.953 $\pm$ 0.007                   | 0.950 $\pm$ 0.006                   |

(continued on next page)

| Model | Layer | Routing: DirAcc@1 (five reseeds, mean $\pm$ std, SIF+ABTT) |                        |                    |                        |                   |                   |
|-------|-------|------------------------------------------------------------|------------------------|--------------------|------------------------|-------------------|-------------------|
|       |       | DirAcc@1<br>(base)                                         | DirAcc@1<br>(SIF+ABTT) | Existing<br>(base) | Existing<br>(SIF+ABTT) | New<br>(base)     | New<br>(SIF+ABTT) |
|       | 24    | 0.848 $\pm$ 0.009                                          | 0.856 $\pm$ 0.004      | 0.749 $\pm$ 0.017  | 0.807 $\pm$ 0.006      | 0.975 $\pm$ 0.007 | 0.918 $\pm$ 0.009 |

## L Cluster Geometry

This appendix shows the effect of ABTT on the whole corpus at the routing layers of Table 3. Figures 10 and 11 give the t-SNE (van der Maaten and Hinton, 2008) and UMAP (McInnes et al., 2018) panels for LaTa, PhilTa, and mT5-base, and Figures 12 and 13 extend them to LaBSE, Qwen3-0.6B, and KaLM-mini. All four draw all 1,705 witnesses, color the same six largest directories, and compare baseline mean pooling with ABTT-only at  $D = 10$ . Each model sits at the layer chosen on the training split by DirAcc@1 under ABTT, the routing ABTT layer of Table 20. In both maps, ABTT draws the witnesses of each colored directory into tighter groups than baseline mean pooling does. The maps are two-dimensional, so we read them as illustrations and compute no statistic on them.

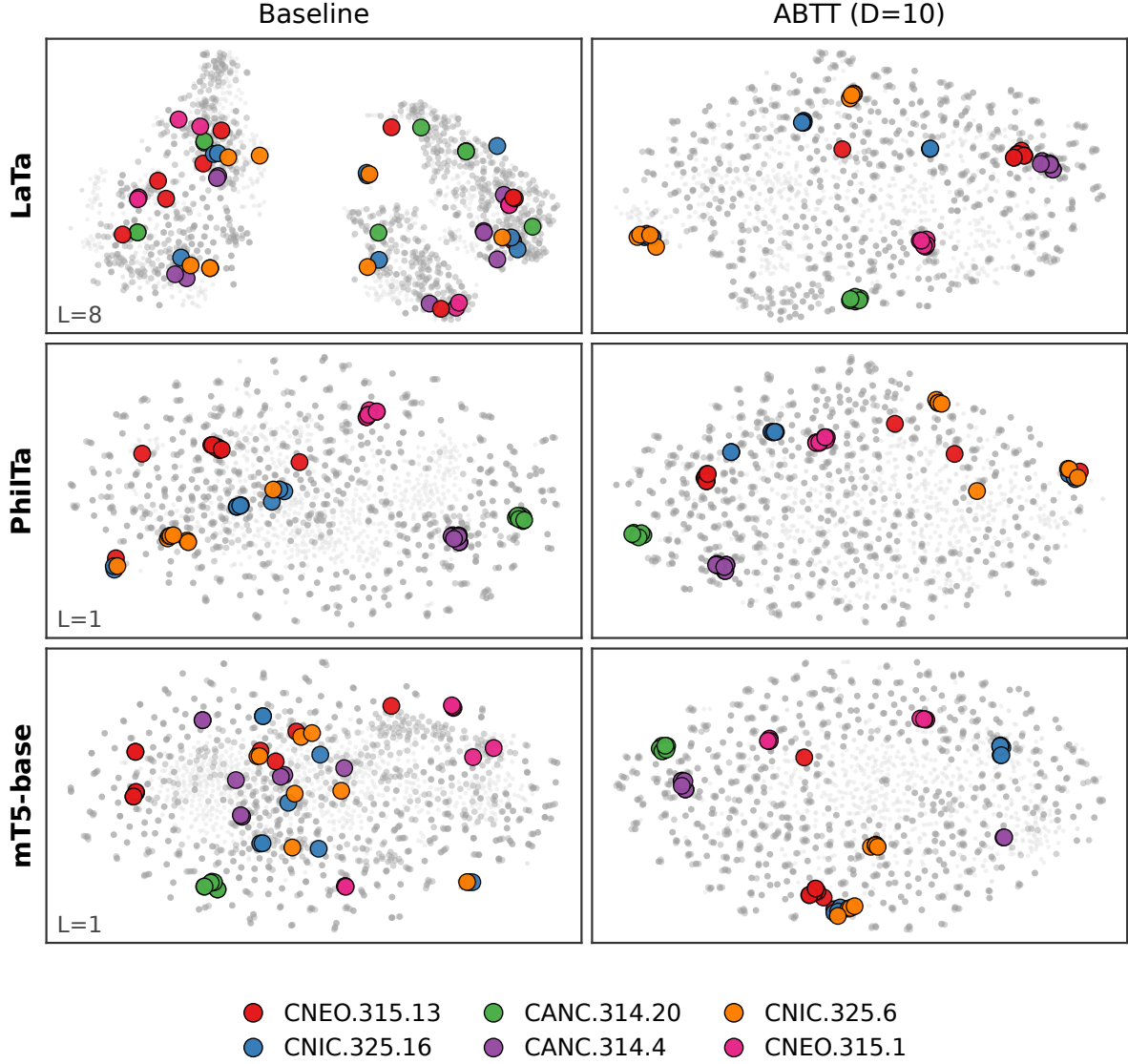


Figure 10: t-SNE maps of all 1,705 manuscript witnesses for LaTa, PhilTa, and mT5-base, each at the layer chosen on the training split: the layer with the highest training-set DirAcc@1 under ABTT, printed in each panel as L and equal to the routing ABTT layer in Table 20. Columns compare baseline mean pooling with ABTT-only using  $D = 10$ , fit on the training embeddings. Colors mark the six largest directories; other witnesses of directories with at least two witnesses are mid gray, and singletons light gray.

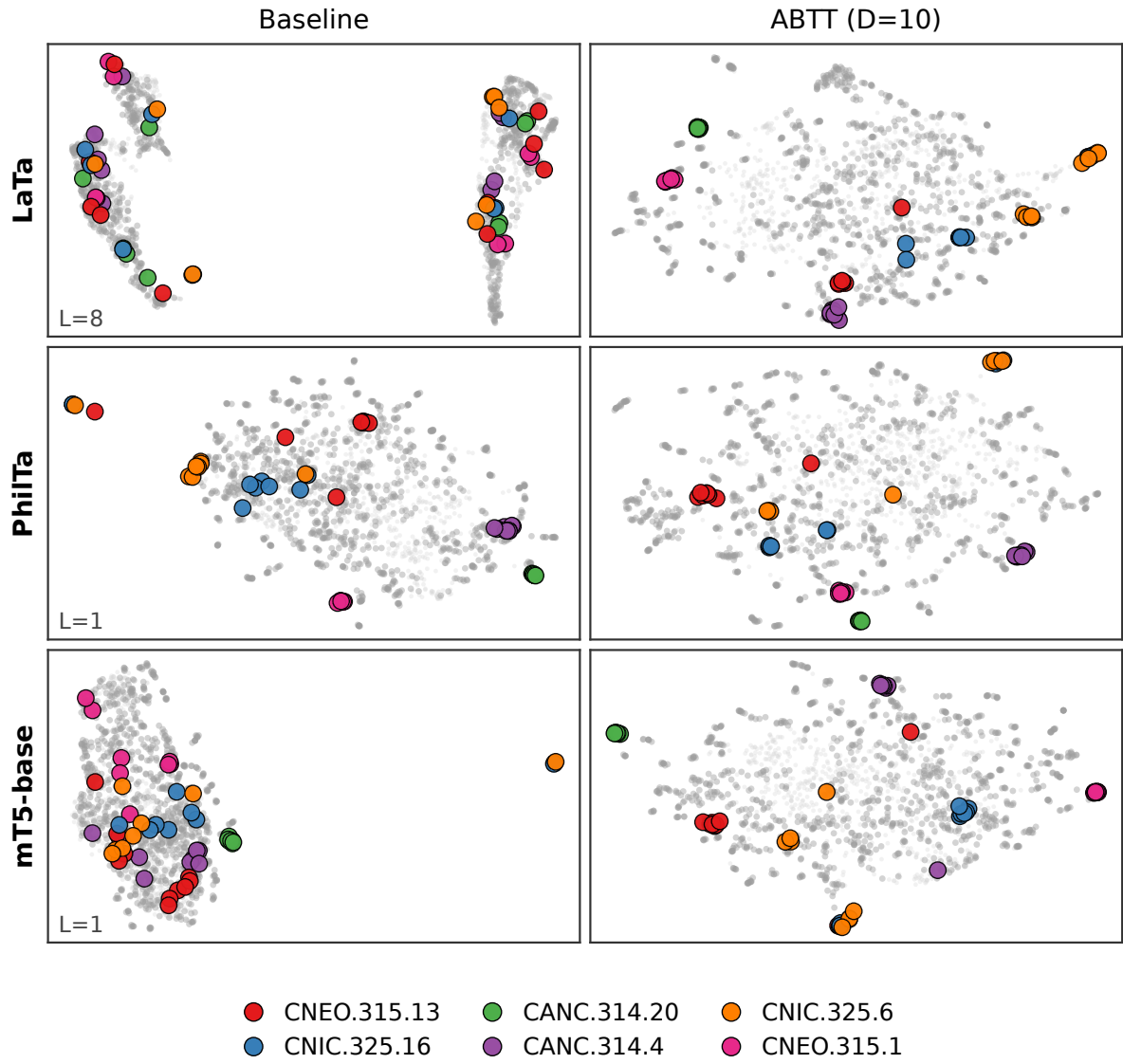


Figure 11: UMAP maps with the same configuration as Figure 10. The same cluster-tightening pattern appears under a different mapping method.

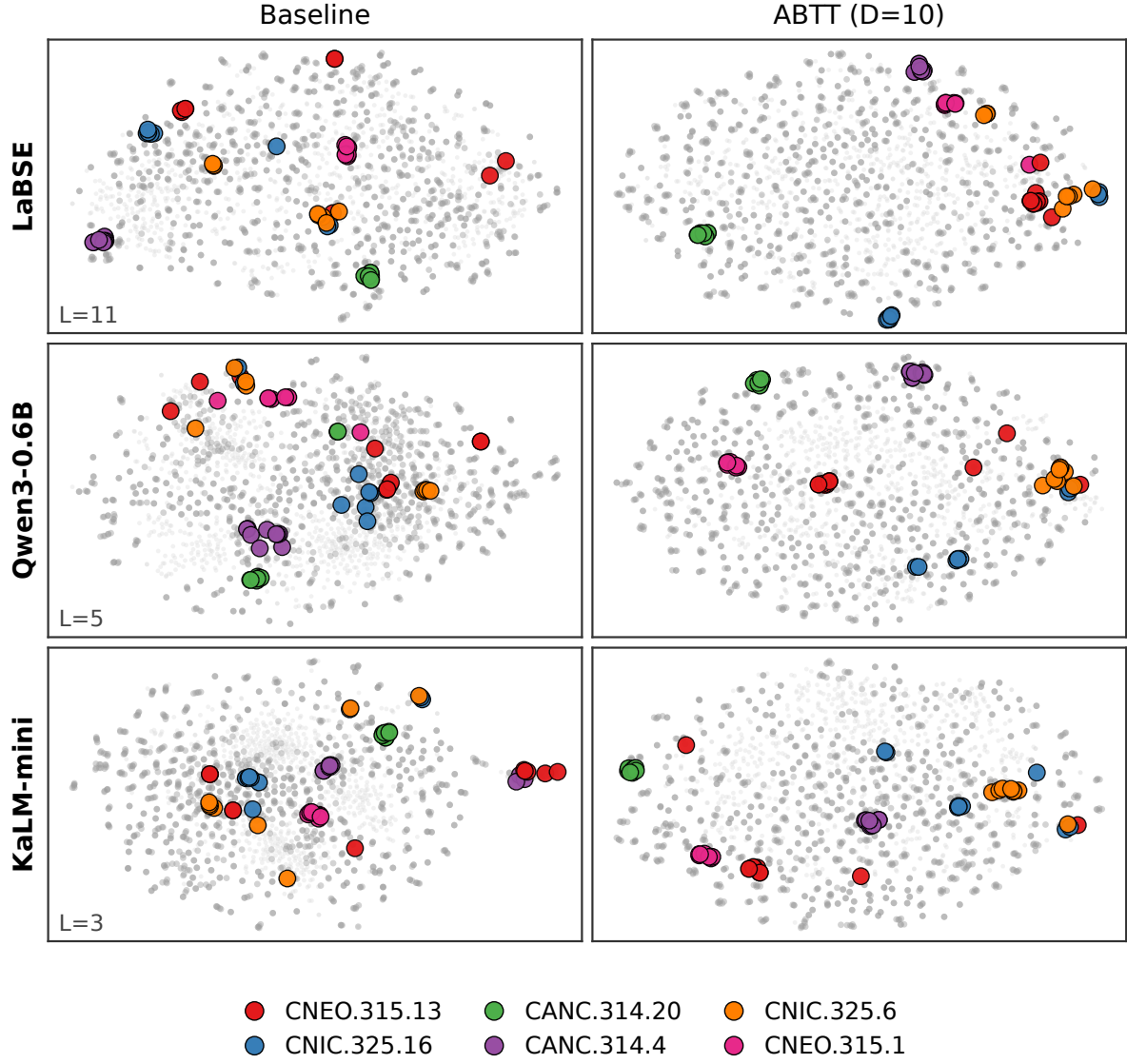


Figure 12: t-SNE maps of all 1,705 manuscript witnesses for LaBSE, Qwen3-0.6B, and KaLM-mini under the layer rule of Figure 10: the layer with the highest training-set DirAcc@1 under ABTT, printed in each panel and equal to the routing ABTT layer in Table 20. Columns compare baseline mean pooling with ABTT-only using  $D = 10$ . Colors mark the same six largest directories as Figure 10.

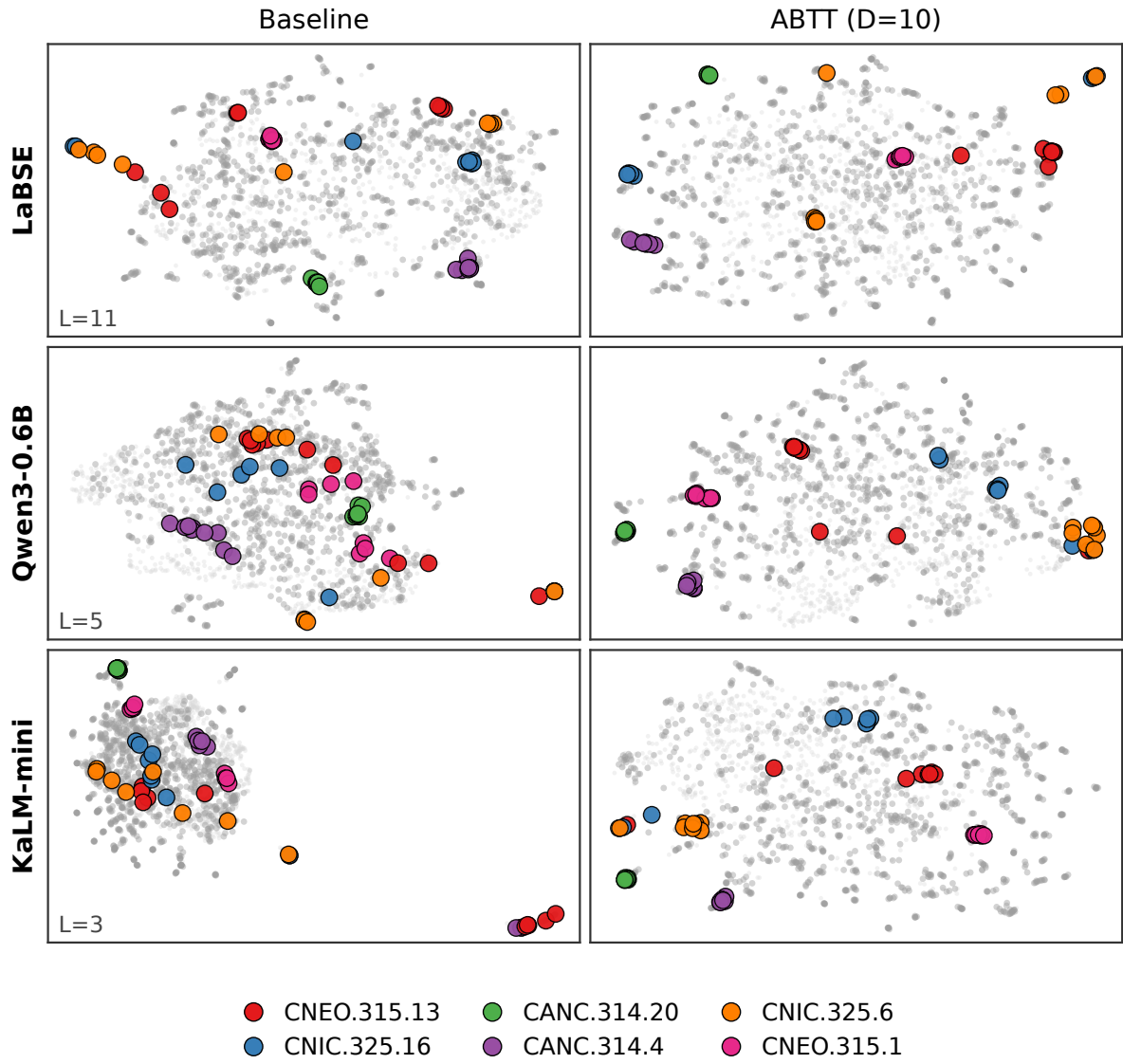


Figure 13: UMAP maps matching Figure 12. The cluster-tightening pattern appears under a second mapping method.